

Team Members:

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GitHub Repo URL : [Repo URL](#)

Removing Existing Python Version

1. **history -c && history -w**: Clear all the history of the specific user

```
swayam@swayam:~$ history -c && history -w
```

```
swayam@swayam:~$ history -c && history -w  
swayam@swayam:~$
```

2. **history**: Displays a list of previously executed commands in the terminal.

```
swayam@swayam:~$ history -c && history -w
swayam@swayam:~$ history
 1  history
swayam@swayam:~$
```

3. **python3 --version**: Checks the version of Python 3 currently installed on the system.

```
swayam@swayam:~$ history -c && history -w
swayam@swayam:~$ history
 1  history
swayam@swayam:~$ python3 --version
```

```
swayam@swayam:~$ history -c && history -w
swayam@swayam:~$ history
1  history
swayam@swayam:~$ python3 --version
Python 3.12.3
swayam@swayam:~$ _
```

4. `sudo apt list | grep -i python3.10`: If the package for Python 3.10 is installed, then it is removed

```
swayamiit@hpvictus:~$ sudo apt list | grep -i python3.10
[sudo] password for swayamiit:

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

swayamiit@hpvictus:~$
```

(This command doesn't work)

5. `sudo apt remove python3`: Removes the Python 3 package from the system using the apt package manager.

```
swayam@swayam:~$ history -c && history -w
swayam@swayam:~$ history
 1 history
swayam@swayam:~$ python3 --version
Python 3.12.3
swayam@swayam:~$ sudo apt remove python3

Removing python3-persistent:amd64 (0.20.0-1build2) ...
Removing python3-serial (3.5-2) ...
Removing python3-sysvend (235-1build4) ...
Removing python3-tz (2024.1-2) ...
Removing python3-xkit (0.5.0ubuntu6) ...
Removing xfsprogs (6.6.0-1ubuntu2) ...
Removing python3-update-manager (1:24.04.6) ...
Removing apport (2.28.1-0ubuntu3.1) ...
Removing apport-core-dump-handler (2.28.1-0ubuntu3.1) ...
Removing grub-pc (2.12-1ubuntu7.1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
Removing grub-gfxpayload-lists (0.7build2) ...
Removing grub2-common (2.12-1ubuntu7.1) ...
Removing grub-ec2-bin (2.12-1ubuntu7.1) ...
Removing grub-common (2.12-1ubuntu7.1) ...
Removing python3-apport (2.28.1-0ubuntu3.1) ...
Removing python3-problem-report (2.28.1-0ubuntu3.1) ...
Removing python3-launchpadlib (1.11.0-6) ...
Removing python3-lazr.restfulclient (0.14.6-1) ...
Removing python3-wadllib (1.3.6-5) ...
Removing python3-distupgrader (1:24.04.22) ...
Removing python3-apt (2.7.7ubuntu3) ...
Removing python3-oauthlib (3.2.2-1) ...
Removing python3-blinker (1.7.0-1) ...
Removing python3-requests (2.31.0+dfsg-1ubuntu1) ...
Removing python3-certifi (2023.11.17-1) ...
Removing python3-cryptography (41.0.7-4ubuntu0.1) ...
Removing python3-cffi-backend:amd64 (1.16.0-2build1) ...
Removing python3-chardet (5.2.0+dfsg-1) ...
Removing python3-dbus (1.3.2-5build3) ...
Removing python3-distro (1.9.0-1) ...
Removing python3-distro-info (1.7build1) ...
Removing python3-gi (3.48.2-1) ...
Removing python3-httplib2 (0.20.4-3) ...
Removing python3-idna (3.6-2ubuntu0.1) ...
Removing python3-jwt (2.7.0-1) ...
Removing python3-lazr.uri (1.0.6-3) ...
Removing python3-pkg-resources (68.1.2-2ubuntu1.1) ...
Removing python3-pyparsing (3.1.1-1) ...
Removing python3-six (1.16.0-4) ...
Removing python3-urllib3 (2.0.7-1ubuntu0.1) ...
Removing python3-yaml (6.0.1-2build2) ...
Removing python3 (3.12.3-0ubuntu2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ _
```

6. `sudo apt autoremove`: Removes unnecessary packages and dependencies from the system.

```

Removing python3-persistent:amd64 (0.20.0-1build2) ...
Removing python3-serial (3.5-2) ...
Removing python3-systemd (235-1build4) ...
Removing python3-tz (2024.1-2) ...
Removing python3-xkit (0.5.0ubuntu6) ...
Removing xfsprogs (6.6.0-1ubuntu2) ...
Removing python3-update-manager (1:24.04.6) ...
Removing apport (2.28.1-0ubuntu3.1) ...
Removing apport-core-dump-handler (2.28.1-0ubuntu3.1) ...
Removing grub-pc (2.12-1ubuntu7.1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
Removing grub-xfsinstall-lists (0.7build2) ...
Removing grub2-common (2.12-1ubuntu7.1) ...
Removing grub-pc-bin (2.12-1ubuntu7.1) ...
Removing grub-common (2.12-1ubuntu7.1) ...
Removing python3-apport (2.28.1-0ubuntu3.1) ...
Removing python3-problem-report (2.28.1-0ubuntu3.1) ...
Removing python3-launchpadlib (1.11.0-6) ...
Removing python3-lazr.restfulclient (0.14.6-1) ...
Removing python3-wadllib (1.3.6-5) ...
Removing python3-distupgrader (1:24.04.22) ...
Removing python3-apt (2.7.7ubuntu3) ...
Removing python3-cauthlib (3.2.2-1) ...
Removing python3-blinker (1.7.0-1) ...
Removing python3-requests (2.31.0+dfsg-1ubuntu1) ...
Removing python3-certifi (2023.11.17-1) ...
Removing python3-cryptography (41.0.7-4ubuntu0.1) ...
Removing python3-cffi-backend:amd64 (1.16.0-2build1) ...
Removing python3-chardet (5.2.0+dfsg-1) ...
Removing python3-dbus (1.3.2-5build3) ...
Removing python3-distro (1.9.0-1) ...
Removing python3-distro-info (1.7build1) ...
Removing python3-gi (3.40.2-1) ...
Removing python3-httplib2 (0.20.4-3) ...
Removing python3-idna (3.6-2ubuntu0.1) ...
Removing python3-jwt (2.7.0-1) ...
Removing python3-lazr.uri (1.0.6-3) ...
Removing python3-pkg-resources (68.1.2-2ubuntu1.1) ...
Removing python3-pyparsing (3.1.1-1) ...
Removing python3-slix (1.16.0-4) ...
Removing python3-urllib3 (2.0.7-1ubuntu0.1) ...
Removing python3-yaml (6.0.1-2build2) ...
Removing python3 (3.12.3-0ubuntu2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt autoremove

```

```

Removing iproute2 (6.1.0-1ubuntu6) ...
Removing iso-codes (4.16.0-1) ...
Removing packagekit-tools (1.2.0-2build3) ...
Removing packagekit (1.2.0-2build3) ...
Removing libaststream5:amd64 (1.0.2-build6) ...
Removing libatm1t64:amd64 (1:2.5.1-5.1build1) ...
Removing libgststreamer1.0-0:amd64 (1.24.2-1ubuntu0.1) ...
Removing libcap2-bin (1:2.66-5ubuntu2) ...
Removing libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Removing libeatmydata1:amd64 (131-1ubuntu1) ...
Removing libefiboot1t64:amd64 (38-3.1build1) ...
Removing libefivar1t64:amd64 (38-3.1build1) ...
Removing libgirepository-1.0-1:amd64 (1.80.1-1) ...
Removing libglib2.0-bin (2.80.0-6ubuntu3.2) ...
Removing libinih1:amd64 (55-1ubuntu2) ...
Removing libmnl0:amd64 (1.0.5-2build1) ...
Removing netplan-generator (1.0.1-1ubuntu2)24.04.1) ...
dpkg: warning: while removing netplan-generator, directory '/etc/netplan' not empty so not removed
Removing libnetplan1:amd64 (1.0.1-1ubuntu2)24.04.1) ...
Removing libpackagekit-glib2-18:amd64 (1:2.0-2build3) ...
Removing libpam-cap:amd64 (1:2.66-5ubuntu2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
Removing libpython3-stdlib:amd64 (3.12.3-0ubuntu2) ...
Removing libstemmer0d:amd64 (2.2.0-4build1) ...
Removing libtirpc3t64:amd64 (1.3.4+ds-1.1build1) ...
Removing 'diversion of /lib/x86_64-linux-gnu/libtirpc.so.3 to /lib/x86_64-linux-gnu/libtirpc.so.3.usr-is-merged by libtirpc3t64'
Removing 'diversion of /lib/x86_64-linux-gnu/libtirpc.so.3.0.0 to /lib/x86_64-linux-gnu/libtirpc.so.3.0.0.usr-is-merged by libtirpc3t64'
Removing libtirpc-common (1.3.4+ds-1.1build1) ...
Removing libuchardet0:amd64 (0.0.8-1build1) ...
Removing libxmlb2:amd64 (0.3.18-1) ...
Removing libxtables12:amd64 (1.8.10-3ubuntu2) ...
Removing libyaml-0-2:amd64 (0.2.5-1build1) ...
Removing pciutils (1:3.10.0-2build1) ...
Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmu1) ...
Removing ucftools (1:0.17-3build1) ...
Processing triggers for systemd (255.4-1ubuntu0.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ _

```

Installing Python3.10 using tarball

7. **sudo apt install wget:** Installs the wget package, a command-line tool for downloading files from the internet.

```

Removing iproute2 (6.1.0-1ubuntu6) ...
Removing iso-codes (4.16.0-1) ...
Removing packagekit-tools (1.2.8-2build3) ...
Removing packagekit (1.2.8-2build3) ...
Removing libappstream5:amd64 (1.0.2-1build6) ...
Removing libatm1t64:amd64 (1:2.5.1-5.1build1) ...
Removing libgstreamer1.0-0:amd64 (1.24.2-1ubuntu0.1) ...
Removing libcap2-bin (1:2.66-5ubuntu2) ...
Removing libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Removing libeatmydata1:amd64 (131-1ubuntu1) ...
Removing libefivar1t64:amd64 (38-3.1build1) ...
Removing libefivar1t64:amd64 (38-3.1build1) ...
Removing libgnupg1.0-1:amd64 (1.40.0-1) ...
Removing libglib2.0-bin (2.80.0-6ubuntu3.2) ...
Removing libinih1:amd64 (55-1ubuntu2) ...
Removing libnm0:amd64 (1.0.5-2build1) ...
Removing netplan-generator (1.0.1-1ubuntu2~24.04.1) ...
dpkg: warning: while removing netplan-generator, directory '/etc/netplan' not empty so not removed
Removing libnetplan1:amd64 (1.0.1-1ubuntu2~24.04.1) ...
Removing libpackagekit-glib2-10:amd64 (1.2.8-2build3) ...
Removing libpam-cap:amd64 (1:2.66-5ubuntu2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
Removing libpython3-stdlib:amd64 (3.12.3-0ubuntu2) ...
Removing libstemmer0d:amd64 (2.2.0-4build1) ...
Removing libtirpc3t64:amd64 (1.3.0+ds-1.1build1) ...
Removing 'diversion of /lib/x86_64-linux-gnu/libtirpc.so.3 to /lib/x86_64-linux-gnu/libtirpc.so.3.usr-is-merged by libtirpc3t64'
Removing 'diversion of /lib/x86_64-linux-gnu/libtirpc.so.3.0.0 to /lib/x86_64-linux-gnu/libtirpc.so.3.0.0.usr-is-merged by libtirpc3t64'
Removing libtirpc-common (1.3.4+ds-1.1build1) ...
Removing libuchardet0:amd64 (0.0.8-1build1) ...
Removing libxmlb2:amd64 (0.3.18-1) ...
Removing libxtables12:amd64 (1.8.10-3ubuntu2) ...
Removing libyam1-0-2:amd64 (0.2.5-1build1) ...
Removing pciutils (1:3.10.0-2build1) ...
Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmu1) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget_

```

```

Removing libyam1-0-2:amd64 (0.2.5-1build1) ...
Removing pciutils (1:3.10.0-2build1) ...
Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmu1) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 338 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$

```

8. `mkdir ~/libraries ~/local`: Creates two new directories, "libraries" and "local", in the user's home directory.

```

Removing libyam1-0-2:amd64 (0.2.5-1build1) ...
Removing pciutils (1:3.10.0-2build1) ...
Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmui) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local

Removing pciutils (1:3.10.0-2build1) ...
Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmui) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local
swayam@swayam:~$

```

9. `cd ~/libraries`: Changes the current working directory to the newly created "libraries" directory.

```

Removing pciutils (1:3.10.0-2build1) ...
Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmu1) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local
swayam@swayam:~$ cd ~/libraries_

```

```

Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmu1) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local
swayam@swayam:~$ cd ~/libraries
swayam@swayam:~/libraries$

```

10. `wget https://www.python.org/ftp/python/3.10.16/Python-3.10.16.tgz`: Downloads the Python 3.10.16 source code tarball from the official Python website.


```

Removing python-apt-common (2.7.7ubuntu3) ...
Removing python-babel-localedata (2.10.3-3build1) ...
Removing python3-minimal (3.12.3-0ubuntu2) ...
Removing python3.12 (3.12.3-1ubuntu0.4) ...
Removing python3.12-minimal (3.12.3-1ubuntu0.4) ...
Unlinking and removing bytecode for runtime python3.12
Removing ucf (3.0043+nmui) ...
Removing usbutils (1:017-3build1) ...
Processing triggers for systemd (255.4-1ubuntu8.4) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget.1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local
swayam@swayam:~$ cd ~/libraries
swayam@swayam:~/libraries$ wget https://www.python.org/ftp/python/3.10.16/Python-3.10.16.tgz_

```

```

swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62946 files and directories currently installed.)
Preparing to unpack .../wget.1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local
swayam@swayam:~$ cd ~/libraries
swayam@swayam:~/libraries$ wget https://www.python.org/ftp/python/3.10.16/Python-3.10.16.tgz
--2025-01-25 17:45:34-- https://www.python.org/ftp/python/3.10.16/Python-3.10.16.tgz
Resolving www.python.org (www.python.org)... 151.101.192.223, 151.101.128.223, 151.101.0.223, ...
Connecting to www.python.org (www.python.org)[151.101.192.223]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 25942994 (25M) [application/octet-stream]
Saving to: 'Python-3.10.16.tgz'

Python-3.10.16.tgz      100%[=====] 24.74M  14.7MB/s   in 1.7s

2025-01-25 17:45:36 (14.7 MB/s) - 'Python-3.10.16.tgz' saved [25942994/25942994]

swayam@swayam:~/libraries$ _

```

11. `tar -xvf Python-3.10.16.tgz`: Extracts the contents of the Python source code tarball.

```

swayam@swayam:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wget
0 upgraded, 1 newly installed, 0 to remove and 45 not upgraded.
Need to get 334 kB of archives.
After this operation, 938 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 2s (161 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package wget.
(Reading database ... 62546 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  swayam @ session #1: login[1119]
  swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ mkdir ~/libraries ~/local
swayam@swayam:~$ cd ~/libraries
swayam@swayam:~/libraries$ wget https://www.python.org/ftp/python/3.10.16/Python-3.10.16.tgz
--2025-01-25 17:45:34-- https://www.python.org/ftp/python/3.10.16/Python-3.10.16.tgz
Resolving www.python.org (www.python.org)... 151.101.192.223, 151.101.128.223, 151.101.0.223, ...
Connecting to www.python.org (www.python.org)|151.101.192.223|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 25942994 (25M) [application/octet-stream]
Saving to: 'Python-3.10.16.tgz'

Python-3.10.16.tgz      100%[=====] 24.74M  14.7MB/s   in 1.7s

2025-01-25 17:45:36 (14.7 MB/s) - 'Python-3.10.16.tgz' saved [25942994/25942994]

swayam@swayam:~/libraries$ tar -xvf Python-3.10.16.tgz_
Python-3.10.16/Tools/stringbench/stringbench.py
Python-3.10.16/Tools/test2to3/
Python-3.10.16/Tools/test2to3/README
Python-3.10.16/Tools/test2to3/main_test.py
Python-3.10.16/Tools/test2to3/setup.py
Python-3.10.16/Tools/test2to3/test/
Python-3.10.16/Tools/test2to3/test/runtests.py
Python-3.10.16/Tools/test2to3/test/test_foo.py
Python-3.10.16/Tools/test2to3/test2to3/
Python-3.10.16/Tools/test2to3/test2to3/__init__.py
Python-3.10.16/Tools/test2to3/test2to3/hello.py
Python-3.10.16/Tools/tz/
Python-3.10.16/Tools/tz/zdump.py
Python-3.10.16/Tools/unicode/
Python-3.10.16/Tools/unicode/Makefile
Python-3.10.16/Tools/unicode/comparecodecs.py
Python-3.10.16/Tools/unicode/gen_cjkcodecs.py
Python-3.10.16/Tools/unicode/gencodecs.py
Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodecs.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$

```

12. ./configure: Runs the configuration script for the Python source code, preparing it for compilation. (Wrong working directory)

```

Python-3.10.16/Tools/stringbench/stringbench.py
Python-3.10.16/Tools/test2to3/
Python-3.10.16/Tools/test2to3/README
Python-3.10.16/Tools/test2to3/mainTest.py
Python-3.10.16/Tools/test2to3/setup.py
Python-3.10.16/Tools/test2to3/test/
Python-3.10.16/Tools/test2to3/test/runtests.py
Python-3.10.16/Tools/test2to3/test/test_foo.py
Python-3.10.16/Tools/test2to3/test2to3/
Python-3.10.16/Tools/test2to3/test2to3/__init__.py
Python-3.10.16/Tools/test2to3/test2to3/hello.py
Python-3.10.16/Tools/tz/
Python-3.10.16/Tools/tz/zdump.py
Python-3.10.16/Tools/unicode/
Python-3.10.16/Tools/unicode/Makefile
Python-3.10.16/Tools/unicode/comparecodecs.py
Python-3.10.16/Tools/unicode/genjcodecs.py
Python-3.10.16/Tools/unicode/gencode.py
Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodecs.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$ ./configure_

```

13. `cd Python-3.10.16`: Changes the current working directory to the extracted Python source code directory.

```

Python-3.10.16/Tools/test2to3/README
Python-3.10.16/Tools/test2to3/mainTest.py
Python-3.10.16/Tools/test2to3/setup.py
Python-3.10.16/Tools/test2to3/test/
Python-3.10.16/Tools/test2to3/test/runtests.py
Python-3.10.16/Tools/test2to3/test/test_foo.py
Python-3.10.16/Tools/test2to3/test2to3/
Python-3.10.16/Tools/test2to3/test2to3/__init__.py
Python-3.10.16/Tools/test2to3/test2to3/hello.py
Python-3.10.16/Tools/tz/
Python-3.10.16/Tools/tz/zdump.py
Python-3.10.16/Tools/unicode/
Python-3.10.16/Tools/unicode/Makefile
Python-3.10.16/Tools/unicode/comparecodecs.py
Python-3.10.16/Tools/unicode/genjcodecs.py
Python-3.10.16/Tools/unicode/gencode.py
Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodecs.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$ ./configure
-bash: ./configure: No such file or directory
swayam@swayam:~/libraries$ cd Python-3.10.16

```

```

Python-3.10.16/Tools/test2to3/maintest.py
Python-3.10.16/Tools/test2to3/setup.py
Python-3.10.16/Tools/test2to3/test/
Python-3.10.16/Tools/test2to3/test/runtests.py
Python-3.10.16/Tools/test2to3/test/test_foo.py
Python-3.10.16/Tools/test2to3/test2to3/
Python-3.10.16/Tools/test2to3/test2to3/__init__.py
Python-3.10.16/Tools/test2to3/test2to3/hello.py
Python-3.10.16/Tools/tz/
Python-3.10.16/Tools/tz/zdump.py
Python-3.10.16/Tools/unicode/
Python-3.10.16/Tools/unicode/Makefile
Python-3.10.16/Tools/unicode/comparecodecs.py
Python-3.10.16/Tools/unicode/genjkcdecoders.py
Python-3.10.16/Tools/unicode/gencodec.py
Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodec.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$ ./configure
-bash: ./configure: No such file or directory
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$

```

14. `./configure`: Runs the configuration script again, this time with the correct directory context.

```

Python-3.10.16/Tools/test2to3/maintest.py
Python-3.10.16/Tools/test2to3/setup.py
Python-3.10.16/Tools/test2to3/test/
Python-3.10.16/Tools/test2to3/test/runtests.py
Python-3.10.16/Tools/test2to3/test/test_foo.py
Python-3.10.16/Tools/test2to3/test2to3/
Python-3.10.16/Tools/test2to3/test2to3/__init__.py
Python-3.10.16/Tools/test2to3/test2to3/hello.py
Python-3.10.16/Tools/tz/
Python-3.10.16/Tools/tz/zdump.py
Python-3.10.16/Tools/unicode/
Python-3.10.16/Tools/unicode/Makefile
Python-3.10.16/Tools/unicode/comparecodecs.py
Python-3.10.16/Tools/unicode/genjkcdecoders.py
Python-3.10.16/Tools/unicode/gencodec.py
Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodec.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$ ./configure
-bash: ./configure: No such file or directory
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$ ./configure

```

```

Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodecs.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$ ./configure
-bash: ./configure: No such file or directory
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$ ./configure
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for python3.10... no
checking for python3... no
checking for python... no
checking for --enable-universalsdk... no
checking for --with-universal-archs... no
checking MACHDEP... "linux"
checking for gcc... no
checking for cc... no
checking for cl.exe... no
configure: error: in '/home/swayam/libraries/Python-3.10.16':
configure: error: no acceptable C compiler found in $PATH
See 'config.log' for more details
swayam@swayam:~/libraries/Python-3.10.16$

```

We faced the first major error here – No acceptable C compiler found in \$PATH.

A C compiler is needed to install Python because a significant portion of the core Python interpreter itself is written in C.

We need to install build-essential package to resolve this.

Resolving errors by installing missing dependencies

15. `sudo apt install build-essential`: Installs the build-essential package, which provides necessary tools for compiling software.

```

Python-3.10.16/Tools/unicode/genmap_japanese.py
Python-3.10.16/Tools/unicode/genmap_korean.py
Python-3.10.16/Tools/unicode/genmap_schinese.py
Python-3.10.16/Tools/unicode/genmap_support.py
Python-3.10.16/Tools/unicode/genwincodecs.py
Python-3.10.16/Tools/unicode/genwincodecs.bat
Python-3.10.16/Tools/unicode/listcodecs.py
Python-3.10.16/Tools/unicode/makeunicodedata.py
Python-3.10.16/Tools/unicode/mkstringprep.py
Python-3.10.16/Tools/unicode/python-mappings/
Python-3.10.16/Tools/unicode/python-mappings/CP1140.TXT
Python-3.10.16/Tools/unicode/python-mappings/CP273.TXT
Python-3.10.16/Tools/unicode/python-mappings/GB2312.TXT
Python-3.10.16/Tools/unicode/python-mappings/KOI8-U.TXT
Python-3.10.16/Tools/unicode/python-mappings/TIS-620.TXT
Python-3.10.16/Tools/unicode/python-mappings/diff/
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2000-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/diff/jisx0213-2004-std.txt.diff
Python-3.10.16/Tools/unicode/python-mappings/gb-18030-2000.xml
Python-3.10.16/Tools/unicode/python-mappings/jisx0213-2004-std.txt
Python-3.10.16/Tools/unittestgui/
Python-3.10.16/Tools/unittestgui/README.txt
Python-3.10.16/Tools/unittestgui/unittestgui.py
Python-3.10.16/aclocal.m4
Python-3.10.16/config.guess
Python-3.10.16/config.sub
Python-3.10.16/configure
Python-3.10.16/configure.ac
Python-3.10.16/install-sh
Python-3.10.16/pyconfig.h.in
Python-3.10.16/setup.py
swayam@swayam:~/libraries$ ./configure
-bash: ./configure: No such file or directory
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$ ./configure
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for python3.10... no
checking for python3... no
checking for python... no
checking for --enable-universalsdk... no
checking for --with-universal-archs... no
checking MACHDEP... "linux"
checking for gcc... no
checking for cc... no
checking for cl.exe... no
configure: error: in '/home/swayam/libraries/Python-3.10.16':
configure: error: no acceptable C compiler found in $PATH
See 'config.log' for more details
swayam@swayam:~/libraries/Python-3.10.16$ sudo apt install build-essential

Setting up libbinutils:amd64 (2.42-4ubuntu2.3) ...
Setting up libisl23:amd64 (0.26-3build1) ...
Setting up libalgorithm-diff-xs-perl:amd64 (0.04-8build3) ...
Setting up libcc1-0:amd64 (14.2.0-4ubuntu2~24.04) ...
Setting up liblsan0:amd64 (14.2.0-4ubuntu2~24.04) ...
Setting up libitm1:amd64 (14.2.0-4ubuntu2~24.04) ...
Setting up libalgorithm-merge-perl (0.08-5) ...
Setting up libc6:amd64 (2.42-4ubuntu2.3) ...
Setting up cpp-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up libgprofng0:amd64 (2.42-4ubuntu2.3) ...
Setting up libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Setting up libstdc++-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Setting up binutils-x86-64-linux-gnu (2.42-4ubuntu2.3) ...
Setting up cpp-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up cpp-13 (13.3.0-6ubuntu2~24.04) ...
Setting up gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up binutils (2.42-4ubuntu2.3) ...
Setting up dpkg-dev (1.22.6ubuntu6.1) ...
Setting up gcc-13 (13.3.0-6ubuntu2~24.04) ...
Setting up cpp (4:13.2.0-7ubuntu1) ...
Setting up g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up gcc (4:13.2.0-7ubuntu1) ...
Setting up g++-13 (13.3.0-6ubuntu2~24.04) ...
Setting up g++ (4:13.2.0-7ubuntu1) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/c++.1.gz because associated file /usr/share/man/man1/g++.1.gz (of link group c++) doesn't exist
Setting up build-essential (12.10ubuntu1) ...
Processing triggers for libc-bin (2.39-0ubuntu0.3) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
swayam @ session #1: login[1119]
swayam @ user manager service: systemd[1190]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~/libraries/Python-3.10.16$ _

```

16. `./configure`: Runs the configuration script again, now that the necessary build tools are installed.

```

Setting up libbinutils:amd64 (2.42-4ubuntu2.3) ...
Setting up libisl23:amd64 (0.26-3build1) ...
Setting up libalgorithm-diff-xs-perl:amd64 (0.04-8build3) ...
Setting up libcc1-0:amd64 (14.2.0-4ubuntu2~24.04) ...
Setting up liblsan0:amd64 (14.2.0-4ubuntu2~24.04) ...
Setting up libitm1:amd64 (14.2.0-4ubuntu2~24.04) ...
Setting up libalgorithm-merge-perl (0.08-5) ...
Setting up libctf0:amd64 (2.42-4ubuntu2.3) ...
Setting up cpp-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up libgprofng0:amd64 (2.42-4ubuntu2.3) ...
Setting up libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Setting up libstdc++-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Setting up binutils-x86-64-linux-gnu (2.42-4ubuntu2.3) ...
Setting up cpp-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up cpp-13 (13.3.0-6ubuntu2~24.04) ...
Setting up gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up binutils (2.42-4ubuntu2.3) ...
Setting up dpkg-dev (1.22.6ubuntu6.1) ...
Setting up gcc-13 (13.3.0-6ubuntu2~24.04) ...
Setting up cpp (4:13.2.0-7ubuntu1) ...
Setting up g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up gcc (4:13.2.0-7ubuntu1) ...
Setting up g++-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up g++-13 (13.3.0-6ubuntu2~24.04) ...
Setting up g++ (4:13.2.0-7ubuntu1) ...
update-alternatives: using /usr/bin/c++ to provide /usr/bin/c++ (c++) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/c++.1.gz because associated file /usr/share/man/man1/g++.1.gz (of link group c++) doesn't exist
Setting up build-essential (12.10ubuntu1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
swayam @ session #1: login[1119]
swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~/libraries/Python-3.10.16$ ./configure

checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for -with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... upgrade
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for pkg-config... no
checking for openssl/ssl.h in /usr/local/ssl... no
checking for openssl/ssl.h in /usr/lib/ssl... no
checking for openssl/ssl.h in /usr/ssl... no
checking for openssl/ssl.h in /usr/pkg... no
checking for openssl/ssl.h in /usr/local... no
checking for openssl/ssl.h in /usr... no
checking whether compiling and linking against OpenSSL works... no
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... no
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$

```

We faced the second major error now – The openssl package seems to be missing.

OpenSSL is required to install Python in Ubuntu because Python relies on OpenSSL's cryptographic libraries to provide secure communication features. We need to install the package in order to resolve this.

17. `cd ..`: Changes the current working directory to the parent directory.

18. `wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz`: Downloads the OpenSSL source code tarball.

```
checking for socklen_t... yes
checking for broken nostdcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -D2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... upgrade
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for pkg-config... no
checking for openssl/ssl.h in /usr/local/ssl... no
checking for openssl/ssl.h in /usr/lib/ssl... no
checking for openssl/ssl.h in /usr/ssl... no
checking for openssl/ssl.h in /usr/pkg... no
checking for openssl/ssl.h in /usr/local... no
checking for openssl/ssl.h in /usr... no
checking whether compiling and linking against OpenSSL works... no
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... no
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz_

checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
--2025-01-25 17:50:23-- https://www.openssl.org/source/openssl-1.1.1w.tar.gz
Resolving www.openssl.org (www.openssl.org)... 34.49.79.89, 2600:1901:0:d50b::
Connecting to www.openssl.org (www.openssl.org) [34.49.79.89]:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com) [20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b62&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b62&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com) [185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz 100%[=====>] 9.43M 10.3MB/s in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$
```

19. `tar -xzf openssl-1.1.1w.tar.gz`: Extracts the contents of the OpenSSL source code tarball.


```

checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/_ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
--2025-01-25 17:50:23-- https://www.openssl.org/source/openssl-1.1.1w.tar.gz
Resolving www.openssl.org (www.openssl.org)... 34.49.79.89, 2600:1901:0:d50b::
Connecting to www.openssl.org (www.openssl.org)|34.49.79.89|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e7X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Ffs3%2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de933931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e7X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Ffs3%2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de933931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M  10.3MB/s   in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz

checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/_ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
--2025-01-25 17:50:23-- https://www.openssl.org/source/openssl-1.1.1w.tar.gz
Resolving www.openssl.org (www.openssl.org)... 34.49.79.89, 2600:1901:0:d50b::
Connecting to www.openssl.org (www.openssl.org)|34.49.79.89|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e7X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Ffs3%2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de933931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e7X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Ffs3%2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de933931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M  10.3MB/s   in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$

```

20. cd openssl-1.1.1w: Changes the current working directory to the extracted OpenSSL source code directory.

```

checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
--2025-01-25 17:50:23-- https://www.openssl.org/source/openssl-1.1.1w.tar.gz
Resolving www.openssl.org (www.openssl.org)... 34.49.79.89, 2600:1901:0:d50b::
Connecting to www.openssl.org (www.openssl.org)[34.49.79.89]:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/openssl-1.1.1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/openssl-1.1.1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
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Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M  10.3MB/s   in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$ cd openssl-1.1.1w

```

21. ./config --prefix=\$HOME/local: Configures the OpenSSL source code for compilation, specifying the installation prefix. (Syntax error)

```

checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
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Resolving www.openssl.org (www.openssl.org)... 34.49.79.89, 2600:1901:0:d50b::
Connecting to www.openssl.org (www.openssl.org)[34.49.79.89]:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/openssl-1.1.1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/openssl-1.1.1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=fea6b56a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream [following]
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Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M  10.3MB/s   in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local_

```

```

checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
--2025-01-25 17:50:23-- https://www.openssl.org/source/openssl-1.1.1w.tar.gz
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HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
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--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%32Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b828X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
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Length: 9893384 (9.4M) [application/octet-stream]
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openssl-1.1.1w.tar.gz      100%[=====] 9.43M 10.3MB/s in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local
-bash: ./: Is a directory
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local

```

22. ./config --prefix=\$HOME/local: Runs the configuration script again, ensuring the correct command.

```

checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
--2025-01-25 17:50:23-- https://www.openssl.org/source/openssl-1.1.1w.tar.gz
Resolving www.openssl.org (www.openssl.org)... 34.49.79.89, 2600:1901:0:d50b::
Connecting to www.openssl.org (www.openssl.org)[34.49.79.89]:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/OpenSSL_1_1_1w/openssl-1.1.1w.tar.gz [following]
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Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%32Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b828X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%32Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b828X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M 10.3MB/s in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local
-bash: ./: Is a directory
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local

```

```

HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/openssl_1.1.1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/openssl_1.1.1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
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--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3%2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M  10.3MB/s   in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local
-bash: ./: is a directory
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local
Operating system: x86_64-whatever-linux2
Configuring OpenSSL version 1.1.1w (0x1010117fL) for linux-x86_64
Using os-specific seed configuration
Creating configdata.pm
Creating Makefile

*****
***
***   OpenSSL has been successfully configured
***
***   If you encounter a problem while building, please open an
***   issue on GitHub <https://github.com/openssl/openssl/issues>
***   and include the output from the following command:
***
***       perl configdata.pm --dump
***
***   (If you are new to OpenSSL, you might want to consult the
***   'Troubleshooting' section in the INSTALL file first)
***
***
*****
swayam@swayam:~/libraries/openssl-1.1.1w$ _

```

23. make -j\$(nproc): Compiles the OpenSSL source code using multiple CPU cores.

```

HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/openssl/openssl/releases/download/openssl_1.1.1w/openssl-1.1.1w.tar.gz [following]
--2025-01-25 17:50:23-- https://github.com/openssl/openssl/releases/download/openssl_1.1.1w/openssl-1.1.1w.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
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--2025-01-25 17:50:24-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/7634677/b4c6e1f3-d928-4de0-a502-fc8dae2adf1e?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2F%3%2Faws4_request&X-Amz-Date=20250125T175024Z&X-Amz-Expires=300&X-Amz-Signature=feab656a2427fcf1ea564de9333931e0b4b51e092c5c0506154bc9d1d5284b82&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dopenssl-1.1.1w.tar.gz&response-content-type=application%2Foctet-stream
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Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9893384 (9.4M) [application/octet-stream]
Saving to: 'openssl-1.1.1w.tar.gz'

openssl-1.1.1w.tar.gz      100%[=====] 9.43M  10.3MB/s   in 0.9s

2025-01-25 17:50:26 (10.3 MB/s) - 'openssl-1.1.1w.tar.gz' saved [9893384/9893384]

swayam@swayam:~/libraries$ tar -xzf openssl-1.1.1w.tar.gz
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local
-bash: ./: is a directory
swayam@swayam:~/libraries/openssl-1.1.1w$ ./config --prefix=$HOME/local
Operating system: x86_64-whatever-linux2
Configuring OpenSSL version 1.1.1w (0x1010117fL) for linux-x86_64
Using os-specific seed configuration
Creating configdata.pm
Creating Makefile

*****
***
***   OpenSSL has been successfully configured
***
***   If you encounter a problem while building, please open an
***   issue on GitHub <https://github.com/openssl/openssl/issues>
***   and include the output from the following command:
***
***       perl configdata.pm --dump
***
***   (If you are new to OpenSSL, you might want to consult the
***   'Troubleshooting' section in the INSTALL file first)
***
***
*****
swayam@swayam:~/libraries/openssl-1.1.1w$ make -j$(nproc)_

```

```

rm -f test/versions
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/versions test/versions.o \
-lcrypto -ldl -pthread
rm -f test/wpackettest
rm -f test/x509_check_cert_pkey_test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/wpackettest test/wpackettest.o \
libssl.a test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_check_cert_pkey_test test/x509_check_cert_pkey_test.o \
test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/x509_dup_cert_test
rm -f test/x509_internal_test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_dup_cert_test test/x509_dup_cert_test.o \
test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_internal_test test/x509_internal_test.o \
test/libtestutil.a libcrypto.a -ldl -pthread
rm -f test/x509_time_test
rm -f test/x509aux
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_time_test test/x509_time_test.o \
test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509aux test/x509aux.o \
test/libtestutil.a -lcrypto -ldl -pthread
rm -f apps/openssl
rm -f fuzz/asn1-test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o apps/openssl apps/asn1pars.o apps/ca.o apps/ciphers.o apps/cms.o apps/crl.o apps/crl2p7.o apps/dgst.o apps/dhparam.o apps/dsa.o apps/dsaparam.o apps/
ec.o apps/ecparam.o apps/enc.o apps/engine.o apps/errstr.o apps/gendsa.o apps/genpkey.o apps/genrsa.o apps/nseq.o apps/ocsp.o apps/openssl.o apps/passwd.o apps/
pkcs12.o apps/pkcs7.o apps/pkcs8.o apps/pkey.o apps/pkeyparam.o apps/pkeyutil.o apps/prime.o apps/rand.o apps/rehash.o apps/req.o apps/rsa.o apps/rsautl.o apps/s
_client.o apps/s_server.o apps/s_time.o apps/session.o apps/smime.o apps/speed.o apps/spkac.o apps/srp.o apps/storeutl.o apps/ts.o apps/verify.o apps/version.o
apps/x509.o \
apps/libapps.a -lssl -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/asn1-test fuzz/asn1.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
rm -f fuzz/client-test
rm -f fuzz/server-test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/client-test fuzz/client.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/server-test fuzz/server.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
make[1]: Leaving directory '/home/swayam/libraries/openssl-1.1.1w'
swayam@swayam:~/libraries/openssl-1.1.1w$

```

24. make install: Installs the compiled OpenSSL software.

```

rm -f test/versions
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/versions test/versions.o \
-lcrypto -ldl -pthread
rm -f test/wpackettest
rm -f test/x509_check_cert_pkey_test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/wpackettest test/wpackettest.o \
libssl.a test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_check_cert_pkey_test test/x509_check_cert_pkey_test.o \
test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/x509_dup_cert_test
rm -f test/x509_internal_test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_dup_cert_test test/x509_dup_cert_test.o \
test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_internal_test test/x509_internal_test.o \
test/libtestutil.a libcrypto.a -ldl -pthread
rm -f test/x509_time_test
rm -f test/x509aux
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509_time_test test/x509_time_test.o \
test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/x509aux test/x509aux.o \
test/libtestutil.a -lcrypto -ldl -pthread
rm -f apps/openssl
rm -f fuzz/asn1-test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o apps/openssl apps/asn1pars.o apps/ca.o apps/ciphers.o apps/cms.o apps/crl.o apps/crl2p7.o apps/dgst.o apps/dhparam.o apps/dsa.o apps/dsaparam.o apps/
ec.o apps/ecparam.o apps/enc.o apps/engine.o apps/errstr.o apps/gendsa.o apps/genpkey.o apps/genrsa.o apps/nseq.o apps/ocsp.o apps/openssl.o apps/passwd.o apps/
pkcs12.o apps/pkcs7.o apps/pkcs8.o apps/pkey.o apps/pkeyparam.o apps/pkeyutil.o apps/prime.o apps/rand.o apps/rehash.o apps/req.o apps/rsa.o apps/rsautl.o apps/s
_client.o apps/s_server.o apps/s_time.o apps/session.o apps/smime.o apps/speed.o apps/spkac.o apps/srp.o apps/storeutl.o apps/ts.o apps/verify.o apps/version.o
apps/x509.o \
apps/libapps.a -lssl -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/asn1-test fuzz/asn1.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
rm -f fuzz/client-test
rm -f fuzz/server-test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/client-test fuzz/client.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/server-test fuzz/server.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
make[1]: Leaving directory '/home/swayam/libraries/openssl-1.1.1w'
swayam@swayam:~/libraries/openssl-1.1.1w$ make install

```

25. cd ..: Changes the current working directory to the parent directory.

[illegible]

26. `cd Python-3.10.16`: Changes the current working directory to the Python source code directory.

[illegible]

- ```
27. ./configure --prefix=$HOME/local --with-openssl=$HOME/local --with-ensurepip=install \
LD_FLAGS="-L$HOME/local/lib" \
CPP_FLAGS="-I$HOME/local/include" \
PKG_CONFIG_PATH="$HOME/local/lib/pkgconfig" \
--enable-optimization: Configures the Python source code for compilation, specifying various options and prefixes. (Syntax error)
```





```

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$

```

## 29. make -j\$(nproc): Compiles the Python source code using multiple CPU cores.

```

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ make -j$(nproc)

CC='gcc' LDSHARED='gcc -shared' OPT='-DDEBUG -g -fwrapv -O3 -Wall' _TCLTK_INCLUDES='' _TCLTK_LIBS='' ./python -E ./setup.py build
running build
running build_ext
INFO: Can't locate Tcl/Tk libs and/or headers
building '_ctypes' extension
gcc -fPIC -Wno-unused-result -Wsign-compare -DDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializers -Werror-implicit-function-declaration -fvisibility=hidden -I./Include/internal -I./Include -I./usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/libraries/Python-3.10.16/Modules/_ctypes/_ctypes.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/_ctypes.o -DPy_BUILD_CORE_MODULE
/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/_ctypes.c:107:10: fatal error: ffi.h: No such file or directory
 107 | #include <ffi.h>
 | ^~~~~~
compilation terminated.

```



```

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline zlib
To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

Failed to build these modules:
_ctypes

Following modules built successfully but were removed because they could not be imported:
_hashlib _ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
creating build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz

```

We could not build the ssl module because the necessary bits to build some other dependencies like libffi, zlib etc. were not found.

Let's now install other dependencies starting with libffi ...

30. `cd ..`: Changes the current working directory to the parent directory

31. `wget https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz`:

Downloads the libffi source code tarball.

```

bmodexp/constants.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/context.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/convolute.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/crt.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/difradix2.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/fmt.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/fourstep.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/io.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/mpalloc.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/mpdecimal.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/numbertheory.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/sixstep.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.o -L/home/swayam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lm -o build/lib.linux-x86_64-3.10/_decimal.cpython-310-x86_64-linux-gnu.so
*** WARNING: renaming "_ssl" since importing it failed: libssl.so.1.1: cannot open shared object file: No such file or directory
*** WARNING: renaming "_hashlib" since importing it failed: libcrypto.so.1.1: cannot open shared object file: No such file or directory

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline zlib
To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

Failed to build these modules:
_ctypes

Following modules built successfully but were removed because they could not be imported:
_hashlib _ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
creating build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz

```

```

Failed to build these modules:
_ctypes

Following modules built successfully but were removed because they could not be imported:
_hashlib
_ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
creating build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz
--2025-01-25 18:09:42-- https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/321154/5391f223-be3b-436b-a3e2-f5216589b684?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250125T180942Z&X-Amz-Expires=300&X-Amz-Signature=424e8ff5a0d7afd803ac9751f5385a9165db3f2eddeb07435ce46ad1c228a1&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dlibffi-3.4.5.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 18:09:42-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/321154/5391f223-be3b-436b-a3e2-f5216589b684?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250125T180942Z&X-Amz-Expires=300&X-Amz-Signature=424e8ff5a0d7afd803ac9751f5385a9165db3f2eddeb07435ce46ad1c228a1&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dlibffi-3.4.5.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.108.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.111.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1391548 (1.3M) [application/octet-stream]
Saving to: 'libffi-3.4.5.tar.gz'

libffi-3.4.5.tar.gz 100%[=====] 1.33M --.-KB/s in 0.1s

2025-01-25 18:09:43 (13.7 MB/s) - 'libffi-3.4.5.tar.gz' saved [1391548/1391548]

swayam@swayam:~/libraries$

```

## 32. tar -xvf libffi-3.4.5.tar.gz: Extracts the contents of the libffi source code tarball.

```

Failed to build these modules:
_ctypes

Following modules built successfully but were removed because they could not be imported:
_hashlib
_ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
creating build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz
--2025-01-25 18:09:42-- https://github.com/libffi/libffi/releases/download/v3.4.5/libffi-3.4.5.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/321154/5391f223-be3b-436b-a3e2-f5216589b684?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250125T180942Z&X-Amz-Expires=300&X-Amz-Signature=424e8ff5a0d7afd803ac9751f5385a9165db3f2eddeb07435ce46ad1c228a1&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dlibffi-3.4.5.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-01-25 18:09:42-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/321154/5391f223-be3b-436b-a3e2-f5216589b684?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250125%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250125T180942Z&X-Amz-Expires=300&X-Amz-Signature=424e8ff5a0d7afd803ac9751f5385a9165db3f2eddeb07435ce46ad1c228a1&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dlibffi-3.4.5.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.108.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.111.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1391548 (1.3M) [application/octet-stream]
Saving to: 'libffi-3.4.5.tar.gz'

libffi-3.4.5.tar.gz 100%[=====] 1.33M --.-KB/s in 0.1s

2025-01-25 18:09:43 (13.7 MB/s) - 'libffi-3.4.5.tar.gz' saved [1391548/1391548]

swayam@swayam:~/libraries$ tar -xvf libffi-3.4.5.tar.gz

```

```
libffi-3.4.5/testsuite/libffi.complex/complex_defs_float.inc
libffi-3.4.5/testsuite/libffi.complex/complex_defs_longdouble.inc
libffi-3.4.5/testsuite/libffi.complex/complex_double.c
libffi-3.4.5/testsuite/libffi.complex/complex_float.c
libffi-3.4.5/testsuite/libffi.complex/complex_int.c
libffi-3.4.5/testsuite/libffi.complex/complex_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/ffitest.h
libffi-3.4.5/testsuite/libffi.complex/many_complex.inc
libffi-3.4.5/testsuite/libffi.complex/many_complex_double.c
libffi-3.4.5/testsuite/libffi.complex/many_complex_float.c
libffi-3.4.5/testsuite/libffi.complex/many_complex_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex1.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex1_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex1_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex1_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex2_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_longdouble.c
libffi-3.4.5/testsuite/libffi.go/
libffi-3.4.5/testsuite/libffi.go/aa-direct.c
libffi-3.4.5/testsuite/libffi.go/closure1.c
libffi-3.4.5/testsuite/libffi.go/ffitest.h
libffi-3.4.5/testsuite/libffi.go/go.exp
libffi-3.4.5/testsuite/libffi.go/static-chain.h
libffi-3.4.5/testsuite/Makefile.am
libffi-3.4.5/testsuite/Makefile.in
libffi-3.4.5/man/
libffi-3.4.5/man/Makefile.am
libffi-3.4.5/man/Makefile.in
libffi-3.4.5/man/ffi.3
libffi-3.4.5/man/ffi_call.3
libffi-3.4.5/man/ffi_prep_cif.3
libffi-3.4.5/man/ffi_prep_cif_var.3
libffi-3.4.5/doc/
libffi-3.4.5/doc/Makefile.am
libffi-3.4.5/doc/version.texi
libffi-3.4.5/doc/stamp-vti
libffi-3.4.5/doc/Makefile.in
libffi-3.4.5/doc/mdate-sh
libffi-3.4.5/doc/texinfo.tex
libffi-3.4.5/doc/libffi.texi
libffi-3.4.5/doc/libffi.info
libffi-3.4.5/doc/libffi.pdf
libffi-3.4.5/ChangeLog
swayam@swayam:~/libraries$
```

33. `cd libffi-3.4.5`: Changes the current working directory to the extracted libffi source code directory.

```
libffi-3.4.5/testsuite/libffi.complex/complex_defs_longdouble.inc
libffi-3.4.5/testsuite/libffi.complex/complex_double.c
libffi-3.4.5/testsuite/libffi.complex/complex_float.c
libffi-3.4.5/testsuite/libffi.complex/complex_int.c
libffi-3.4.5/testsuite/libffi.complex/complex_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/ffitest.h
libffi-3.4.5/testsuite/libffi.complex/many_complex.inc
libffi-3.4.5/testsuite/libffi.complex/many_complex_double.c
libffi-3.4.5/testsuite/libffi.complex/many_complex_float.c
libffi-3.4.5/testsuite/libffi.complex/many_complex_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex1.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex1_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex1_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex1_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex2_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_longdouble.c
libffi-3.4.5/testsuite/libffi.go/
libffi-3.4.5/testsuite/libffi.go/aa-direct.c
libffi-3.4.5/testsuite/libffi.go/closure1.c
libffi-3.4.5/testsuite/libffi.go/ffitest.h
libffi-3.4.5/testsuite/libffi.go/go.exp
libffi-3.4.5/testsuite/libffi.go/static-chain.h
libffi-3.4.5/testsuite/Makefile.am
libffi-3.4.5/testsuite/Makefile.in
libffi-3.4.5/man/
libffi-3.4.5/man/Makefile.am
libffi-3.4.5/man/Makefile.in
libffi-3.4.5/man/ffi.3
libffi-3.4.5/man/ffi_call.3
libffi-3.4.5/man/ffi_prep_cif.3
libffi-3.4.5/man/ffi_prep_cif_var.3
libffi-3.4.5/doc/
libffi-3.4.5/doc/Makefile.am
libffi-3.4.5/doc/version.texi
libffi-3.4.5/doc/stamp-vti
libffi-3.4.5/doc/Makefile.in
libffi-3.4.5/doc/mdate-sh
libffi-3.4.5/doc/texinfo.tex
libffi-3.4.5/doc/libffi.texi
libffi-3.4.5/doc/libffi.info
libffi-3.4.5/doc/libffi.pdf
libffi-3.4.5/ChangeLog
swayam@swayam:~/libraries$ cd libffi-3.4.5
swayam@swayam:~/libraries/libffi-3.4.5$
```

34. `./configure --prefix=$HOME/local`: Configures the libffi source code for compilation, specifying the installation prefix.

```

libffi-3.4.5/testsuite/libffi.complex/complex_defs_longdouble.inc
libffi-3.4.5/testsuite/libffi.complex/complex_double.c
libffi-3.4.5/testsuite/libffi.complex/complex_float.c
libffi-3.4.5/testsuite/libffi.complex/complex_int.c
libffi-3.4.5/testsuite/libffi.complex/complex_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/ffitest.h
libffi-3.4.5/testsuite/libffi.complex/many_complex.inc
libffi-3.4.5/testsuite/libffi.complex/many_complex_double.c
libffi-3.4.5/testsuite/libffi.complex/many_complex_float.c
libffi-3.4.5/testsuite/libffi.complex/many_complex_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex1.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex1_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex1_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex1_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2.inc
libffi-3.4.5/testsuite/libffi.complex/return_complex2_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex2_longdouble.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_double.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_float.c
libffi-3.4.5/testsuite/libffi.complex/return_complex_longdouble.c
libffi-3.4.5/testsuite/libffi.go/
libffi-3.4.5/testsuite/libffi.go/aa-direct.c
libffi-3.4.5/testsuite/libffi.go/closure1.c
libffi-3.4.5/testsuite/libffi.go/ffitest.h
libffi-3.4.5/testsuite/libffi.go/go.exp
libffi-3.4.5/testsuite/libffi.go/static-chain.h
libffi-3.4.5/testsuite/Makefile.am
libffi-3.4.5/testsuite/Makefile.in
libffi-3.4.5/man/
libffi-3.4.5/man/Makefile.am
libffi-3.4.5/man/Makefile.in
libffi-3.4.5/man/ffi.3
libffi-3.4.5/man/ffi_call.3
libffi-3.4.5/man/ffi_prep_cif.3
libffi-3.4.5/man/ffi_prep_cif_var.3
libffi-3.4.5/doc/
libffi-3.4.5/doc/Makefile.am
libffi-3.4.5/doc/Version.texi
libffi-3.4.5/doc/stamp-vti
libffi-3.4.5/doc/Makefile.in
libffi-3.4.5/doc/mdate.sh
libffi-3.4.5/doc/texinfo.tex
libffi-3.4.5/doc/libffi.texi
libffi-3.4.5/doc/libffi.info
libffi-3.4.5/doc/libffi.pdf
libffi-3.4.5/ChangeLog
swayam@swayam:~/libraries$ cd libffi-3.4.5
swayam@swayam:~/libraries/libffi-3.4.5$./configure --prefix=$HOME/local
checking how to hardcode library paths into programs... immediate
checking for readelf... readelf
checking size of size_t... 8
checking for C compiler vendor... gnu
checking whether C compiler accepts -fstrict-aliasing... yes
checking whether C compiler accepts -ffast-math... yes
checking for gcc architecture flag...
checking for x86 cpuid 0 output... 16:756e6547:6c65746e:49656e69
checking for x86 cpuid 1 output... 906a3:b0c0800:c2da2203:178bfbff
checking whether C compiler accepts -march=core2... yes
checking for gcc architecture flag... -march=core2
checking whether C compiler accepts -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2... yes
checking CFLAGS for most reasonable warnings... -Wall
checking whether to enable maintainer-specific portions of Makefiles... no
checking for sys/memfd.h... no
checking for memfd_create... yes
checking for egrep... (cached) /usr/bin/grep -E
checking for memcpu... yes
checking for alloca.h... yes
checking size of double... 8
checking size of long double... 16
checking whether byte ordering is bigendian... no
checking assembler .cfi pseudo-op support... yes
checking assembler supports pc related relocs... yes
checking whether compiler supports pointer authentication... no
checking for _ prefix in compiled symbols... no
checking toolchain supports unwind section type... yes
checking whether C compiler accepts -fno-lto... yes
checking whether .eh_frame section should be read-only... yes
checking for __attribute__((visibility("hidden")))... yes
checking for shared libgcc... yes
configure: versioning on shared library symbols is gnu
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating include/Makefile
config.status: creating include/ffi.h
config.status: creating Makefile
config.status: creating testsuite/Makefile
config.status: creating man/Makefile
config.status: creating doc/Makefile
config.status: creating libffi.pc
config.status: creating ffigenconfig.h
config.status: executing buildin commands
config.status: create top_srcdir/Makefile guessed from local Makefile
config.status: build in x86_64-pc-linux-gnu (HOST=)
config.status: executing depfiles commands
config.status: executing libtool commands
config.status: executing include commands
config.status: executing src commands
swayam@swayam:~/libraries/libffi-3.4.5$

```

35. `make -j$(nproc)`: Compiles the libffi source code using multiple CPU cores.

```

checking how to hardcode library paths into programs... immediate
checking for readelf... readelf
checking size of size_t... 8
checking for C compiler vendor... gnu
checking whether C compiler accepts -fstrict-aliasing... yes
checking whether C compiler accepts -ffast-math... yes
checking for gcc architecture flag...
checking for x86 cpuid 0 output... 16:756e6547:6c65746e:49656e69
checking for x86 cpuid 1 output... 906a3:b0c0800:c2da2203:178bfbff
checking whether C compiler accepts -march=core2... yes
checking for gcc architecture flag... -march=core2
checking whether C compiler accepts -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2... yes
checking CFLAGS for most reasonable warnings... -Wall
checking whether to enable maintainer-specific portions of Makefiles... no
checking for sys/memfd.h... no
checking for memfd_create... yes
checking for egrep... (cached) /usr/bin/grep -E
checking for mcpu... yes
checking for alloca.h... yes
checking size of double... 8
checking size of long double... 16
checking whether byte ordering is bigendian... no
checking assembler .cfi pseudo-op support... yes
checking assembler supports pc related relocs... yes
checking whether compiler supports pointer authentication... no
checking for _ prefix in compiled symbols... no
checking toolchain supports unwind section type... yes
checking whether C compiler accepts -fno-lto... yes
checking whether .eh_frame section should be read-only... yes
checking for __attribute__((visibility("hidden")))... yes
checking for shared libgcc... yes
configure: versioning on shared library symbols is gnu
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating include/Makefile
config.status: creating include/ffi.h
config.status: creating Makefile
config.status: creating testsuite/Makefile
config.status: creating man/Makefile
config.status: creating doc/Makefile
config.status: creating libffi.pc
config.status: creating fficonfig.h
config.status: executing buildir commands
config.status: create top_srcdir/Makefile guessed from local Makefile
config.status: build in x86_64-pc-linux-gnu (HOST=)
config.status: executing depfiles commands
config.status: executing libtool commands
config.status: executing include commands
config.status: executing src commands
swayam@swayam:~/libraries/libffi-3.4.5$ make -j$(nproc)_
45 | ffi_java_raw_size (ffi_cif *cif)
...
../src/java_raw_api.c:331:3: warning: 'ffi_java_ptrnarray_to_raw' is deprecated [-Wdeprecated-declarations]
331 | ffi_java_ptrnarray_to_raw (cif, avalue, raw);
...
../src/java_raw_api.c:157:1: note: declared here
157 | ffi_java_ptrnarray_to_raw (ffi_cif *cif, void **args, ffi_java_raw *raw)
...
../src/java_raw_api.c: In function 'ffi_prep_java_raw_closure':
../src/java_raw_api.c:369:3: warning: 'ffi_prep_java_raw_closure_loc' is deprecated [-Wdeprecated-declarations]
369 | return ffi_prep_java_raw_closure_loc (cl, cif, fun, user_data, cl);
...
../src/java_raw_api.c:337:1: note: declared here
337 | ffi_prep_java_raw_closure_loc (ffi_java_raw_closure* cl,
...
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/raw_api.lo -MD -MP -MF src/deps/raw_api.Tpo -c ../src/raw_api.c -o src/raw_api.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/prep_cif.lo -MD -MP -MF src/deps/prep_cif.Tpo -c ../src/prep_cif.c -o src/prep_cif.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/x86/ffi64.lo -MD -MP -MF src/x86/deps/ffi64.Tpo -c ../src/x86/ffi64.c -o src/x86/ffi64.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/java_raw_api.lo -MD -MP -MF src/deps/java_raw_api.Tpo -c ../src/java_raw_api.c -o src/java_raw_api.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/tramp.lo -MD -MP -MF src/deps/tramp.Tpo -c ../src/tramp.c -o src/tramp.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/closures.lo -MD -MP -MF src/deps/closures.Tpo -c ../src/closures.c -o src/closures.o >/dev/null 2>&1
/bin/bash ./libtool --tag=CC --mode=link gcc -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -o libffi_convenience .la src/prep_cif.lo src/types.lo src/raw_api.lo src/java_raw_api.lo src/closures.lo src/tramp.lo src/x86/ffi64.lo src/x86/unix64.lo src/x86/ffi64.lo src/x86/win64.lo
/bin/bash ./libtool --tag=CC --mode=link gcc -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -no-undefined -version-info 9:0:0 -Wl,--version-script,libffi.map -o libffi.la -rpath /home/swayam/local/lib/.lib src/prep_cif.lo src/types.lo src/raw_api.lo src/java_raw_api.lo src/closures.lo src/tramp.lo src/x86/ffi64.lo src/x86/unix64.lo src/x86/win64.lo
libtool: link: ar cr .libs/libffi_convenience.a src/prep_cif.o src/types.o src/raw_api.o src/java_raw_api.o src/closures.o src/tramp.o src/x86/ffi64.o src/x86/unix64.o src/x86/win64.o
libtool: link: gcc -shared -fPIC -DPIC -o .libs/libffi.so.8.1.3 src/prep_cif.o src/types.o src/raw_api.o src/java_raw_api.o src/closures.o src/tramp.o src/x86/ffi64.o src/x86/unix64.o src/x86/win64.o
libtool: link: ranlib .libs/libffi_convenience.a
libtool: link: (cd ".libs" && rm -f "libffi_convenience.la" && ln -s "../libffi_convenience.la" "libffi_convenience.la")
libtool: link: (cd ".libs" && rm -f "libffi.so.8" && ln -s "libffi.so.8.1.3" "libffi.so.8")
libtool: link: (cd ".libs" && rm -f "libffi.so" && ln -s "libffi.so.8.1.3" "libffi.so")
libtool: link: ar cr .libs/libffi.a src/prep_cif.o src/types.o src/raw_api.o src/java_raw_api.o src/closures.o src/tramp.o src/x86/ffi64.o src/x86/unix64.o src/x86/win64.o
libtool: link: ranlib .libs/libffi.a
libtool: link: (cd ".libs" && rm -f "libffi.la" && ln -s "../libffi.la" "libffi.la")
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
swayam@swayam:~/libraries/libffi-3.4.5$

```

### 36. make install: Installs the compiled libffi software.

```

45 | ffi_java_raw_size (ffi_cif *cif)
.../src/java_raw_api.c:331:3: warning: 'ffi_java_ptrarray_to_raw' is deprecated [-Wdeprecated-declarations]
331 | ffi_java_ptrarray_to_raw (cif, avalue, raw);
.../src/java_raw_api.c:157:1: note: declared here
157 | ffi_java_ptrarray_to_raw (ffi_cif *cif, void **args, ffi_java_raw *raw)
.../src/java_raw_api.c: In function 'ffi_prep_java_raw_closure':
.../src/java_raw_api.c:369:3: warning: 'ffi_prep_java_raw_closure_loc' is deprecated [-Wdeprecated-declarations]
369 | return ffi_prep_java_raw_closure_loc (cl, cif, fun, user_data, cl);
.../src/java_raw_api.c:337:1: note: declared here
337 | ffi_prep_java_raw_closure_loc (ffi_java_raw_closure* cl,

libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/raw_api.lo -MD -MP -MF src/.deps/raw_api.Tpo -c ../src/raw_api.c -o src/raw_api.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/prepare_cif.lo -MD -MP -MF src/.deps/prepare_cif.Tpo -c ../src/prepare_cif.c -o src/prepare_cif.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/x86/ffi64.lo -MD -MP -MF src/x86/.deps/ffi64.Tpo -c ../src/x86/ffi64.c -o src/x86/ffi64.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/java_raw_api.lo -MD -MP -MF src/.deps/java_raw_api.Tpo -c ../src/java_raw_api.c -o src/java_raw_api.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/tramp.lo -MD -MP -MF src/.deps/tramp.Tpo -c ../src/tramp.c -o src/tramp.o >/dev/null 2>&1
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I. -I. -I./include -Iinclude -I./src -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -MT src/closures.lo -MD -MP -MF src/.deps/closures.Tpo -c ../src/closures.c -o src/closures.o >/dev/null 2>&1
/bin/bash ./libtool --tag=CC --mode=link gcc -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -fexceptions -no-undefined -version-info 3:0:0 -Wl,-version-script,libffi.map -o libffi.so.8.1.3 src/prepare_cif.lo src/types.lo src/raw_api.lo src/java_raw_api.lo src/closures.lo src/tramp.lo src/x86/ffi64.lo src/x86/unix64.lo src/x86/ffi64.lo src/x86/unix64.lo
/bin/bash ./libtool --tag=CC --mode=link gcc -Wall -O3 -fomit-frame-pointer -fstrict-aliasing -ffast-math -march=core2 -Wl,-version-script,libffi.map -Wl,-soname,libffi.so.8 -o libffi.so.8.1.3 src/prepare_cif.lo src/types.lo src/raw_api.lo src/java_raw_api.lo src/closures.lo src/tramp.lo src/x86/ffi64.lo src/x86/unix64.lo src/x86/ffi64.lo src/x86/unix64.lo
libtool: link: ranlib libffi.so.8.1.3
libtool: link: (cd ".libs" && rm -f "libffi_convenience.a" && ln -s "../libffi_convenience.a" "libffi_convenience.a")
libtool: link: (cd ".libs" && rm -f "libffi.so.8" && ln -s "libffi.so.8.1.3" "libffi.so.8")
libtool: link: (cd ".libs" && rm -f "libffi.so" && ln -s "libffi.so.8.1.3" "libffi.so")
libtool: link: ar cr .libs/libffi.a src/prepare_cif.o src/types.o src/raw_api.o src/java_raw_api.o src/closures.o src/tramp.o src/x86/ffi64.o src/x86/unix64.o src/x86/ffi64.o src/x86/unix64.o
libtool: link: ranlib .libs/libffi.a
libtool: link: (cd ".libs" && rm -f "libffi.la" && ln -s "../libffi.la" "libffi.la")
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
swayam@swayam:~/libraries/libffi-3.4.5$ make install

make[3]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p /home/swayam/local/share/man/man3
/usr/bin/install -c -m 644 ../man/ffi.3 ../man/ffi_call.3 ../man/ffi_prep_cif.3 ../man/ffi_prep_cif_var.3 /home/swayam/local/share/man/man3
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
Making install in doc
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p /home/swayam/local/share/info
/usr/bin/install -c -m 644 ../doc/libffi.info /home/swayam/local/share/info
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
/usr/bin/mkdir -p /home/swayam/local/lib/.libs
/bin/bash ./libtool --mode=install /usr/bin/install -c libffi.la /home/swayam/local/lib/.libs
libtool: install: /usr/bin/install -c .libs/libffi.so.8.1.3 /home/swayam/local/lib/.libs/libffi.so.8.1.3
libtool: install: (cd /home/swayam/local/lib/.libs && { ln -s -f libffi.so.8.1.3 libffi.so.8 || { rm -f libffi.so.8 && ln -s libffi.so.8.1.3 libffi.so.8; }; })
libtool: install: (cd /home/swayam/local/lib/.libs && { ln -s -f libffi.so.8.1.3 libffi.so || { rm -f libffi.so && ln -s libffi.so.8.1.3 libffi.so; }; })
libtool: install: /usr/bin/install -c .libs/libffi.lai /home/swayam/local/lib/.libs/libffi.la
libtool: install: /usr/bin/install -c .libs/libffi.a /home/swayam/local/lib/.libs/libffi.a
libtool: install: chmod 644 /home/swayam/local/lib/.libs/libffi.a
libtool: install: ranlib /home/swayam/local/lib/.libs/libffi.a
libtool: finish: PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin" ldconfig -n /home/swayam/local/lib/.libs

Libraries have been installed in:
/home/swayam/local/lib/.libs

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p /home/swayam/local/lib/pkgconfig
/usr/bin/install -c -m 644 libffi.pc /home/swayam/local/lib/pkgconfig
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
swayam@swayam:~/libraries/libffi-3.4.5$

```

37. `cd ..` Changes the current working directory to the parent directory.

```

/usr/bin/mkdir -p '/home/swayam/local/share/man/man3'
/usr/bin/install -c -m 644 ../man/ffi.3 ../man/ffi_call.3 ../man/ffi_prep_cif.3 ../man/ffi_prep_cif_var.3 '/home/swayam/local/share/man/man3'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
Making install in doc
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/home/swayam/local/share/info'
/usr/bin/install -c -m 644 ../doc/libffi.info '/home/swayam/local/share/info'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
/usr/bin/mkdir -p '/home/swayam/local/lib/./lib'
/bin/bash ./libtool --mode=install /usr/bin/install -c libffi.la '/home/swayam/local/lib/./lib'
libtool: install: /usr/bin/install -c .libs/libffi.so.8.1.3 /home/swayam/local/lib/./lib/libffi.so.8.1.3
libtool: install: (cd /home/swayam/local/lib/./lib && { ln -s -f libffi.so.8.1.3 libffi.so.8 || { rm -f libffi.so.8 && ln -s libffi.so.8.1.3 libffi.so.8; }; })
libtool: install: (cd /home/swayam/local/lib/./lib && { ln -s -f libffi.so.8.1.3 libffi.so || { rm -f libffi.so && ln -s libffi.so.8.1.3 libffi.so; }; })
libtool: install: /usr/bin/install -c .libs/libffi.la /home/swayam/local/lib/./lib/libffi.la
libtool: install: /usr/bin/install -c .libs/libffi.a /home/swayam/local/lib/./lib/libffi.a
libtool: install: chmod 644 /home/swayam/local/lib/./lib/libffi.a
libtool: install: ranlib /home/swayam/local/lib/./lib/libffi.a
libtool: finish: PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin ldconfig -n /home/swayam/local/lib/./lib

Libraries have been installed in:
/home/swayam/local/lib/./lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
 during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
 during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p '/home/swayam/local/lib/pkgconfig'
/usr/bin/install -c -m 644 libffi.pc '/home/swayam/local/lib/pkgconfig'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
swayam@swayam:~/libraries/libffi-3.4.5$ cd ..
swayam@swayam:~/libraries$

```

### 38. cd Python-3.10.16: Changes the current working directory to the Python source code directory.

```

/usr/bin/install -c -m 644 ../man/ffi.3 ../man/ffi_call.3 ../man/ffi_prep_cif.3 ../man/ffi_prep_cif_var.3 '/home/swayam/local/share/man/man3'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
Making install in doc
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/home/swayam/local/share/info'
/usr/bin/install -c -m 644 ../doc/libffi.info '/home/swayam/local/share/info'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
/usr/bin/mkdir -p '/home/swayam/local/lib/./lib'
/bin/bash ./libtool --mode=install /usr/bin/install -c libffi.la '/home/swayam/local/lib/./lib'
libtool: install: /usr/bin/install -c .libs/libffi.so.8.1.3 /home/swayam/local/lib/./lib/libffi.so.8.1.3
libtool: install: (cd /home/swayam/local/lib/./lib && { ln -s -f libffi.so.8.1.3 libffi.so.8 || { rm -f libffi.so.8 && ln -s libffi.so.8.1.3 libffi.so.8; }; })
libtool: install: (cd /home/swayam/local/lib/./lib && { ln -s -f libffi.so.8.1.3 libffi.so || { rm -f libffi.so && ln -s libffi.so.8.1.3 libffi.so; }; })
libtool: install: /usr/bin/install -c .libs/libffi.la /home/swayam/local/lib/./lib/libffi.la
libtool: install: /usr/bin/install -c .libs/libffi.a /home/swayam/local/lib/./lib/libffi.a
libtool: install: chmod 644 /home/swayam/local/lib/./lib/libffi.a
libtool: install: ranlib /home/swayam/local/lib/./lib/libffi.a
libtool: finish: PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin ldconfig -n /home/swayam/local/lib/./lib

Libraries have been installed in:
/home/swayam/local/lib/./lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
 during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
 during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p '/home/swayam/local/lib/pkgconfig'
/usr/bin/install -c -m 644 libffi.pc '/home/swayam/local/lib/pkgconfig'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
swayam@swayam:~/libraries/libffi-3.4.5$ cd ..
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$

```

### 39. ./configure --prefix=\$HOME/local --with-openssl=\$HOME/local --with-ensurepip=install --LDLDFLAGS="-L\$HOME/local/lib" --CPPFLAGS="-I\$HOME/local/include" --PKG\_CONFIG\_PATH="\$HOME/local/lib/pkgconfig" --enable-optimization: Configures the Python source code for compilation, specifying various options and prefixes.



```

make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/man'
Making install in doc
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[3]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/home/swayam/local/share/info'
/usr/bin/install -c -m 644 ../doc/libffi.info '/home/swayam/local/share/info'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu/doc'
make[2]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[3]: Entering directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
/usr/bin/mkdir -p '/home/swayam/local/lib/./lib'
./bin/bash ./libtool --mode=install /usr/bin/install -c libffi.la '/home/swayam/local/lib/./lib'
libtool: install: /usr/bin/install -c .libs/libffi.so.8.1.3 /home/swayam/local/lib/./lib/libffi.so.8.1.3
libtool: install: (cd /home/swayam/local/lib/./lib && { ln -s -f libffi.so.8.1.3 libffi.so.8 || { rm -f libffi.so.8 && ln -s libffi.so.8.1.3 libffi.so.8; }; })
libtool: install: (cd /home/swayam/local/lib/./lib && { ln -s -f libffi.so.8.1.3 libffi.so || { rm -f libffi.so && ln -s libffi.so.8.1.3 libffi.so; }; })
libtool: install: /usr/bin/install -c .libs/libffi.lai /home/swayam/local/lib/./lib/libffi.lai
libtool: install: /usr/bin/install -c .libs/libffi.a /home/swayam/local/lib/./lib/libffi.a
libtool: install: chmod 644 /home/swayam/local/lib/./lib/libffi.a
libtool: install: ranlib /home/swayam/local/lib/./lib/libffi.a
libtool: finish: PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin" ldconfig -n /home/swayam/local/lib/./lib

Libraries have been installed in:
/home/swayam/local/lib/./lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
 during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
 during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p '/home/swayam/local/lib/pkgconfig'
/usr/bin/install -c -m 644 libffi.pc '/home/swayam/local/lib/pkgconfig'
make[3]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[2]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
make[1]: Leaving directory '/home/swayam/libraries/libffi-3.4.5/x86_64-pc-linux-gnu'
swayam@swayam:~/libraries/libffi-3.4.5$ cd ..
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$./configure --prefix=$HOME/local --with-openssl=$HOME/local --with-ensurepip=install --LDLFLAGS="-L$HOME/local/lib" --CPPLFLAGS="-I$HOME/local/include" --PKG_CONFIG_PATH="$HOME/local/lib/pkgconfig" --enable-optimization_

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/rtc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ansipoint... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5, sha1, sha256, sha512, sha3, blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$

```

40. `make -j$(nproc)`: Compiles the Python source code using multiple CPU cores.



```

configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5, sha1, sha256, sha512, sha3, blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ make -j$(nproc)_

gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swayam/local/include -I./Include -I./home/swayam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/librar
ies/Python-3.10.16/Modules/_ctypes/cfield.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/cfield.o -DPy_BUILD_CORE_MODULE
-DHAVE_FFI_PREP_CIF_VAR=1 -DHAVE_FFI_PREP_CLOSURE_LOC=1 -DHAVE_FFI_CLOSURE_ALLOC=1
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swayam/local/include -I./Include -I./home/swayam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/librar
ies/Python-3.10.16/Modules/_ctypes/stdgdict.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/stdgdict.o -DPy_BUILD_CORE_MODULE
-DHAVE_FFI_PREP_CIF_VAR=1 -DHAVE_FFI_PREP_CLOSURE_LOC=1 -DHAVE_FFI_CLOSURE_ALLOC=1
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/_ctypes.o build/temp.linux-x86_64-3.10/hom
e/swayam/libraries/Python-3.10.16/Modules/_ctypes/callbacks.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/callproc.o buil
d/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ctypes/cfield.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules
/_ctypes/stdgdict.o -L/home/swayam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lffi -ldl -o build/lib.linux-x86_64-3.10/_ctypes.cpython-310-x86_64-li
nux-gnu.so
*** WARNING: renaming "_ssl" since importing it failed: libssl.so.1.1: cannot open shared object file: No such file or directory
*** WARNING: renaming "_hashlib" since importing it failed: libcrypto.so.1.1: cannot open shared object file: No such file or directory

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_lzma _sqlite3 _uuid
nis readline zlib
To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

Following modules built successfully but were removed because they could not be imported:
_hashlib _ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$ _

```

41. make install: Installs the compiled Python software.

```

gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror-implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swapam/local/include -I./Include -I -I/home/swapam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swapam/libraries/Python-3.10.16/Include -I/home/swapam/libraries/Python-3.10.16 -c /home/swapam/librar
ies/Python-3.10.16/Modules/_ctypes/cfield.c -o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_ctypes/cfield.o -DPy_BUILD_CORE_MODULE
-DHAVE_FFI_PREP_CIF_VAR=1 -DHAVE_FFI_PREP_CLOSURE_LOCC=1 -DHAVE_FFI_CLOSURE_ALLOC=1
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror-implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swapam/local/include -I./Include -I -I/home/swapam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swapam/libraries/Python-3.10.16/Include -I/home/swapam/libraries/Python-3.10.16 -c /home/swapam/librar
ies/Python-3.10.16/Modules/_ctypes/stgdict.c -o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_ctypes/stgdict.o -DPy_BUILD_CORE_MODULE
-DHAVE_FFI_PREP_CIF_VAR=1 -DHAVE_FFI_PREP_CLOSURE_LOCC=1 -DHAVE_FFI_CLOSURE_ALLOC=1
gcc -shared -L/home/swapam/local/lib build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_ctypes/_ctypes.o build/temp.linux-x86_64-3.10/ho
me/swapam/libraries/Python-3.10.16/Modules/_ctypes/callbacks.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_ctypes/callproc.o buil
d/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_ctypes/cfield.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules
/_ctypes/stgdict.o -L/home/swapam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lffi -ldl -o build/lib.linux-x86_64-3.10/_ctypes.cpython-310-x86_64-li
nux-gnu.so
*** WARNING: renaming "ssl" since importing it failed: libssl.so.1.1: cannot open shared object file: No such file or directory
*** WARNING: renaming "hashlib" since importing it failed: libcrypto.so.1.1: cannot open shared object file: No such file or directory

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline zlib
To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

Following modules built successfully but were removed because they could not be imported:
_hashlib _ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
copying and adjusting /home/swapam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swapam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swapam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swapam:~/libraries/Python-3.10.16$ make install

esac; \
./python -E -m ensurepip \
 $ensurepip --root=/ ; \

fi
Traceback (most recent call last):
 File "<frozen zipimport>", line 577, in _get_decompress_func
ModuleNotFoundError: No module named 'zlib'

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<frozen zipimport>", line 625, in _get_data
 File "<frozen zipimport>", line 580, in _get_decompress_func
zipimport.ZipImportError: can't decompress data; zlib not available

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<strings>", line 6, in <module>
 File "/home/swapam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swapam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swapam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 __import__(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swapam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return run_code(code, main_globals, None)
 File "/home/swapam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swapam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip.main())
 File "/home/swapam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap()
 File "/home/swapam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(kwargs, *PACKAGE_NAMES), additional_paths)
 File "/home/swapam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swapam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swapam/libraries/Python-3.10.16/python', '-H', 'ignore:DeprecationWarning', '-c', '\nimport runpy\nimport sys\n
sys.path = [\n/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl\n, \n/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n] + sys.path\nsys.argv[1:] = [\n'install\n,
\n'-no-cache-dir\n, \n'-no-index\n, \n'-find-links\n, \n'/tmp/tmp1adkretg\n, \n'-root\n, \n'\n, \n'setuptools\n, \n'pip\n']\nrunpy.run_module('pip', run_name='__mai
n__', alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swapam:~/libraries/Python-3.10.16$

```

### Another error – ModuleNotFoundError: No module named 'zlib'

Zlib is required to install Python because it is a crucial library used for data compression, which is essential for building and distributing Python itself. Let's install it.

42. `cd ..`: Changes the current working directory to the parent directory.

```

./python -E -m ensurepip \
$ensurepip --root=/ ; \
fi
Traceback (most recent call last):
 File "<frozen zipimport>", line 577, in _get_decompress_func
ModuleNotFoundError: No module named 'zlib'

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<frozen zipimport>", line 625, in _get_data
 File "<frozen zipimport>", line 580, in _get_decompress_func
zipimport.ZipImportError: can't decompress data; zlib not available

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 __import__(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-M', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl\n', '\n/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n'] + sys.path\nsys.argv[1:] = [\n'install\n', '\n--no-cache-dir\n', '\n--no-index\n', '\n--find-links\n', '\n/tmp/tmp1adkretg\n', '\n--root\n', '\n\n', '\nsetuptools\n', '\npip\n']\nrunpy.run_module("pip", run_name="__mai
n__", alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ _

```

#### 43. wget https://www.zlib.net/zlib-1.3.1.tar.gz: Downloads the zlib source code tarball.

```

./python -E -m ensurepip \
$ensurepip --root=/ ; \
fi
Traceback (most recent call last):
 File "<frozen zipimport>", line 577, in _get_decompress_func
ModuleNotFoundError: No module named 'zlib'

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<frozen zipimport>", line 625, in _get_data
 File "<frozen zipimport>", line 580, in _get_decompress_func
zipimport.ZipImportError: can't decompress data; zlib not available

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 __import__(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-M', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl\n', '\n/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n'] + sys.path\nsys.argv[1:] = [\n'install\n', '\n--no-cache-dir\n', '\n--no-index\n', '\n--find-links\n', '\n/tmp/tmp1adkretg\n', '\n--root\n', '\n\n', '\nsetuptools\n', '\npip\n']\nrunpy.run_module("pip", run_name="__mai
n__", alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz

```

```

zipimport.ZipImportError: can't decompress data; zlib not available
During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 __import__(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap(
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-H', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl\n', '\n/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n'] + sys.path\nsys.argv[1:] = [\n'install\n', '\n--no-cache-dir\n', '\n--no-index\n', '\n--find-links\n', '\n/tmp/tmp1adkretg\n', '\n--root\n', '\n\n', '\nsetuptools\n', '\npip\n']\nrunpy.run_module("pip", run_name="__mai\nn__", alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 85.187.148.2
Connecting to www.zlib.net (www.zlib.net)[85.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ _

```

44. tar -xzf zlib-1.3.1.tar.gz: Extracts the contents of the zlib source code tarball.

```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 __import__(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap(
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-H', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl\n', '\n/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n'] + sys.path\nsys.argv[1:] = [\n'install\n', '\n--no-cache-dir\n', '\n--no-index\n', '\n--find-links\n', '\n/tmp/tmp1adkretg\n', '\n--root\n', '\n\n', '\nsetuptools\n', '\npip\n']\nrunpy.run_module("pip", run_name="__mai\nn__", alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 85.187.148.2
Connecting to www.zlib.net (www.zlib.net)[85.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz

```

```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 import_(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap()
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-W', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n"/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl", \n"/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl"] + sys.path\nsys.argv[1:] = [\n"install", \n"--no-cache-dir", \n"--no-index", \n"--find-links", \n"/tmp/tmp1adkretg", \n"--root", \n"/", \n"setuptools", \n"pip"]\nrunpy.run_module("pip", run_name="__mai\nn__", alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 85.187.148.2
Connecting to www.zlib.net (www.zlib.net)[85.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz
swayam@swayam:~/libraries$

```

#### 45. cd zlib-1.3.1: Changes the current working directory to the extracted zlib source code directory.

```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 import_(pkg_name)
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap()
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-W', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n"/tmp/tmp1adkretg/setuptools-65.5.0-py3-none-any.whl", \n"/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl"] + sys.path\nsys.argv[1:] = [\n"install", \n"--no-cache-dir", \n"--no-index", \n"--find-links", \n"/tmp/tmp1adkretg", \n"--root", \n"/", \n"setuptools", \n"pip"]\nrunpy.run_module("pip", run_name="__mai\nn__", alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 85.187.148.2
Connecting to www.zlib.net (www.zlib.net)[85.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz
swayam@swayam:~/libraries$ cd zlib-1.3.1

```

```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 import __pkg_name__
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap(
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-W', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n/tmp/tmpladkretg/setuptools-65.5.0-py3-none-any.whl\n, \n/tmp/tmpladkretg/pip-23.0.1-py3-none-any.whl\n] + sys.path\nsys.argv[1:] = [\n\'install\n\', \n\'--no-cache-dir\n\', \n\'--no-index\n\', \n\'--find-links\n\', \n/tmp/tmpladkretg\n\', \n\'--root\n\', \n\'/\n\', \n\'setuptools\n\', \n\'pip\n\']\nrunpy.run_module(\'pip\', run_name=\'__mai\nn__\', alter_sys=True\n\n\']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 85.187.148.2
Connecting to www.zlib.net (www.zlib.net)[85.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====>] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz
swayam@swayam:~/libraries$ cd zlib-1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$ _

```

#### 46. ./configure --prefix=\$HOME/local: Configures the zlib source code for compilation, specifying the installation prefix.

```

Traceback (most recent call last):
 File "<string>", line 6, in <module>
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 220, in run_module
 mod_name, mod_spec, code = _get_module_details(mod_name)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 146, in _get_module_details
 return _get_module_details(pkg_main_name, error)
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 110, in _get_module_details
 import __pkg_name__
 File "<frozen zipimport>", line 196, in get_code
 File "<frozen zipimport>", line 759, in _get_module_code
 File "<frozen zipimport>", line 627, in _get_data
zipimport.ZipImportError: can't decompress data; zlib not available
Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main__.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 287, in _main
 return _bootstrap(
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES, additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init__.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-W', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n/tmp/tmpladkretg/setuptools-65.5.0-py3-none-any.whl\n, \n/tmp/tmpladkretg/pip-23.0.1-py3-none-any.whl\n] + sys.path\nsys.argv[1:] = [\n\'install\n\', \n\'--no-cache-dir\n\', \n\'--no-index\n\', \n\'--find-links\n\', \n/tmp/tmpladkretg\n\', \n\'--root\n\', \n\'/\n\', \n\'setuptools\n\', \n\'pip\n\']\nrunpy.run_module(\'pip\', run_name=\'__mai\nn__\', alter_sys=True\n\n\']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 85.187.148.2
Connecting to www.zlib.net (www.zlib.net)[85.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====>] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz
swayam@swayam:~/libraries$ cd zlib-1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$./configure --prefix=$HOME/local

```

```

Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main_.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init_.py", line 287, in _main
 return _bootstrap(
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init_.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES), additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init_.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-W', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n\t/tmp/tmp1adkretg/setupools-65.5.0-py3-none-any.whl\n\t, \n\t/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n\t] + sys.path\nsys.argv[1:] = [\n\tinstall\n\t, \n\t--no-cache-dir\n\t, \n\t--no-index\n\t, \n\t--find-links\n\t, \n\t/tmp/tmp1adkretg\n\t, \n\t--root\n\t, \n\t/\n\t, \n\tsetupools\n\t, \n\tpip\n\t]\nrunpy.run_module(\'pip\', run_name=\'__mai\n\t', alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 65.187.148.2
Connecting to www.zlib.net (www.zlib.net)[65.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz
swayam@swayam:~/libraries$ cd zlib-1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$./configure --prefix=$HOME/local
Checking for gcc...
Checking for shared library support...
Building shared library libz.so.1.3.1 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for streerror... Yes.
Checking for unistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
swayam@swayam:~/libraries/zlib-1.3.1$ _

```

#### 47. make -j\$(nproc): Compiles the zlib source code using multiple CPU cores.

```

Traceback (most recent call last):
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 196, in _run_module_as_main
 return _run_code(code, main_globals, None,
 File "/home/swayam/libraries/Python-3.10.16/Lib/runpy.py", line 86, in _run_code
 exec(code, run_globals)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_main_.py", line 5, in <module>
 sys.exit(ensurepip._main())
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init_.py", line 287, in _main
 return _bootstrap(
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init_.py", line 203, in _bootstrap
 return _run_pip(*args, *PACKAGE_NAMES), additional_paths)
 File "/home/swayam/libraries/Python-3.10.16/Lib/ensurepip/_init_.py", line 104, in _run_pip
 return subprocess.run(cmd, check=True).returncode
 File "/home/swayam/libraries/Python-3.10.16/Lib/subprocess.py", line 526, in run
 raise CalledProcessError(retcode, process.args,
subprocess.CalledProcessError: Command '['/home/swayam/libraries/Python-3.10.16/python', '-W', 'ignore::DeprecationWarning', '-c', '\nimport runpy\nimport sys\nsys.path = [\n\t/tmp/tmp1adkretg/setupools-65.5.0-py3-none-any.whl\n\t, \n\t/tmp/tmp1adkretg/pip-23.0.1-py3-none-any.whl\n\t] + sys.path\nsys.argv[1:] = [\n\tinstall\n\t, \n\t--no-cache-dir\n\t, \n\t--no-index\n\t, \n\t--find-links\n\t, \n\t/tmp/tmp1adkretg\n\t, \n\t--root\n\t, \n\t/\n\t, \n\tsetupools\n\t, \n\tpip\n\t]\nrunpy.run_module(\'pip\', run_name=\'__mai\n\t', alter_sys=True)\n']' returned non-zero exit status 1.
make: *** [Makefile:1280: install] Error 1
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.zlib.net/zlib-1.3.1.tar.gz
--2025-01-25 18:15:49-- https://www.zlib.net/zlib-1.3.1.tar.gz
Resolving www.zlib.net (www.zlib.net)... 65.187.148.2
Connecting to www.zlib.net (www.zlib.net)[65.187.148.2]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1512791 (1.4M) [application/x-gzip]
Saving to: 'zlib-1.3.1.tar.gz'

zlib-1.3.1.tar.gz 100%[=====] 1.44M 686KB/s in 2.2s

2025-01-25 18:15:53 (686 KB/s) - 'zlib-1.3.1.tar.gz' saved [1512791/1512791]

swayam@swayam:~/libraries$ tar -xzf zlib-1.3.1.tar.gz
swayam@swayam:~/libraries$ cd zlib-1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$./configure --prefix=$HOME/local
Checking for gcc...
Checking for shared library support...
Building shared library libz.so.1.3.1 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for streerror... Yes.
Checking for unistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
swayam@swayam:~/libraries/zlib-1.3.1$ make -j$(nproc)

```



```

Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
swayam@swayam:~/libraries/zlib-1.3.1$ make -j$(nproc)
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o infback.o infback.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inffast.o inffast.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inflate.o inflate.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inftrees.o inftrees.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o trees.o trees.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o zutil.o zutil.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o compress.o compress.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o uncompress.o uncompress.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzclose.o gzclose.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzlib.o gzlib.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzread.o gzread.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/adler32.o adler32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/crc32.o crc32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/deflate.o deflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/infback.o infback.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inffast.o inffast.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inflate.o inflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/trees.o trees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/zutil.o zutil.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inftrees.o inftrees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/compress.o compress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/uncompress.o uncompress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzclose.o gzclose.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzlib.o gzlib.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzread.o gzread.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
ar rc libz.a adler32.o crc32.o deflate.o infback.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzclose.o gzlib.o gzread.o gzwrite.o
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example example.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip minigzip.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example64 example64.o -L. libz.a
gcc -shared -Wl,-soname,libz.so.1,-version-script,zlib.map -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o libz.so.1.3.1 adler32.lo crc32.lo deflate.lo inftrees.lo inffast.lo inflate.lo inftrees.lo trees.lo zutil.lo compress.lo uncompress.lo gzclose.lo gzlib.lo gzread.lo gzwrite.lo -lc
rm -f libz.so libz.so.1
ln -s libz.so.1.3.1 libz.so
ln -s libz.so.1.3.1 libz.so.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o examplesh example.o -L. libz.so.1.3.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzipsh minigzip.o -L. libz.so.1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$

```

#### 48. make install: Installs the compiled zlib software.

```

Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
swayam@swayam:~/libraries/zlib-1.3.1$ make -j$(nproc)
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o infback.o infback.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inffast.o inffast.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inflate.o inflate.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inftrees.o inftrees.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o trees.o trees.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o zutil.o zutil.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o compress.o compress.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o uncompress.o uncompress.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzclose.o gzclose.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzlib.o gzlib.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzread.o gzread.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/adler32.o adler32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/crc32.o crc32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/deflate.o deflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/infback.o infback.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inffast.o inffast.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inflate.o inflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/trees.o trees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/zutil.o zutil.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inftrees.o inftrees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/compress.o compress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/uncompress.o uncompress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzclose.o gzclose.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzlib.o gzlib.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzread.o gzread.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
ar rc libz.a adler32.o crc32.o deflate.o infback.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzclose.o gzlib.o gzread.o gzwrite.o
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example example.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip minigzip.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example64 example64.o -L. libz.a
gcc -shared -Wl,-soname,libz.so.1,-version-script,zlib.map -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o libz.so.1.3.1 adler32.lo crc32.lo deflate.lo inftrees.lo inffast.lo inflate.lo inftrees.lo trees.lo zutil.lo compress.lo uncompress.lo gzclose.lo gzlib.lo gzread.lo gzwrite.lo -lc
rm -f libz.so libz.so.1
ln -s libz.so.1.3.1 libz.so
ln -s libz.so.1.3.1 libz.so.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o examplesh example.o -L. libz.so.1.3.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzipsh minigzip.o -L. libz.so.1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$ make install

```



```

gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzcrc32.o gzcrc32.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzlib.o gzlib.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzread.o gzread.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/adler32.o adler32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/crc32.o crc32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/deflate.o deflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inffast.o inffast.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inflate.o inflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/trees.o trees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/zutil.o zutil.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inftrees.o inftrees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/compress.o compress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/uncompress.o uncompress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzlib.o gzlib.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzread.o gzread.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
ar rc libz.a adler32.o crc32.o deflate.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzcrc32.o gzlib.o gzread.o gzwrite.o
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example example.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip minigzip.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example64 example64.o -L. libz.a
gcc -shared -Wl,-soname,libz.so.1,-version-script,zlib.map -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o libz.so.1.3.1 adler32.o crc32.o deflate.o inffast.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzcrc32.o gzlib.o gzread.o gzwrite.o -lc
rm -f libz.so libz.so.1
ln -s libz.so.1.3.1 libz.so
ln -s libz.so.1.3.1 libz.so.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o examplesh example.o -L. libz.so.1.3.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzipsh minigzip.o -L. libz.so.1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$ make install
rm -f /home/swayam/local/lib/libz.a
cp libz.a /home/swayam/local/lib
chmod 644 /home/swayam/local/lib/libz.a
cp libz.so.1.3.1 /home/swayam/local/lib
chmod 755 /home/swayam/local/lib/libz.so.1.3.1
rm -f /home/swayam/local/share/man/man3/zlib.3
cp zlib.3 /home/swayam/local/share/man/man3
chmod 644 /home/swayam/local/share/man/man3/zlib.3
rm -f /home/swayam/local/lib/pkgconfig/zlib.pc
cp zlib.pc /home/swayam/local/lib/pkgconfig
chmod 644 /home/swayam/local/lib/pkgconfig/zlib.pc
rm -f /home/swayam/local/include/zlib.h /home/swayam/local/include/zconf.h
cp zlib.h zconf.h /home/swayam/local/include
chmod 644 /home/swayam/local/include/zlib.h /home/swayam/local/include/zconf.h
swayam@swayam:~/libraries/zlib-1.3.1$

```

49. `cd ..`: Changes the current working directory to the parent directory.

50. `cd Python-3.10.16`: Changes the current working directory to the Python source code directory

```

gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzread.o gzread.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/adler32.o adler32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/crc32.o crc32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/deflate.o deflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inffast.o inffast.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inflate.o inflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/trees.o trees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/zutil.o zutil.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inftrees.o inftrees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/compress.o compress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/uncompress.o uncompress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzlib.o gzlib.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzread.o gzread.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
ar rc libz.a adler32.o crc32.o deflate.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzcrc32.o gzlib.o gzread.o gzwrite.o
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example example.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip minigzip.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example64 example64.o -L. libz.a
gcc -shared -Wl,-soname,libz.so.1,-version-script,zlib.map -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o libz.so.1.3.1 adler32.o crc32.o deflate.o inffast.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzcrc32.o gzlib.o gzread.o gzwrite.o -lc
rm -f libz.so libz.so.1
ln -s libz.so.1.3.1 libz.so
ln -s libz.so.1.3.1 libz.so.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o examplesh example.o -L. libz.so.1.3.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzipsh minigzip.o -L. libz.so.1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$ make install
rm -f /home/swayam/local/lib/libz.a
cp libz.a /home/swayam/local/lib
chmod 644 /home/swayam/local/lib/libz.a
cp libz.so.1.3.1 /home/swayam/local/lib
chmod 755 /home/swayam/local/lib/libz.so.1.3.1
rm -f /home/swayam/local/share/man/man3/zlib.3
cp zlib.3 /home/swayam/local/share/man/man3
chmod 644 /home/swayam/local/share/man/man3/zlib.3
rm -f /home/swayam/local/lib/pkgconfig/zlib.pc
cp zlib.pc /home/swayam/local/lib/pkgconfig
chmod 644 /home/swayam/local/lib/pkgconfig/zlib.pc
rm -f /home/swayam/local/include/zlib.h /home/swayam/local/include/zconf.h
cp zlib.h zconf.h /home/swayam/local/include
chmod 644 /home/swayam/local/include/zlib.h /home/swayam/local/include/zconf.h
swayam@swayam:~/libraries/zlib-1.3.1$ cd ..
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$

```

51. `./configure --prefix=$HOME/local --with-openssl=$HOME/local --with-ensurepip=install LDFLAGS="-L$HOME/local/lib" CPPFLAGS="-I$HOME/local/include" PKG_CONFIG_PATH="$HOME/local/lib/pkgconfig" --enable-optimization`: Configures the Python source code for compilation, specifying various options and prefixes.

```

gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/adler32.o adler32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/crc32.o crc32.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/deflate.o deflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inffast.o inffast.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inflate.o inflate.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/trees.o trees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/zutil.o zutil.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/inftrees.o inftrees.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/compress.o compress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/uncompress.o uncompress.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzclose.o gzclose.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzlib.o gzlib.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzread.o gzread.c
gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -DPIC -c -o objs/gzwrite.o gzwrite.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
ar rc libz.a adler32.o crc32.o deflate.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzclose.o gzlib.o gzread.o gzwrite.o
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example example.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip minigzip.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o example64 example64.o -L. libz.a
gcc -shared -Wl,-soname,libz.so.1,-version-script,zlib.map -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o libz.so.1.3.1 adler32.o crc32.o deflate.o inffast.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzclose.o gzlib.o gzread.o gzwrite.o -lc
rm -f libz.so libz.so.1
ln -s libz.so.1.3.1 libz.so
ln -s libz.so.1.3.1 libz.so.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o examplesh example.o -L. libz.so.1.3.1
gcc -O3 -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzipsh minigzip.o -L. libz.so.1.3.1
swayam@swayam:~/libraries/zlib-1.3.1$ make install
rm -f /home/swayam/local/lib/libz.a
cp libz.a /home/swayam/local/lib
chmod 644 /home/swayam/local/lib/libz.a
cp libz.so.1.3.1 /home/swayam/local/lib
chmod 755 /home/swayam/local/lib/libz.so.1.3.1
rm -f /home/swayam/local/share/man/man3/zlib.3
cp zlib.3 /home/swayam/local/share/man/man3
chmod 644 /home/swayam/local/share/man/man3/zlib.3
rm -f /home/swayam/local/lib/pkgconfig/zlib.pc
cp zlib.pc /home/swayam/local/lib/pkgconfig
chmod 644 /home/swayam/local/lib/pkgconfig/zlib.pc
rm -f /home/swayam/local/include/zlib.h /home/swayam/local/include/zconf.h
cp zlib.h zconf.h /home/swayam/local/include
chmod 644 /home/swayam/local/include/zlib.h /home/swayam/local/include/zconf.h
swayam@swayam:~/libraries/zlib-1.3.1$ cd ..
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$./configure --prefix=$HOME/local --with-openssl=$HOME/local --with-ensurepip=install \
LD_FLAGS="-L$HOME/local/lib" \
CPP_FLAGS="-I$HOME/local/include" \
PKG_CONFIG_PATH="$HOME/local/lib/pkgconfig" \
--enable-optimization

checking for typeahead... no
checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc_FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5, sha1, sha256, sha512, sha3, blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ _

```

52. `make -j$(nproc)`: Compiles the Python source code using multiple CPU cores.

```

checking for typeahead... no
checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc lib-__pure_const bug... no
checking for statatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a_d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5, sha1, sha256, sha512, sha3, blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/_ld_so_aix
config.status: creating pyconfig.h
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ make -j$(nproc)_

```

```

ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -DCONFIG_64=1 -DASM=1 -I/home/swayam/libraries/Python-3.10.16/Modules/_decimal/
libmpdec -I./Include -I./home/swayam/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/
home/swayam/libraries/Python-3.10.16 -c /home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.c -o build/temp.linux-x86_64-3.10/home/swayam/
libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/_decimal.o build/temp.linux-x86_64-3.10/
home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/basearith.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/li
bmpdec/constants.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/context.o build/temp.linux-x86_64-3.10/home/swaya
m/libraries/Python-3.10.16/Modules/_decimal/libmpdec/convolute.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/crt
.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/diffradix2.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Py
thon-3.10.16/Modules/_decimal/libmpdec/fnt.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/fourstep.o build/temp.l
inux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/io.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_
decimal/libmpdec/mpaalloc.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/mpdecimal.o build/temp.linux-x86_64-3.10/
home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/numbertheory.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal
/libmpdec/sixstep.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.o -L/home/swayam/local/lib -L/usr/lib/
x86_64-linux-gnu -L/usr/local/lib -lm -o build/lib.linux-x86_64-3.10/_decimal.cpython-310-x86_64-linux-gnu.so
*** WARNING: renaming '_ssl' since importing it failed: libssl.so.1.1: cannot open shared object file: No such file or directory
*** WARNING: renaming '_hashlib' since importing it failed: libcrypto.so.1.1: cannot open shared object file: No such file or directory

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline

To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

Following modules built successfully but were removed because they could not be imported:
_hashlib _ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$

```

53. make install: Installs the compiled Python software.

```

ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -DCONFIG_64=1 -DASM=1 -I/home/swapam/libraries/Python-3.10.16/Modules/_decimal/
libmpdec -I./Include -I./I/home/swapam/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swapam/libraries/Python-3.10.16/Include -I/
home/swapam/libraries/Python-3.10.16 -c /home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.c -o build/temp.linux-x86_64-3.10/home/swapam/
libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.o
gcc -shared -L/home/swapam/local/lib build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal.o build/temp.linux-x86_64-3.10/
home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/basearith.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/li
bmpdec/constants.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/context.o build/temp.linux-x86_64-3.10/home/swapa
m/libraries/Python-3.10.16/Modules/_decimal/libmpdec/convolute.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/crt
.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/diffradix2.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Py
thon-3.10.16/Modules/_decimal/libmpdec/fnt.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/fourstep.o build/temp.l
inux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/lo.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_
decimal/libmpdec/mpalloc.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/mpdecimal.o build/temp.linux-x86_64-3.10/
home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/numbertheory.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal
/libmpdec/sixstep.o build/temp.linux-x86_64-3.10/home/swapam/libraries/Python-3.10.16/Modules/_decimal/libmpdec/transpose.o -L/home/swapam/local/lib -L/usr/lib/
x86_64-linux-gnu -L/usr/local/lib -lm -o build/lib.linux-x86_64-3.10/_decimal.cpython-310-x86_64-linux-gnu.so
*** WARNING: renaming "_ssl" since importing it failed: libssl.so.1.1: cannot open shared object file: No such file or directory
*** WARNING: renaming "_hashlib" since importing it failed: libcrypto.so.1.1: cannot open shared object file: No such file or directory

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline

To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

Following modules built successfully but were removed because they could not be imported:
_hashlib _ssl

Could not build the ssl module!
Python requires a OpenSSL 1.1.1 or newer
Custom linker flags may require --with-openssl-rpath=auto

running build_scripts
copying and adjusting /home/swapam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swapam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swapam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
swayam@swayam:~/libraries/Python-3.10.16$ make install

then rm -f /home/swapam/local/bin/python3; \
else true; \
fi
(cd /home/swapam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swapam/local/bin/python3.10-config; \
 (cd /home/swapam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swapam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swapam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swapam/local/bin/python3-config
(cd /home/swapam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swapam/local/lib/pkgconfig/python3.pc
(cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swapam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swapam/local/bin/idle3
(cd /home/swapam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swapam/local/bin/pydoc3
(cd /home/swapam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swapam/local/bin/2to3
(cd /home/swapam/local/bin; ln -s 2to3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swapam/local/bin/python3-32; \
 (cd /home/swapam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swapam/local/bin/python3-intel64; \
 (cd /home/swapam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swapam/local/share/man/man1/python3.1
(cd /home/swapam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuputils-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuputils, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swapam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuputils-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$

```

#### 54. python3 --version: Verifies the installation of Python by checking its version

```

then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version

```

```

fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$

```

At this point, we faced our third major error/warning: pip scripts not on PATH.

Let's quickly fix it ...

55. `cd ..`: Changes the current working directory to the parent directory.

```
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn-fjlr
Processing /tmp/tmpnbn-fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn-fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$
```

56. `export PATH=$HOME/local/bin:$PATH`: Adds the local bin directory to the system's PATH environment variable.

```
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn-fjlr
Processing /tmp/tmpnbn-fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn-fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
```

57. `echo $PATH`: Displays the updated PATH environment variable.

```

if test "3.10" != "3.10"; then \
 rm -f /home/swapam/local/bin/python3.10-config; \
 (cd /home/swapam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swapam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swapam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swapam/local/bin/python3-config
(cd /home/swapam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swapam/local/lib/pkgconfig/python3.pc
(cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swapam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swapam/local/bin/idle3
(cd /home/swapam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swapam/local/bin/pydoc3
(cd /home/swapam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swapam/local/bin/2to3
(cd /home/swapam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swapam/local/bin/python3-32; \
 (cd /home/swapam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swapam/local/bin/python3-intel64; \
 (cd /home/swapam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swapam/local/share/man/man1/python3.1
(cd /home/swapam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swapam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swapam@swapam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swapam@swapam:~/libraries/Python-3.10.16$ cd ..
swapam@swapam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swapam@swapam:~/libraries$ echo $PATH

 (cd /home/swapam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swapam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swapam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swapam/local/bin/python3-config
(cd /home/swapam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swapam/local/lib/pkgconfig/python3.pc
(cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swapam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swapam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swapam/local/bin/idle3
(cd /home/swapam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swapam/local/bin/pydoc3
(cd /home/swapam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swapam/local/bin/2to3
(cd /home/swapam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swapam/local/bin/python3-32; \
 (cd /home/swapam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swapam/local/bin/python3-intel64; \
 (cd /home/swapam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swapam/local/share/man/man1/python3.1
(cd /home/swapam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swapam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swapam@swapam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swapam@swapam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swapam@swapam:~/libraries$ echo $PATH
/home/swapam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swapam@swapam:~/libraries$

```

58. `python3 --version`: Verifies that the Python installation is accessible via the updated PATH.



```

(cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
 WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
 Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$

```

```

(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
 WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
 Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$

```

We successfully installed the Python 3.10.16!

Now let's try for openssl again ...

59. `cd openssl-1.1.1w`: Changes the current working directory to the OpenSSL source code directory.



```

(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$ cd openssl-1.1.1w

rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$ _

```

60. `./config --prefix=$HOME/openssl --openssldir=$HOME/openssl`: Reconfigures the OpenSSL source code for compilation, specifying a new installation prefix.

```

rm -f /home/swayam/local/lib/pkgconfig/python3-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$./config --prefix=$HOME/openssl --openssldir=$HOME/openssl_
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpnbn_fjlr
Processing /tmp/tmpnbn_fjlr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tmpnbn_fjlr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$./config --prefix=$HOME/openssl --openssldir=$HOME/openssl
Operating system: x86_64-whatever-linux2
Configuring OpenSSL version 1.1.1w (0x1010117fL) for linux-x86_64
Using os-specific seed configuration
Creating configdata.pm
Creating Makefile

*** OpenSSL has been successfully configured

*** If you encounter a problem while building, please open an
*** issue on GitHub <https://github.com/openssl/openssl/issues>
*** and include the output from the following command:

*** perl configdata.pm --dump

*** (If you are new to OpenSSL, you might want to consult the
*** 'Troubleshooting' section in the INSTALL file first)

swayam@swayam:~/libraries/openssl-1.1.1w$

```

61. `make -j$(nproc)`: Compiles the OpenSSL source code using multiple CPU cores.

```

(cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 &ensurepip --root=/ ; \
 fi
Looking in links: /tmp/tm0bnf_filr
Processing /tmp/tm0bnf_filr/setuptools-65.5.0-py3-none-any.whl
Processing /tmp/tm0bnf_filr/pip-23.0.1-py3-none-any.whl
Installing collected packages: setuptools, pip
WARNING: The scripts pip3 and pip3.10 are installed in '/home/swayam/local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.0.1 setuptools-65.5.0
swayam@swayam:~/libraries/Python-3.10.16$ python3 --version
-bash: /usr/bin/python3: No such file or directory
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ export PATH=$HOME/local/bin:$PATH
swayam@swayam:~/libraries$ echo $PATH
/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
swayam@swayam:~/libraries$ python3 --version
Python 3.10.16
swayam@swayam:~/libraries$ cd openssl-1.1.1w
swayam@swayam:~/libraries/openssl-1.1.1w$./config --prefix=$HOME/openssl --openssldir=$HOME/openssl
Operating system: x86_64-whatever-linux2
Configuring OpenSSL version 1.1.1w (0x1010117fL) for linux-x86_64
Using os-specific seed configuration
Creating configdata.pm
Creating Makefile

*** OpenSSL has been successfully configured

*** If you encounter a problem while building, please open an
*** issue on GitHub <https://github.com/openssl/openssl/issues>
*** and include the output from the following command:

*** perl configdata.pm --dump

*** (If you are new to OpenSSL, you might want to consult the
*** 'Troubleshooting' section in the INSTALL file first)

swayam@swayam:~/libraries/openssl-1.1.1w$ make -j$(nproc)

-lssl test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/sslappitest test/sslappitest.o test/ssltestlib.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/sslbuffertest
rm -f test/sslcorrupttest
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/sslbuffertest test/sslbuffertest.o test/ssltestlib.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/sslcorrupttest test/sslcorrupttest.o test/ssltestlib.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/ssltest_old
rm -f test/sysdefaulttest
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/ssltest_old test/ssltest_old.o \
-lssl -lcrypto -ldl -pthread
rm -f test/tls13ccctest
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/sysdefaulttest test/sysdefaulttest.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/tls13secretstest
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/tls13ccctest test/ssltestlib.o test/tls13ccctest.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/tls13secretstest ssl/packet.o ssl/tls13_enc.o test/tls13secretstest.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/uitest
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o test/uitest test/uitest.o \
apps/libapps.a -lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f apps/openssl
rm -f fuzz/asn1-test
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o apps/openssl apps/asn1pars.o apps/ca.o apps/ciphers.o apps/cms.o apps/crl.o apps/crl2p7.o apps/dgst.o apps/dhparam.o apps/dsa.o apps/dsaparam.o apps/
ec.o apps/ecparam.o apps/enc.o apps/engine.o apps/errstr.o apps/genrsa.o apps/genkey.o apps/genrsa.o apps/nseq.o apps/ocsp.o apps/openssl.o apps/passwd.o apps/
pkcs12.o apps/pkcs7.o apps/pkcs8.o apps/pkey.o apps/pkeyparam.o apps/pkeyutil.o apps/prime.o apps/rand.o apps/rehash.o apps/req.o apps/rsa.o apps/rsautl.o apps/s
_client.o apps/s_server.o apps/s_time.o apps/session.o apps/smime.o apps/speed.o apps/spkac.o apps/srp.o apps/storeutil.o apps/ts.o apps/verify.o apps/version.o
apps/x509.o \
apps/libapps.a -lssl -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o fuzz/asn1-test fuzz/asn1.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
rm -f fuzz/client-test
${LDCMD:-gcc} -pthread -m64 -Ma,-noexecstack -Wall -O3 -L. \
-o fuzz/client-test fuzz/client.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
make[1]: Leaving directory '/home/swayam/libraries/openssl-1.1.1w'
swayam@swayam:~/libraries/openssl-1.1.1w$

```

62. make install: Installs the compiled OpenSSL software.

```

-lssl test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/sslapiptest test/sslapiptest.o test/ssltestlib.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/sslbuffertest
rm -f test/sslcorrupttest
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/sslbuffertest test/sslbuffertest.o test/ssltestlib.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/sslcorrupttest test/sslcorrupttest.o test/ssltestlib.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/ssltest_old
rm -f test/sysdefaulttest
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/ssltest_old test/ssltest_old.o \
-lssl -lcrypto -ldl -pthread
rm -f test/tls13ccstest
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/sysdefaulttest test/sysdefaulttest.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/tls13secretstest
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/tls13ccstest test/ssltestlib.o test/tls13ccstest.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/tls13secretstest ssl/packet.o ssl/tls13_enc.o test/tls13secretstest.o \
-lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f test/uitest
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o test/uitest test/uitest.o \
apps/libapps.a -lssl test/libtestutil.a -lcrypto -ldl -pthread
rm -f apps/openssl
rm -f fuzz/asn1-test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o apps/openssl apps/asnipars.o apps/ca.o apps/ciphers.o apps/cms.o apps/crl.o apps/crl2p7.o apps/dgst.o apps/dhparam.o apps/dsa.o apps/dsaparam.o apps/
ec.o apps/ecparam.o apps/enc.o apps/engine.o apps/errstr.o apps/gendsa.o apps/genpkey.o apps/genrsa.o apps/nseq.o apps/ocsp.o apps/openssl.o apps/passwd.o apps/
pkcs12.o apps/pkcs7.o apps/pkcs8.o apps/pkey.o apps/pkeyparam.o apps/pkeyutl.o apps/prime.o apps/rand.o apps/rehash.o apps/req.o apps/rsa.o apps/rsautil.o apps/s
_client.o apps/s_server.o apps/s_time.o apps/ssl.o apps/smime.o apps/speed.o apps/spkac.o apps/srp.o apps/storeutl.o apps/ts.o apps/verify.o apps/version.o
apps/x509.o \
apps/libapps.a -lssl -lcrypto -ldl -pthread
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/asn1-test fuzz/asn1.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
rm -f fuzz/client-test
${LDCMD:-gcc} -pthread -m64 -Wa,--noexecstack -Wall -O3 -L. \
-o fuzz/client-test fuzz/client.o fuzz/test-corpus.o \
-lssl -lcrypto -ldl -pthread
make[1]: Leaving directory '/home/swayam/libraries/openssl-1.1.1w'
swayam@swayam:~/libraries/openssl-1.1.1w$ make install
/home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_OBJ.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_critical.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509v3_delete_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509v3_add_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_get_ext_count.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_get_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_get_ext_by_NID.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_get_ext_by_OBJ.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_get_ext_by_critical.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_delete_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_add_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_get_ext_count.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_get_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_get_ext_by_NID.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_get_ext_by_OBJ.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_get_ext_by_critical.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_delete_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_CRL_add_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_get_ext_count.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_get_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_get_ext_by_NID.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_get_ext_by_OBJ.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_get_ext_by_critical.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_delete_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man3/X509_REVOKED_add_ext.html -> /home/swayam/openssl/share/doc/openssl/html/man3/X509v3_get_ext_by_NID.html
/home/swayam/openssl/share/doc/openssl/html/man5/config.html
/home/swayam/openssl/share/doc/openssl/html/man7/bio.html
/home/swayam/openssl/share/doc/openssl/html/man7/crypt.html
/home/swayam/openssl/share/doc/openssl/html/man7/ct.html
/home/swayam/openssl/share/doc/openssl/html/man7/des_modes.html
/home/swayam/openssl/share/doc/openssl/html/man7/Ed25519.html
/home/swayam/openssl/share/doc/openssl/html/man7/Ed448.html -> /home/swayam/openssl/share/doc/openssl/html/man7/Ed25519.html
/home/swayam/openssl/share/doc/openssl/html/man7/evp.html
/home/swayam/openssl/share/doc/openssl/html/man7/ssl_store-file.html
/home/swayam/openssl/share/doc/openssl/html/man7/ssl_store.html
/home/swayam/openssl/share/doc/openssl/html/man7/passphrase-encoding.html
/home/swayam/openssl/share/doc/openssl/html/man7/proxy-certificates.html
/home/swayam/openssl/share/doc/openssl/html/man7/RAND.html
/home/swayam/openssl/share/doc/openssl/html/man7/RAND_DRBG.html
/home/swayam/openssl/share/doc/openssl/html/man7/RSA-PSS.html
/home/swayam/openssl/share/doc/openssl/html/man7/scrpt.html
/home/swayam/openssl/share/doc/openssl/html/man7/SM2.html
/home/swayam/openssl/share/doc/openssl/html/man7/ssl.html
/home/swayam/openssl/share/doc/openssl/html/man7/X25519.html
/home/swayam/openssl/share/doc/openssl/html/man7/X448.html -> /home/swayam/openssl/share/doc/openssl/html/man7/X25519.html
/home/swayam/openssl/share/doc/openssl/html/man7/x509.html
swayam@swayam:~/libraries/openssl-1.1.1w$

```

63. `cd ..`: Changes the current working directory to the parent directory.

64. `export PATH=$HOME/openssl/bin:$PATH`: Adds the OpenSSL bin directory to the system's PATH environment variable.

65. `export LD_LIBRARY_PATH=$HOME/openssl/lib:$LD_LIBRARY_PATH`: Adds the OpenSSL lib directory to the system's LD\_LIBRARY\_PATH environment variable.



[illegible]

66. `cd Python-3.10.16`: Changes the current working directory to the Python source code directory.



```

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$

```

## 68. make: Compiles the Python source code

```

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ make_

```



```

 echo "generate-posix-vars failed" ; \
 rm -f ./pybuilddir.txt ; \
 exit 1 ; \
fi
CC='gcc' LDSHARED='gcc -shared -L/home/swayam/local/lib' OPT='-DNDEBUG -g -fwrapv -O3 -Wall' _TCLTK_INCLUDES='' _TCLTK_LIBS='' ./python -E ./se
tup.py build
running build
running build_ext
INFO: Can't locate Tcl/Tk libs and/or headers
building '_ssl' extension
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swayam/local/include -I./Include -I. -I/home/swayam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/librar
ies/Python-3.10.16/Modules/_ssl.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ssl.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ssl.o -L/home/swayam/local/lib -L/home/swayam/lo
cal/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lssl -lcrypto -o build/lib.linux-x86_64-3.10/_ssl.cpython-310-x86_64-linux-gnu.so
building '_hashlib' extension
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swayam/local/include -I./Include -I. -I/home/swayam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/librar
ies/Python-3.10.16/Modules/_hashopenssl.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_hashopenssl.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_hashopenssl.o -L/home/swayam/local/lib -L/home/s
wayam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lssl -lcrypto -o build/lib.linux-x86_64-3.10/_hashlib.cpython-310-x86_64-linux-gnu.so

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline

To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

running build_scripts
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
gcc -L/home/swayam/local/lib -Xlinker -export-dynamic -o Programs/_testembed Programs/_testembed.o libpython3.10.a -lcrypt -ldl -lm -lm
sed -e 's,${EXECNAME},/home/swayam/local/bin/python3.10,' < ./Misc/python-config.in >python-config.py
LC_ALL=C sed -e 's,${(A-Za-z0-9_*)},\${\1},g' < Misc/python-config.sh >python-config
swayam@swayam:~/libraries/Python-3.10.16$ _

```

## 69. make install: Installs the compiled Python software.

```

 echo "generate-posix-vars failed" ; \
 rm -f ./pybuilddir.txt ; \
 exit 1 ; \
fi
CC='gcc' LDSHARED='gcc -shared -L/home/swayam/local/lib' OPT='-DNDEBUG -g -fwrapv -O3 -Wall' _TCLTK_INCLUDES='' _TCLTK_LIBS='' ./python -E ./se
tup.py build
running build
running build_ext
INFO: Can't locate Tcl/Tk libs and/or headers
building '_ssl' extension
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swayam/local/include -I./Include -I. -I/home/swayam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/librar
ies/Python-3.10.16/Modules/_ssl.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ssl.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_ssl.o -L/home/swayam/local/lib -L/home/swayam/lo
cal/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lssl -lcrypto -o build/lib.linux-x86_64-3.10/_ssl.cpython-310-x86_64-linux-gnu.so
building '_hashlib' extension
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -I/home/swayam/local/include -I./Include -I. -I/home/swayam/local/include -I/
usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c /home/swayam/librar
ies/Python-3.10.16/Modules/_hashopenssl.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_hashopenssl.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_hashopenssl.o -L/home/swayam/local/lib -L/home/s
wayam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lssl -lcrypto -o build/lib.linux-x86_64-3.10/_hashlib.cpython-310-x86_64-linux-gnu.so

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_sqlite3 _tkinter _uuid
nis readline

To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

running build_scripts
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
gcc -L/home/swayam/local/lib -Xlinker -export-dynamic -o Programs/_testembed Programs/_testembed.o libpython3.10.a -lcrypt -ldl -lm -lm
sed -e 's,${EXECNAME},/home/swayam/local/bin/python3.10,' < ./Misc/python-config.in >python-config.py
LC_ALL=C sed -e 's,${(A-Za-z0-9_*)},\${\1},g' < Misc/python-config.sh >python-config
swayam@swayam:~/libraries/Python-3.10.16$ make install_

```

```

 echo "Creating directory /home/swayam/local/lib/pkgconfig"; \
 /usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \
 fi
 if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
 then rm -f /home/swayam/local/bin/python3; \
 else true; \
 fi
 (cd /home/swayam/local/bin; ln -s python3.10 python3)
 if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10.pc python3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10-embed.pc python3.10-embed.pc); \
 fi
 rm -f /home/swayam/local/bin/python3-config
 (cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
 rm -f /home/swayam/local/lib/pkgconfig/python3.pc
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10.pc python3.pc)
 rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10-embed.pc python3-embed.pc)
 rm -f /home/swayam/local/bin/idle3
 (cd /home/swayam/local/bin; ln -s idle3.10 idle3)
 rm -f /home/swayam/local/bin/pydoc3
 (cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
 rm -f /home/swayam/local/bin/2to3
 (cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
 if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
 fi
 if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
 fi
 rm -f /home/swayam/local/share/man/man1/python3.1
 (cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
 if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
 fi
 Looking in links: /tmp/tmpi9ggfmr
 Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
 Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
 swayam@swayam:~/libraries/Python-3.10.16$

```

## Creating a virtual environment and installing label-studio

70. `python3.10 -m venv my`: Creates a new virtual environment named "my" using the Python 3.10 installation.

```

 echo "Creating directory /home/swayam/local/lib/pkgconfig"; \
 /usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \
 fi
 if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
 then rm -f /home/swayam/local/bin/python3; \
 else true; \
 fi
 (cd /home/swayam/local/bin; ln -s python3.10 python3)
 if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10.pc python3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10-embed.pc python3.10-embed.pc); \
 fi
 rm -f /home/swayam/local/bin/python3-config
 (cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
 rm -f /home/swayam/local/lib/pkgconfig/python3.pc
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10.pc python3.pc)
 rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
 (cd /home/swayam/local/lib/pkgconfig; ln -s python3.10-embed.pc python3-embed.pc)
 rm -f /home/swayam/local/bin/idle3
 (cd /home/swayam/local/bin; ln -s idle3.10 idle3)
 rm -f /home/swayam/local/bin/pydoc3
 (cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
 rm -f /home/swayam/local/bin/2to3
 (cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
 if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
 fi
 if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
 fi
 rm -f /home/swayam/local/share/man/man1/python3.1
 (cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
 if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
 fi
 Looking in links: /tmp/tmpi9ggfmr
 Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
 Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
 swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my

```

```

/usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \
fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
rm -f /home/swayam/local/bin/python3.10-config; \
(cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-32; \
(cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-intel64; \
(cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
case install in \
upgrade) ensurepip="--upgrade" ;; \
install|*) ensurepip="" ;; \
esac; \
./python -E -m ensurepip \
$ensurepip --root=/; \
fi
Looking in links: /tmp/tmpi6ggfmr
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my
swayam@swayam:~/libraries/Python-3.10.16$ _

```

## 71. source my/bin/activate: Activates the virtual environment.

```

/usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \
fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
rm -f /home/swayam/local/bin/python3.10-config; \
(cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-32; \
(cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-intel64; \
(cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
case install in \
upgrade) ensurepip="--upgrade" ;; \
install|*) ensurepip="" ;; \
esac; \
./python -E -m ensurepip \
$ensurepip --root=/; \
fi
Looking in links: /tmp/tmpi6ggfmr
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my
swayam@swayam:~/libraries/Python-3.10.16$ source my/bin/activate

```

```

fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
rm -f /home/swayam/local/bin/python3.10-config; \
(cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-32; \
(cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-intel64; \
(cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
case install in \
upgrade) ensurepip="--upgrade" ;; \
install|*) ensurepip="" ;; \
esac; \
./python -E -m ensurepip \
$ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpiy6ggfmr
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv mj
swayam@swayam:~/libraries/Python-3.10.16$ source mj/bin/activate
(mj) swayam@swayam:~/libraries/Python-3.10.16$

```

## 72. pip install label-studio: Installs the label-studio package using pip.

```

fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
rm -f /home/swayam/local/bin/python3.10-config; \
(cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-32; \
(cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-intel64; \
(cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno"; then \
case install in \
upgrade) ensurepip="--upgrade" ;; \
install|*) ensurepip="" ;; \
esac; \
./python -E -m ensurepip \
$ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpiy6ggfmr
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv mj
swayam@swayam:~/libraries/Python-3.10.16$ source mj/bin/activate
(mj) swayam@swayam:~/libraries/Python-3.10.16$ pip install label-studio_

```

```

ted, cffi, bleach, black, asgiref, anyio, user-agents, rich, requests-mock, pydantic, pandas, opentelemetry-api, jsonschema-specifications, httpx, grpcio-status
, google-cloud-audit-log, google-auth, faker, Django, cryptography, botocore, azure-core, s3transfer, pyboxen, openai, jsonschema, grpc-google-iam-v1, google-ap
i-core, drf-generators, django-rest-framework, django-user-agents, django-storages, django-rq, django-model-utils, django-migration-linter, django-filter, django-ex
tensions, django-debug-toolbar, django-csp, django-cors-headers, django-anoying, azure-storage-blob, jsf, humansignal-drif-yasg, google-cloud-core, datamodel-
code-generator, boto3, label-studio-sdk, google-cloud-storage, google-cloud-appengine-logging, google-cloud-logging, label-studio
DEPRECATION: drf-flex-fields is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' packag
e is not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https
://github.com/pypa/pip/issues/8559
Running setup.py install for drf-flex-fields ... done
DEPRECATION: django-ranged-fileresponse is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wh
eel' package is not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be fou
nd at https://github.com/pypa/pip/issues/8559
Running setup.py install for django-ranged-fileresponse ... done
DEPRECATION: attr is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' package is not in
stalled. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https://github.c
om/pypa/pip/issues/8559
Running setup.py install for attr ... done
Attempting uninstall: setuptools
Found existing installation: setuptools 65.5.0
Uninstalling setuptools-65.5.0:
Successfully uninstalled setuptools-65.5.0
DEPRECATION: ordered-set is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' package is
not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https://g
ithub.com/pypa/pip/issues/8559
Running setup.py install for ordered-set ... done
DEPRECATION: drf-generators is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' package
is not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https:
//github.com/pypa/pip/issues/8559
Running setup.py install for drf-generators ... done
Successfully installed Django-4.2.18 MarkupSafe-3.0.2 Pillow-11.1.0 annotated-types-0.7.0 anyio-4.8.0 appdirs-1.4.4 argcomplete-3.5.3 asgiref-3.8.1 attr-0.3.1 a
ttrs-25.1.0 azure-core-1.32.0 azure-storage-blob-12.24.1 black-24.10.0 bleach-5.0.1 boto-2.49.0 boto3-1.36.6 botocore-1.36.6 cachetools-5.5.1 certifi-2024.12.14
cffi-1.17.1 charset-normalizer-3.4.1 click-8.1.8 colorama-0.4.6 cryptography-44.0.0 datamodel-code-generator-0.26.1 defusedxml-0.7.1 deprecated-1.2.17 distro-1
.9.0 django-anoying-0.10.6 django-cors-headers-3.6.0 django-csp-3.7 django-debug-toolbar-3.2.1 django-enviro-0.10.0 django-extensions-3.2.3 django-filter-2.4.
0 django-migration-linter-5.1.0 django-model-utils-4.1.1 django-ranged-fileresponse-0.1.2 django-rq-2.5.1 django-storages-1.12.3 django-user-agents-0.4.0 django
restframework-3.15.2 dnspython-2.7.0 drf-dynamic-fields-0.3.0 drf-flex-fields-0.9.5 drf-generators-0.3.0 email-validator-2.2.0 exceptiongroup-1.2.2 expiringdict
-1.2.2 faker-35.0.0 genson-1.3.0 google-api-core-2.24.0 google-auth-2.38.0 google-cloud-appengine-logging-1.5.0 google-cloud-audit-log-0.3.0 google-cloud-core-2
.4.1 google-cloud-logging-3.11.3 google-cloud-storage-2.19.0 google-crc32c-1.6.0 google-resumable-media-2.7.2 googleapis-common-protos-1.66.0 grpc-google-iam-v1
-0.14.0 grpcio-1.70.0 grpcio-status-1.70.0 h11-0.14.0 httpcore-1.0.7 httpx-0.28.1 humansignal-drif-yasg-1.21.10.post1 idna-3.10 ljson-3.3.0 importlib-metadata-8.
5.0 inflect-5.6.2 inflection-0.5.1 isodate-0.7.2 isort-5.13.2 Jinja2-3.1.5 Jiter-0.8.2 jmespath-1.0.1 joblib-1.4.2 jsf-0.11.2 jsonschema-4.23.0 jsonschema-speci
fications-2024.10.1 label-studio-1.15.0 label-studio-sdk-1.0.8 launchdarkly-server-sdk-8.2.1 lockfile-0.12.2 lxml-5.3.0 markdown-it-py-3.0.0 mdurl-0.1.2 mypy-ex
tensions-1.0.0 nltk-3.9.1 numpy-1.26.4 openai-1.60.1 opentelemetry-api-1.29.0 ordered-set-4.0.2 packaging-24.2 pandas-2.2.3 pathspec-0.12.1 platformdirs-4.3.6 p
oto-plus-1.25.0 protobuf-5.29.3 pycpg2-binary-2.9.10 pyRFC3339-2.0.1 pyasn1-0.6.1 pyasn1-modules-0.4.1 pyboxen-1.3.0 pycparser-2.22 pydantic-2.10.6 pydantic-
core-2.27.2 pygments-2.19.1 python-dateutil-2.9.0.post0 python-json-logger-2.0.4 pytz-2022.7.1 pyyaml-6.0.2 redis-3.5.3 referencing-0.36.2 regex-2024.11.6 requ
sts-2.32.3 requests-mock-1.12.1 rich-13.9.4 rpsd-py-0.22.3 rq-1.10.1 rsa-4.9 rstr-3.2.2 rules-3.4 s3transfer-0.11.2 semver-3.0.4 sentry-sdk-2.20.0 setuptools-75
.8.0 six-1.17.0 smart-open-7.1.0 sniffio-1.3.1 sqlalchemy-0.5.3 toml-0.10.2 tomlit-2.2.1 tqdm-4.67.1 typing_extensions-4.12.2 tzdata-2025.1 ua-parser-1.0.0 ua-pars
er-builtins-0.18.0.post1 ujson-5.10.0 uritemplate-4.1.1 urllib3-1.26.20 user-agents-2.2.0 webencodings-0.5.1 wheel-0.40.0 wrapt-1.17.2 xmljson-0.2.1 zipp-3.21.0

[notice] A new release of pip is available: 23.0.1 -> 24.3.1
[notice] To update, run: pip install --upgrade pip
(my) swayam@swayam:~/libraries/Python-3.10.16$

```

### 73. label-studio: Runs the label-studio application.

```

ted, cffi, bleach, black, asgiref, anyio, user-agents, rich, requests-mock, pydantic, pandas, opentelemetry-api, jsonschema-specifications, httpx, grpcio-status
, google-cloud-audit-log, google-auth, faker, Django, cryptography, botocore, azure-core, s3transfer, pyboxen, openai, jsonschema, grpc-google-iam-v1, google-ap
i-core, drf-generators, django-rest-framework, django-user-agents, django-storages, django-rq, django-model-utils, django-migration-linter, django-filter, django-ex
tensions, django-debug-toolbar, django-csp, django-cors-headers, django-anoying, azure-storage-blob, jsf, humansignal-drif-yasg, google-cloud-core, datamodel-
code-generator, boto3, label-studio-sdk, google-cloud-storage, google-cloud-appengine-logging, google-cloud-logging, label-studio
DEPRECATION: drf-flex-fields is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' packag
e is not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https
://github.com/pypa/pip/issues/8559
Running setup.py install for drf-flex-fields ... done
DEPRECATION: django-ranged-fileresponse is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wh
eel' package is not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be fou
nd at https://github.com/pypa/pip/issues/8559
Running setup.py install for django-ranged-fileresponse ... done
DEPRECATION: attr is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' package is not in
stalled. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https://github.c
om/pypa/pip/issues/8559
Running setup.py install for attr ... done
Attempting uninstall: setuptools
Found existing installation: setuptools 65.5.0
Uninstalling setuptools-65.5.0:
Successfully uninstalled setuptools-65.5.0
DEPRECATION: ordered-set is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' package is
not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https://g
ithub.com/pypa/pip/issues/8559
Running setup.py install for ordered-set ... done
DEPRECATION: drf-generators is being installed using the legacy 'setup.py install' method, because it does not have a 'pyproject.toml' and the 'wheel' package
is not installed. pip 23.1 will enforce this behaviour change. A possible replacement is to enable the '--use-pep517' option. Discussion can be found at https:
//github.com/pypa/pip/issues/8559
Running setup.py install for drf-generators ... done
Successfully installed Django-4.2.18 MarkupSafe-3.0.2 Pillow-11.1.0 annotated-types-0.7.0 anyio-4.8.0 appdirs-1.4.4 argcomplete-3.5.3 asgiref-3.8.1 attr-0.3.1 a
ttrs-25.1.0 azure-core-1.32.0 azure-storage-blob-12.24.1 black-24.10.0 bleach-5.0.1 boto-2.49.0 boto3-1.36.6 botocore-1.36.6 cachetools-5.5.1 certifi-2024.12.14
cffi-1.17.1 charset-normalizer-3.4.1 click-8.1.8 colorama-0.4.6 cryptography-44.0.0 datamodel-code-generator-0.26.1 defusedxml-0.7.1 deprecated-1.2.17 distro-1
.9.0 django-anoying-0.10.6 django-cors-headers-3.6.0 django-csp-3.7 django-debug-toolbar-3.2.1 django-enviro-0.10.0 django-extensions-3.2.3 django-filter-2.4.
0 django-migration-linter-5.1.0 django-model-utils-4.1.1 django-ranged-fileresponse-0.1.2 django-rq-2.5.1 django-storages-1.12.3 django-user-agents-0.4.0 django
restframework-3.15.2 dnspython-2.7.0 drf-dynamic-fields-0.3.0 drf-flex-fields-0.9.5 drf-generators-0.3.0 email-validator-2.2.0 exceptiongroup-1.2.2 expiringdict
-1.2.2 faker-35.0.0 genson-1.3.0 google-api-core-2.24.0 google-auth-2.38.0 google-cloud-appengine-logging-1.5.0 google-cloud-audit-log-0.3.0 google-cloud-core-2
.4.1 google-cloud-logging-3.11.3 google-cloud-storage-2.19.0 google-crc32c-1.6.0 google-resumable-media-2.7.2 googleapis-common-protos-1.66.0 grpc-google-iam-v1
-0.14.0 grpcio-1.70.0 grpcio-status-1.70.0 h11-0.14.0 httpcore-1.0.7 httpx-0.28.1 humansignal-drif-yasg-1.21.10.post1 idna-3.10 ljson-3.3.0 importlib-metadata-8.
5.0 inflect-5.6.2 inflection-0.5.1 isodate-0.7.2 isort-5.13.2 Jinja2-3.1.5 Jiter-0.8.2 jmespath-1.0.1 joblib-1.4.2 jsf-0.11.2 jsonschema-4.23.0 jsonschema-speci
fications-2024.10.1 label-studio-1.15.0 label-studio-sdk-1.0.8 launchdarkly-server-sdk-8.2.1 lockfile-0.12.2 lxml-5.3.0 markdown-it-py-3.0.0 mdurl-0.1.2 mypy-ex
tensions-1.0.0 nltk-3.9.1 numpy-1.26.4 openai-1.60.1 opentelemetry-api-1.29.0 ordered-set-4.0.2 packaging-24.2 pandas-2.2.3 pathspec-0.12.1 platformdirs-4.3.6 p
oto-plus-1.25.0 protobuf-5.29.3 pycpg2-binary-2.9.10 pyRFC3339-2.0.1 pyasn1-0.6.1 pyasn1-modules-0.4.1 pyboxen-1.3.0 pycparser-2.22 pydantic-2.10.6 pydantic-
core-2.27.2 pygments-2.19.1 python-dateutil-2.9.0.post0 python-json-logger-2.0.4 pytz-2022.7.1 pyyaml-6.0.2 redis-3.5.3 referencing-0.36.2 regex-2024.11.6 requ
sts-2.32.3 requests-mock-1.12.1 rich-13.9.4 rpsd-py-0.22.3 rq-1.10.1 rsa-4.9 rstr-3.2.2 rules-3.4 s3transfer-0.11.2 semver-3.0.4 sentry-sdk-2.20.0 setuptools-75
.8.0 six-1.17.0 smart-open-7.1.0 sniffio-1.3.1 sqlalchemy-0.5.3 toml-0.10.2 tomlit-2.2.1 tqdm-4.67.1 typing_extensions-4.12.2 tzdata-2025.1 ua-parser-1.0.0 ua-pars
er-builtins-0.18.0.post1 ujson-5.10.0 uritemplate-4.1.1 urllib3-1.26.20 user-agents-2.2.0 webencodings-0.5.1 wheel-0.40.0 wrapt-1.17.2 xmljson-0.2.1 zipp-3.21.0

[notice] A new release of pip is available: 23.0.1 -> 24.3.1
[notice] To update, run: pip install --upgrade pip
(my) swayam@swayam:~/libraries/Python-3.10.16$ label-studio_

```

```

_setup_env()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/label_studio/server.py", line 35, in _setup_env
get_wsgi_application()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/core/wsgi.py", line 12, in get_wsgi_application
django.setup(set_prefix=False)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/_init_.py", line 24, in setup
apps.populate(settings.INSTALLED_APPS)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/apps/registry.py", line 116, in populate
app_config.import_models()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/apps/config.py", line 269, in import_models
self.models_module = import_module(models_module_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
return _bootstrap._gcd_import(name[level:], package, level)
File "<frozen importlib._bootstrap>", line 1050, in _gcd_import
File "<frozen importlib._bootstrap>", line 1027, in _find_and_load
File "<frozen importlib._bootstrap>", line 1006, in _find_and_load_unlocked
File "<frozen importlib._bootstrap>", line 688, in _load_unlocked
File "<frozen importlib._bootstrap_external>", line 883, in exec_module
File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
from django.contrib.auth.base_user import AbstractBaseUser, BaseUserManager
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/base_user.py", line 57, in <module>
class AbstractBaseUser(models.Model):
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 143, in __new__
new_class.add_to_class("_meta", Options(meta, app_label))
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in __getattr__
return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in __getitem__
conn = self.create_connection(alias)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 193, in create_connection
backend = load_backend(db["ENGINE"])
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 113, in load_backend
return import_module("%s.base" % backend_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
return _bootstrap._gcd_import(name[level:], package, level)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/backends/sqlite3/base.py", line 9, in <module>
from sqlite3 import dbapi2 as Database
File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$

```

Again a new error...

SQLite is required to run Label Studio because it acts as the default database to store project data, user information, and labeling task annotations. Let's install sqlite then.

#### 74. deactivate: Deactivates the virtual environment.

```

_setup_env()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/label_studio/server.py", line 35, in _setup_env
get_wsgi_application()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/core/wsgi.py", line 12, in get_wsgi_application
django.setup(set_prefix=False)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/_init_.py", line 24, in setup
apps.populate(settings.INSTALLED_APPS)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/apps/registry.py", line 116, in populate
app_config.import_models()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/apps/config.py", line 269, in import_models
self.models_module = import_module(models_module_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
return _bootstrap._gcd_import(name[level:], package, level)
File "<frozen importlib._bootstrap>", line 1050, in _gcd_import
File "<frozen importlib._bootstrap>", line 1027, in _find_and_load
File "<frozen importlib._bootstrap>", line 1006, in _find_and_load_unlocked
File "<frozen importlib._bootstrap>", line 688, in _load_unlocked
File "<frozen importlib._bootstrap_external>", line 883, in exec_module
File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
from django.contrib.auth.base_user import AbstractBaseUser, BaseUserManager
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/base_user.py", line 57, in <module>
class AbstractBaseUser(models.Model):
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 143, in __new__
new_class.add_to_class("_meta", Options(meta, app_label))
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in __getattr__
return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in __getitem__
conn = self.create_connection(alias)
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backend = load_backend(db["ENGINE"])
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File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
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File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/backends/sqlite3/base.py", line 9, in <module>
from sqlite3 import dbapi2 as Database
File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate_

```



```

File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/label_studio/server.py", line 35, in _setup_env
 get_wsgi_application()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/core/wsgi.py", line 12, in get_wsgi_application
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File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
 value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
 self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in _getattr__
 return getattr(self._connections[self._alias], item)
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 from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$

```

75. `cd ..`: Changes the current working directory to the parent directory.

76. `wget https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz`: Downloads the SQLite source code tarball.

```

 get_wsgi_application()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/core/wsgi.py", line 12, in get_wsgi_application
 django.setup(set_prefix=False)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/_init_.py", line 24, in setup
 apps.populate(settings.INSTALLED_APPS)
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File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/apps/config.py", line 269, in import_models
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File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
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File "<frozen importlib._bootstrap>", line 1027, in _find_and_load
File "<frozen importlib._bootstrap>", line 1006, in _find_and_load_unlocked
File "<frozen importlib._bootstrap>", line 688, in _load_unlocked
File "<frozen importlib._bootstrap_external>", line 883, in exec_module
File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
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File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
 value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
 self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in _getattr__
 return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in _getitem__
 conn = self.create_connection(alias)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 193, in create_connection
 backend = load_backend(db['ENGINE'])
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 113, in load_backend
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File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
 from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz

```

77. `tar xzf sqlite-autoconf-3420000.tar.gz`: Extracts the contents of the SQLite source code tarball.



```

File "<frozen importlib._bootstrap>" line 1027, in _find_and_load
File "<frozen importlib._bootstrap>" line 1006, in _find_and_load_unlocked
File "<frozen importlib._bootstrap>" line 688, in _load_unlocked
File "<frozen importlib._bootstrap_external>" line 883, in exec_module
File "<frozen importlib._bootstrap>" line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
 from django.contrib.auth.base_user import AbstractBaseUser, BaseUserManager
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/base_user.py", line 57, in <module>
 class AbstractBaseUser(models.Model):
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 143, in __new__
 new_class.add_to_class("_meta", Options(meta, app_label))
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
 value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
 self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in __getattr__
 return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in __getitem__
 conn = self.create_connection(alias)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 193, in create_connection
 backend = load_backend(db["ENGINE"])
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 113, in load_backend
 return import_module("%s.base" % backend_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
 return _bootstrap._gcd_import(name[level:], package, level)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/backends/sqlite3/base.py", line 9, in <module>
 from sqlite3 import dbapi2 as Database
File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
 from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
--2025-01-25 18:40:07-- https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
Resolving www.sqlite.org (www.sqlite.org)... 45.33.6.223, 2600:3c00::f03c:91fff:fe96:b959
Connecting to www.sqlite.org (www.sqlite.org)[45.33.6.223]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3148813 (3.0M) [application/x-gzip]
Saving to: 'sqlite-autoconf-3420000.tar.gz'

sqlite-autoconf-3420000.tar.gz 100%[=====] 3.00M 692KB/s in 4.4s

2025-01-25 18:40:13 (692 KB/s) - 'sqlite-autoconf-3420000.tar.gz' saved [3148813/3148813]

swayam@swayam:~/libraries$ tar xzf sqlite-autoconf-3420000.tar.gz

File "<frozen importlib._bootstrap>" line 1006, in _find_and_load_unlocked
File "<frozen importlib._bootstrap>" line 688, in _load_unlocked
File "<frozen importlib._bootstrap_external>" line 883, in exec_module
File "<frozen importlib._bootstrap>" line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
 from django.contrib.auth.base_user import AbstractBaseUser, BaseUserManager
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/base_user.py", line 57, in <module>
 class AbstractBaseUser(models.Model):
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 143, in __new__
 new_class.add_to_class("_meta", Options(meta, app_label))
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
 value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
 self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in __getattr__
 return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in __getitem__
 conn = self.create_connection(alias)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 193, in create_connection
 backend = load_backend(db["ENGINE"])
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 113, in load_backend
 return import_module("%s.base" % backend_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
 return _bootstrap._gcd_import(name[level:], package, level)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/backends/sqlite3/base.py", line 9, in <module>
 from sqlite3 import dbapi2 as Database
File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
 from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
--2025-01-25 18:40:07-- https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
Resolving www.sqlite.org (www.sqlite.org)... 45.33.6.223, 2600:3c00::f03c:91fff:fe96:b959
Connecting to www.sqlite.org (www.sqlite.org)[45.33.6.223]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3148813 (3.0M) [application/x-gzip]
Saving to: 'sqlite-autoconf-3420000.tar.gz'

sqlite-autoconf-3420000.tar.gz 100%[=====] 3.00M 692KB/s in 4.4s

2025-01-25 18:40:13 (692 KB/s) - 'sqlite-autoconf-3420000.tar.gz' saved [3148813/3148813]

swayam@swayam:~/libraries$ tar xzf sqlite-autoconf-3420000.tar.gz
swayam@swayam:~/libraries$ _

```

78. `cd sqlite-autoconf-3420000`: Changes the current working directory to the extracted SQLite source code directory.

```

File "<frozen importlib._bootstrap>", line 688, in _load_unlocked
File "<frozen importlib._bootstrap_external>", line 883, in exec_module
File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
 from django.contrib.auth.base_user import AbstractBaseUser, BaseUserManager
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/base_user.py", line 57, in <module>
 class AbstractBaseUser(models.Model):
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 143, in _new__
 new_class.add_to_class("_meta", Options(meta, app_label))
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
 value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
 self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in __getattr__
 return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in __getitem__
 conn = self.create_connection(alias)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 193, in create_connection
 backend = load_backend(db["ENGINE"])
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 113, in load_backend
 return import_module("%s.base" % backend_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
 return _bootstrap._gcd_import(name[level:], package, level)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/backends/sqlite3/base.py", line 9, in <module>
 from sqlite3 import dbapi2 as Database
File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
 from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
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Resolving www.sqlite.org (www.sqlite.org)... 45.33.6.223, 2600:3c00::f03c:91ff:fe96:b959
Connecting to www.sqlite.org (www.sqlite.org)|45.33.6.223|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3148813 (3.0M) [application/x-gzip]
Saving to: 'sqlite-autoconf-3420000.tar.gz'

sqlite-autoconf-3420000.tar.gz 100%[=====] 3.00M 692KB/s in 4.4s

2025-01-25 18:40:13 (692 KB/s) - 'sqlite-autoconf-3420000.tar.gz' saved [3148813/3148813]

swayam@swayam:~/libraries$ tar xzf sqlite-autoconf-3420000.tar.gz
swayam@swayam:~/libraries$ cd sqlite-autoconf-3420000
swayam@swayam:~/libraries/sqlite-autoconf-3420000$

```

## 79. ./configure --prefix=\$HOME/local: Configures the SQLite source code for compilation, specifying the installation prefix.

```

File "<frozen importlib._bootstrap_external>", line 883, in exec_module
File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/models.py", line 3, in <module>
 from django.contrib.auth.base_user import AbstractBaseUser, BaseUserManager
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/contrib/auth/base_user.py", line 57, in <module>
 class AbstractBaseUser(models.Model):
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 143, in _new__
 new_class.add_to_class("_meta", Options(meta, app_label))
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/base.py", line 371, in add_to_class
 value.contribute_to_class(cls, name)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/models/options.py", line 243, in contribute_to_class
 self.db_table, connection.ops.max_name_length()
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 15, in __getattr__
 return getattr(self._connections[self._alias], item)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/utils/connection.py", line 62, in __getitem__
 conn = self.create_connection(alias)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 193, in create_connection
 backend = load_backend(db["ENGINE"])
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/utils.py", line 113, in load_backend
 return import_module("%s.base" % backend_name)
File "/home/swayam/local/lib/python3.10/importlib/_init_.py", line 126, in import_module
 return _bootstrap._gcd_import(name[level:], package, level)
File "/home/swayam/libraries/Python-3.10.16/my/lib/python3.10/site-packages/django/db/backends/sqlite3/base.py", line 9, in <module>
 from sqlite3 import dbapi2 as Database
File "/home/swayam/local/lib/python3.10/sqlite3/_init_.py", line 57, in <module>
 from sqlite3.dbapi2 import *
File "/home/swayam/local/lib/python3.10/sqlite3/dbapi2.py", line 27, in <module>
 from _sqlite3 import *
ModuleNotFoundError: No module named '_sqlite3'
Sentry is attempting to send 2 pending events
Waiting up to 2 seconds
Press Ctrl-C to quit
(my) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ wget https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
--2025-01-25 18:40:07-- https://www.sqlite.org/2023/sqlite-autoconf-3420000.tar.gz
Resolving www.sqlite.org (www.sqlite.org)... 45.33.6.223, 2600:3c00::f03c:91ff:fe96:b959
Connecting to www.sqlite.org (www.sqlite.org)|45.33.6.223|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3148813 (3.0M) [application/x-gzip]
Saving to: 'sqlite-autoconf-3420000.tar.gz'

sqlite-autoconf-3420000.tar.gz 100%[=====] 3.00M 692KB/s in 4.4s

2025-01-25 18:40:13 (692 KB/s) - 'sqlite-autoconf-3420000.tar.gz' saved [3148813/3148813]

swayam@swayam:~/libraries$ tar xzf sqlite-autoconf-3420000.tar.gz
swayam@swayam:~/libraries$ cd sqlite-autoconf-3420000
swayam@swayam:~/libraries/sqlite-autoconf-3420000$./configure --prefix=$HOME/local

```

```

checking forunistd.h... yes
checking fordlfcn.h... yes
checking forobjdir... libs
checking if gcc supports -fno-rtti -fno-exceptions... no
checking for gcc option to produce PIC... -fPIC -DPIC
checking if gcc PIC flag -fPIC -DPIC works... yes
checking if gcc static flag -static works... yes
checking if gcc supports -c -o file.o... yes
checking if gcc supports -c -o file.o... (cached) yes
checking whether the gcc linker (/usr/bin/ld) supports shared libraries... yes
checking whether -lc should be explicitly linked in... no
checking dynamic linker characteristics... GNU/Linux ld.so
checking how to hardcode library paths into programs... immediate
checking whether stripping libraries is possible... yes
checking if libtool supports shared libraries... yes
checking whether to build shared libraries... yes
checking whether to build static libraries... yes
checking for fdatasync... yes
checking for usleep... yes
checking for fullsync... no
checking for localtime_r... yes
checking for gmtime_r... yes
checking for gcc options needed to detect all undeclared functions... none needed
checking whether strerror_r is declared... yes
checking whether strerror_r returns char *... no
checking for editline/readline.h... no
checking for readline/readline.h... no
checking for library containing pthread_create... none required
checking for library containing pthread_mutexattr_t... none required
checking for library containing dlopen... none required
checking for whether to support dynamic extensions... yes
checking SQL math functions... enabled
checking for library containing ceil... -lm
checking FTS4 extension... enabled
checking FTS3 extension... disabled
checking FTS5 extension... enabled
checking for library containing log... none required
checking RTREE extension... enabled
checking Session extension... disabled
checking Build type... release
checking for posix_fallocate... yes
checking for zlib.h... no
checking for library containing system... none required
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating sqlite3.pc
config.status: executing depfiles commands
config.status: executing libtool commands
swayam@swayam:~/libraries/sqlite-autoconf-3420000$

```

## 80. make -j\$(nproc): Compiles the SQLite source code using multiple CPU cores.

```

checking forunistd.h... yes
checking fordlfcn.h... yes
checking forobjdir... libs
checking if gcc supports -fno-rtti -fno-exceptions... no
checking for gcc option to produce PIC... -fPIC -DPIC
checking if gcc PIC flag -fPIC -DPIC works... yes
checking if gcc static flag -static works... yes
checking if gcc supports -c -o file.o... yes
checking if gcc supports -c -o file.o... (cached) yes
checking whether the gcc linker (/usr/bin/ld) supports shared libraries... yes
checking whether -lc should be explicitly linked in... no
checking dynamic linker characteristics... GNU/Linux ld.so
checking how to hardcode library paths into programs... immediate
checking whether stripping libraries is possible... yes
checking if libtool supports shared libraries... yes
checking whether to build shared libraries... yes
checking whether to build static libraries... yes
checking for fdatasync... yes
checking for usleep... yes
checking for fullsync... no
checking for localtime_r... yes
checking for gmtime_r... yes
checking for gcc options needed to detect all undeclared functions... none needed
checking whether strerror_r is declared... yes
checking whether strerror_r returns char *... no
checking for editline/readline.h... no
checking for readline/readline.h... no
checking for library containing pthread_create... none required
checking for library containing pthread_mutexattr_t... none required
checking for library containing dlopen... none required
checking for whether to support dynamic extensions... yes
checking SQL math functions... enabled
checking for library containing ceil... -lm
checking FTS4 extension... enabled
checking FTS3 extension... disabled
checking FTS5 extension... enabled
checking for library containing log... none required
checking RTREE extension... enabled
checking Session extension... disabled
checking Build type... release
checking for posix_fallocate... yes
checking for zlib.h... no
checking for library containing system... none required
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating sqlite3.pc
config.status: executing depfiles commands
config.status: executing libtool commands
swayam@swayam:~/libraries/sqlite-autoconf-3420000$ make -j$(nproc)

```

81. `make install`: Installs the compiled SQLite software.

```
gcc -DPACKAGE_NAME='sqlite' -DPACKAGE_TARNAME='sqlite' -DPACKAGE_VERSION='3.42.0' -DPACKAGE_STRING='sqlite 3.42.0' -DPACKAGE_BUGREPORT='http://www.sqlite.org' -DPACKAGE_URL='' -DPACKAGE='sqlite' -OVERSON='3.42.0' -DHAVE_STDLIB_H=1 -DHAVE_STDLIB_H=1 -DHAVE_STRING_H=1 -DHAVE_INTTYPES_H=1 -DHAVE_STDINT_H=1 -DHAVE_STRINGS_H=1 -DHAVE_SYS_STAT_H=1 -DHAVE_SYS_TYPES_H=1 -DHAVE_UNISTD_H=1 -DOSTDC_HEADERS=1 -DHAVE_DLFCN_H=1 -DLT_OBJDIR='' -libs/ -DHAVE_FDATASYNC=1 -DHAVE_USLEEP=1 -DHAVE_LOCALTIME_R=1 -DHAVE_GMTIME_R=1 -DHAVE_DECL_STREERRR_R=1 -DHAVE_STRERROR_R=1 -DHAVE_POSIX_FALLOCCATE=1 -I. -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -DSQLITE_ENABLE_EXPLAIN_COMMENTS -D_SQLITE_ENABLE_DBPAGE_VTAB -DSQLITE_ENABLE_STMTVTAB -DSQLITE_ENABLE_DBSTAT_VTAB -g -O2 -MT sqlite3-shell.o -MD -MP -MF .deps/sqlite3-shell.Tpo -c -o sqlite3-shell.o test -f shell.c || echo './shell.c'
```

```
gcc -DPACKAGE_NAME='sqlite' -DPACKAGE_TARNAME='sqlite' -DPACKAGE_VERSION='3.42.0' -DPACKAGE_STRING='sqlite 3.42.0' -DPACKAGE_BUGREPORT='http://www.sqlite.org' -DPACKAGE_URL='' -DPACKAGE='sqlite' -OVERSON='3.42.0' -DHAVE_STDLIB_H=1 -DHAVE_STDLIB_H=1 -DHAVE_STRING_H=1 -DHAVE_INTTYPES_H=1 -DHAVE_STDINT_H=1 -DHAVE_STRINGS_H=1 -DHAVE_SYS_STAT_H=1 -DHAVE_SYS_TYPES_H=1 -DHAVE_UNISTD_H=1 -DOSTDC_HEADERS=1 -DHAVE_DLFCN_H=1 -DLT_OBJDIR='' -libs/ -DHAVE_FDATASYNC=1 -DHAVE_USLEEP=1 -DHAVE_LOCALTIME_R=1 -DHAVE_GMTIME_R=1 -DHAVE_DECL_STREERRR_R=1 -DHAVE_STRERROR_R=1 -DHAVE_POSIX_FALLOCCATE=1 -I. -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -DSQLITE_ENABLE_EXPLAIN_COMMENTS -D_SQLITE_ENABLE_DBPAGE_VTAB -DSQLITE_ENABLE_STMTVTAB -DSQLITE_ENABLE_DBSTAT_VTAB -g -O2 -MT sqlite3-sqlite3.o -MD -MP -MF .deps/sqlite3-sqlite3.Tpo -c -o sqlite3-sqlite3.o test -f sqlite3.c || echo './sqlite3.c'
```

```
/bin/bash ./libtool --tag=CC --mode=compile gcc -DPACKAGE_NAME='sqlite' -DPACKAGE_TARNAME='sqlite' -DPACKAGE_VERSION='3.42.0' -DPACKAGE_STRING='sqlite 3.42.0' -DPACKAGE_BUGREPORT='http://www.sqlite.org' -DPACKAGE_URL='' -DPACKAGE='sqlite' -OVERSON='3.42.0' -DHAVE_STDLIB_H=1 -DHAVE_STDLIB_H=1 -DHAVE_STRING_H=1 -DHAVE_INTTYPES_H=1 -DHAVE_STDINT_H=1 -DHAVE_STRINGS_H=1 -DHAVE_SYS_STAT_H=1 -DHAVE_SYS_TYPES_H=1 -DHAVE_UNISTD_H=1 -DOSTDC_HEADERS=1 -DHAVE_DLFCN_H=1 -DLT_OBJDIR='' -libs/ -DHAVE_FDATASYNC=1 -DHAVE_USLEEP=1 -DHAVE_LOCALTIME_R=1 -DHAVE_GMTIME_R=1 -DHAVE_DECL_STREERRR_R=1 -DHAVE_STRERROR_R=1 -DHAVE_POSIX_FALLOCCATE=1 -I. -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -g -O2 -MT sqlite3.lo -MD -MP -MF .deps/sqlite3.Tpo -c -o sqlite3.c -fpic -DPIC -o .libs/sqlite3.o
```

```
mv -f .deps/sqlite3-shell.Tpo .deps/sqlite3-shell.Po
```

```
libtool: compile: gcc -DPACKAGE_NAME='sqlite' -DPACKAGE_TARNAME='sqlite' -DPACKAGE_VERSION='3.42.0' -DPACKAGE_STRING='sqlite 3.42.0' -DPACKAGE_BUGREPORT='http://www.sqlite.org' -DPACKAGE_URL='' -DPACKAGE='sqlite' -OVERSON='3.42.0' -DHAVE_STDLIB_H=1 -DHAVE_STDLIB_H=1 -DHAVE_STRING_H=1 -DHAVE_INTTYPES_H=1 -DHAVE_STDINT_H=1 -DHAVE_STRINGS_H=1 -DHAVE_SYS_STAT_H=1 -DHAVE_SYS_TYPES_H=1 -DHAVE_UNISTD_H=1 -DOSTDC_HEADERS=1 -DHAVE_DLFCN_H=1 -DLT_OBJDIR='' -libs/ -DHAVE_FDATASYNC=1 -DHAVE_USLEEP=1 -DHAVE_LOCALTIME_R=1 -DHAVE_GMTIME_R=1 -DHAVE_DECL_STREERRR_R=1 -DHAVE_STRERROR_R=1 -DHAVE_POSIX_FALLOCCATE=1 -I. -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -g -O2 -MT sqlite3.lo -MD -MP -MF .deps/sqlite3.Tpo -c -o sqlite3.c -dev/null 2>&1
```

```
mv -f .deps/sqlite3-sqlite3.Tpo .deps/sqlite3-sqlite3.Po
```

```
/bin/bash ./libtool --tag=CC --mode=link gcc -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -DSQLITE_ENABLE_EXPLAIN_COMMENTS -DSQLITE_ENABLE_DBPAGE_VTAB -DSQLITE_ENABLE_STMTVTAB -DSQLITE_ENABLE_DBSTAT_VTAB -g -O2 -o sqlite3-sqlite3-shell.o -lmsvcrt -L. -lsqlite3 -lstdc++ -lm
```

```
libtool: link: gcc -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -DSQLITE_ENABLE_EXPLAIN_COMMENTS -DSQLITE_ENABLE_DBPAGE_VTAB -DSQLITE_ENABLE_STMTVTAB -DSQLITE_ENABLE_DBSTAT_VTAB -g -O2 -o sqlite3-sqlite3-shell.o -lmsvcrt -L. -lsqlite3 -lstdc++ -lm
```

```
mv -f .deps/sqlite3.Tpo .deps/sqlite3.Plo
```

```
/bin/bash ./libtool --tag=CC --mode=link gcc -O_REENTRANT=1 -DSQLITE_THREADSAFE=1 -DSQLITE_ENABLE_MATH_FUNCTIONS -DSQLITE_ENABLE_FT54 -DSQLITE_ENABLE_FT55 -DSQLITE_ENABLE_RTREE -DSQLITE_ENABLE_GEOPOLY -g -O2 -no-undefined -version_info 6:6:8 -o libsqlite3.la -rpath /home/swagun/local/lib sqlite3.lo -lm
```

```
libtool: link: gcc -shared -fpic -fPIE -pie -Wl,-soname,libsqlite3.so.0 -Wl,-as-needed -o libsqlite3.so.0.8.6 -L/home/swagun/local/lib -lsqlite3.lo -lm
```

```
libtool: link: (cd ".libs" && rm -f "libsqlite3.so.0" && ln -s "libsqlite3.so.0.8.6" "libsqlite3.so.0")
```

```
libtool: link: (cd ".libs" && rm -f "libsqlite3.a" && ln -s "libsqlite3.so.0" "libsqlite3.a")
```

```
libtool: link: ar cr .libs/libsqlite3.a sqlite3.o
```

```
libtool: link: ranlib .libs/libsqlite3.a
```

```
libtool: link: (cd ".libs" && rm -f "libsqlite3.la" && ln -s "../libsqlite3.la" "libsqlite3.la")
```

```
swagun@swagun:~/libraries/sqlite-autocgen-3420009$ make install.
```

```

libtool: link: gcc -shared -fPIC -DPIC .libs/sqlite3.o -lm -g -O2 -Wl,-soname -Wl,libsqlite3.so.0 -o .libs/libsqlite3.so.0.8.6
libtool: link: (cd ".libs" && rm -f "libsqlite3.so.0" && ln -s "libsqlite3.so.0.8.6" "libsqlite3.so.0")
libtool: link: (cd ".libs" && rm -f "libsqlite3.so" && ln -s "libsqlite3.so.0.8.6" "libsqlite3.so")
libtool: link: ar cr .libs/libsqlite3.a sqlite3.o
libtool: link: ranlib .libs/libsqlite3.a
libtool: link: (cd ".libs" && rm -f "libsqlite3.la" && ln -s "../libsqlite3.la" "libsqlite3.la")
swayam@swayam:~/libraries/sqlite-autoconf-3420000$ make install
make[1]: Entering directory '/home/swayam/libraries/sqlite-autoconf-3420000'
/usr/bin/mkdir -p '/home/swayam/local/lib'
/bin/bash ./libtool --mode=install /usr/bin/install -c libsqlite3.la '/home/swayam/local/lib'
libtool: install: /usr/bin/install -c .libs/libsqlite3.so.0.8.6 /home/swayam/local/lib/libsqlite3.so.0.8.6
libtool: install: (cd /home/swayam/local/lib && { ln -s -f libsqlite3.so.0.8.6 libsqlite3.so.0 || { rm -f libsqlite3.so.0 && ln -s libsqlite3.so.0.8.6 libsqlite3.so.0; }; })
libtool: install: (cd /home/swayam/local/lib && { ln -s -f libsqlite3.so.0.8.6 libsqlite3.so || { rm -f libsqlite3.so && ln -s libsqlite3.so.0.8.6 libsqlite3.so; }; })
libtool: install: /usr/bin/install -c .libs/libsqlite3.lai /home/swayam/local/lib/libsqlite3.la
libtool: install: /usr/bin/install -c .libs/libsqlite3.a /home/swayam/local/lib/libsqlite3.a
libtool: install: chmod 644 /home/swayam/local/lib/libsqlite3.a
libtool: install: ranlib /home/swayam/local/lib/libsqlite3.a
libtool: finish: PATH="/home/swayam/openssl/bin:/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin" ldconfig -n /home/swayam/local/lib

Libraries have been installed in:
/home/swayam/local/lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
 during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
 during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p '/home/swayam/local/bin'
/bin/bash ./libtool --mode=install /usr/bin/install -c sqlite3 '/home/swayam/local/bin'
libtool: install: /usr/bin/install -c sqlite3 /home/swayam/local/bin/sqlite3
/usr/bin/mkdir -p '/home/swayam/local/include'
/usr/bin/install -c -m 644 sqlite3.h sqlite3ext.h '/home/swayam/local/include'
/usr/bin/mkdir -p '/home/swayam/local/share/man/man1'
/usr/bin/install -c -m 644 sqlite3.1 '/home/swayam/local/share/man/man1'
/usr/bin/mkdir -p '/home/swayam/local/lib/pkgconfig'
/usr/bin/install -c -m 644 sqlite3.pc '/home/swayam/local/lib/pkgconfig'
make[1]: Leaving directory '/home/swayam/libraries/sqlite-autoconf-3420000'
swayam@swayam:~/libraries/sqlite-autoconf-3420000$

```

82. `cd ..`: Changes the current working directory to the parent directory.

83. `cd Python-3.10.16`: Changes the current working directory to the Python source code directory.

```

libtool: link: (cd ".libs" && rm -f "libsqlite3.so" && ln -s "libsqlite3.so.0.8.6" "libsqlite3.so")
libtool: link: ar cr .libs/libsqlite3.a sqlite3.o
libtool: link: ranlib .libs/libsqlite3.a
libtool: link: (cd ".libs" && rm -f "libsqlite3.la" && ln -s "../libsqlite3.la" "libsqlite3.la")
swayam@swayam:~/libraries/sqlite-autoconf-3420000$ make install
make[1]: Entering directory '/home/swayam/libraries/sqlite-autoconf-3420000'
/usr/bin/mkdir -p '/home/swayam/local/lib'
/bin/bash ./libtool --mode=install /usr/bin/install -c libsqlite3.la '/home/swayam/local/lib'
libtool: install: /usr/bin/install -c .libs/libsqlite3.so.0.8.6 /home/swayam/local/lib/libsqlite3.so.0.8.6
libtool: install: (cd /home/swayam/local/lib && { ln -s -f libsqlite3.so.0.8.6 libsqlite3.so.0 || { rm -f libsqlite3.so.0 && ln -s libsqlite3.so.0.8.6 libsqlite3.so.0; }; })
libtool: install: (cd /home/swayam/local/lib && { ln -s -f libsqlite3.so.0.8.6 libsqlite3.so || { rm -f libsqlite3.so && ln -s libsqlite3.so.0.8.6 libsqlite3.so; }; })
libtool: install: /usr/bin/install -c .libs/libsqlite3.lai /home/swayam/local/lib/libsqlite3.la
libtool: install: /usr/bin/install -c .libs/libsqlite3.a /home/swayam/local/lib/libsqlite3.a
libtool: install: chmod 644 /home/swayam/local/lib/libsqlite3.a
libtool: install: ranlib /home/swayam/local/lib/libsqlite3.a
libtool: finish: PATH="/home/swayam/openssl/bin:/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin" ldconfig -n /home/swayam/local/lib

Libraries have been installed in:
/home/swayam/local/lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
 during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
 during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p '/home/swayam/local/bin'
/bin/bash ./libtool --mode=install /usr/bin/install -c sqlite3 '/home/swayam/local/bin'
libtool: install: /usr/bin/install -c sqlite3 /home/swayam/local/bin/sqlite3
/usr/bin/mkdir -p '/home/swayam/local/include'
/usr/bin/install -c -m 644 sqlite3.h sqlite3ext.h '/home/swayam/local/include'
/usr/bin/mkdir -p '/home/swayam/local/share/man/man1'
/usr/bin/install -c -m 644 sqlite3.1 '/home/swayam/local/share/man/man1'
/usr/bin/mkdir -p '/home/swayam/local/lib/pkgconfig'
/usr/bin/install -c -m 644 sqlite3.pc '/home/swayam/local/lib/pkgconfig'
make[1]: Leaving directory '/home/swayam/libraries/sqlite-autoconf-3420000'
swayam@swayam:~/libraries/sqlite-autoconf-3420000$ cd ..
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$

```

84. `./configure --prefix=$HOME/local ...`: Configures the Python source code for compilation, specifying various options and prefixes.

```

libtool: link: ar cr .libs/libsqlite3.a sqlite3.o
libtool: link: ranlib .libs/libsqlite3.a
libtool: link: (cd ".libs" && rm -f "libsqlite3.la" && ln -s "../libsqlite3.la" "libsqlite3.la")
swayam@swayam:~/libraries/sqlite-autoconf-3420000$ make install
make[1]: Entering directory '/home/swayam/libraries/sqlite-autoconf-3420000'
/usr/bin/mkdir -p '/home/swayam/local/lib'
/bin/bash ./libtool --mode=install /usr/bin/install -c .libs/libsqlite3.la '/home/swayam/local/lib'
libtool: install: /usr/bin/install -c .libs/libsqlite3.so.0.8.6 /home/swayam/local/lib/libsqlite3.so.0.8.6
libtool: install: (cd /home/swayam/local/lib && { ln -s -f libsqlite3.so.0.8.6 libsqlite3.so.0 || { rm -f libsqlite3.so.0 && ln -s libsqlite3.so.0.8.6 libsqlite3.so.0; }; })
libtool: install: (cd /home/swayam/local/lib && { ln -s -f libsqlite3.so.0.8.6 libsqlite3.so || { rm -f libsqlite3.so && ln -s libsqlite3.so.0.8.6 libsqlite3.so; }; })
libtool: install: /usr/bin/install -c .libs/libsqlite3.lai /home/swayam/local/lib/libsqlite3.la
libtool: install: chmod 644 /home/swayam/local/lib/libsqlite3.a
libtool: install: ranlib /home/swayam/local/lib/libsqlite3.a
libtool: finish: PATH="/home/swayam/openssl/bin:/home/swayam/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/sbin" ldconfig -n /home/swayam/local/lib

Libraries have been installed in:
 /home/swayam/local/lib

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the '-LLIBDIR'
flag during linking and do at least one of the following:
- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable
 during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable
 during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.

/usr/bin/mkdir -p '/home/swayam/local/bin'
/bin/bash ./libtool --mode=install /usr/bin/install -c sqlite3 '/home/swayam/local/bin'
libtool: install: /usr/bin/install -c sqlite3 /home/swayam/local/bin/sqlite3
/usr/bin/mkdir -p '/home/swayam/local/include'
/usr/bin/install -c -m 644 sqlite3.h sqlite3ext.h '/home/swayam/local/include'
/usr/bin/mkdir -p '/home/swayam/local/share/man/man1'
/usr/bin/install -c -m 644 sqlite3.1 '/home/swayam/local/share/man/man1'
/usr/bin/mkdir -p '/home/swayam/local/lib/pkgconfig'
/usr/bin/install -c -m 644 sqlite3.pc '/home/swayam/local/lib/pkgconfig'
make[1]: Leaving directory '/home/swayam/libraries/sqlite-autoconf-3420000'
swayam@swayam:~/libraries/sqlite-autoconf-3420000$ cd ..
swayam@swayam:~/libraries$ cd Python-3.10.16
swayam@swayam:~/libraries/Python-3.10.16$./configure --prefix=$HOME/local --with-openssl=$HOME/local --with-ensurepip=install \
LD_FLAGS="-L$HOME/local/lib" \
CPPFLAGS="-I$HOME/local/include" \
PKG_CONFIG_PATH="$HOME/local/lib/pkgconfig" --enable-optimization

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc FORTIFY_SOURCE/memmove bug... no
checking for gcc ioa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ _

```

## 85. make: Compiles the Python source code.



```

checking for use_env... no
configure: checking for device files
checking for /dev/ptmx... yes
checking for /dev/ptc... no
checking for %zd printf() format support... yes
checking for socklen_t... yes
checking for broken mbstowcs... no
checking for --with-computed-gotos... no value specified
checking whether gcc supports computed gotos... yes
checking for build directories... done
checking for -O2... yes
checking for glibc _FORTIFY_SOURCE/memmove bug... no
checking for gcc ipa-pure-const bug... no
checking for stdatomic.h... yes
checking for builtin __atomic_load_n and __atomic_store_n functions... yes
checking for ensurepip... install
checking if the dirent structure of a d_type field... yes
checking for the Linux getrandom() syscall... yes
checking for the getrandom() function... yes
checking for library containing shm_open... none required
checking for sys/mman.h... (cached) yes
checking for shm_open... yes
checking for shm_unlink... yes
checking for openssl/ssl.h in /home/swayam/local... yes
checking whether compiling and linking against OpenSSL works... yes
checking for --with-openssl-rpath...
checking whether OpenSSL provides required APIs... yes
checking for --with-ssl-default-suites... python
checking for --with-builtin-hashlib-hashes... md5,sha1,sha256,sha512,sha3,blake2
checking for --with-experimental-isolated-subinterpreters... no
checking for --with-static-libpython... yes
checking for --disable-test-modules... no
configure: creating ./config.status
config.status: creating Makefile.pre
config.status: creating Misc/python.pc
config.status: creating Misc/python-embed.pc
config.status: creating Misc/python-config.sh
config.status: creating Modules/ld_so_aix
config.status: creating pyconfig.h
config.status: pyconfig.h is unchanged
configure: WARNING: unrecognized options: --enable-optimization
creating Modules/Setup.local
creating Makefile

If you want a release build with all stable optimizations active (PGO, etc),
please run ./configure --enable-optimizations

swayam@swayam:~/libraries/Python-3.10.16$ make

am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/prepare_protocol.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/p
prepare_protocol.o
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -DSQLITE_OMIT_LOAD_EXTENSION=1 -IModules/_sqlite -I./Include -I. -I/home/sway
am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/row.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/row.o
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -DSQLITE_OMIT_LOAD_EXTENSION=1 -IModules/_sqlite -I./Include -I. -I/home/sway
am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/statement.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/statemen
t.o
gcc -fPIC -Wno-unused-result -Wsign-compare -DNDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror=implicit-function-declaration -fvisibility=hidden -I./Include/internal -DSQLITE_OMIT_LOAD_EXTENSION=1 -IModules/_sqlite -I./Include -I. -I/home/sway
am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/util.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/util.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/cache.o build/temp.linux-x86_64-3.10/home
/swayam/libraries/Python-3.10.16/Modules/_sqlite/connection.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/cursor.o build/t
emp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/microprotocols.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Mo
dules/_sqlite/module.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/prepare_protocol.o build/temp.linux-x86_64-3.10/home/sw
ayam/libraries/Python-3.10.16/Modules/_sqlite/row.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/statement.o build/temp.lin
ux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/util.o -L/home/swayam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lsqlite3 -o bu
ild/lib.linux-x86_64-3.10/_sqlite3.cpython-310-x86_64-linux-gnu.so

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm lzma
_tkinter _uuid nis
readline
To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

running build_scripts
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
gcc -L/home/swayam/local/lib -xlinker -export-dynamic -o Programs/_testembed Programs/_testembed.o libpython3.10.a -lcrypt -ldl -lm -lm
sed -e 's_@EXENAME@_home/swayam/local/bin/python3.10_' < Misc/python-config.in >python-config.py
LC_ALL=C sed -e 's_@((A-Za-z0-9_*)@)_@{1}_g' < Misc/python-config.sh >python-config
swayam@swayam:~/libraries/Python-3.10.16$ _

```

86. make install: Installs the compiled Python software.



```

am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/prepare_protocol.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/p
prepare_protocol.o
gcc -fPIC -Wno-unused-result -Wsign-compare -DDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror-implicit-function-declaration -fvisibility=hidden -I./Include/internal -DSQLite_OMIT_LOAD_EXTENSION=1 -IModules/_sqlite -I./Include -I. -I/home/sway
am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/row.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/row.o
gcc -fPIC -Wno-unused-result -Wsign-compare -DDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror-implicit-function-declaration -fvisibility=hidden -I./Include/internal -DSQLite_OMIT_LOAD_EXTENSION=1 -IModules/_sqlite -I./Include -I. -I/home/sway
am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/statement.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/statemen
t.o
gcc -fPIC -Wno-unused-result -Wsign-compare -DDEBUG -g -fwrapv -O3 -Wall -std=c99 -Wextra -Wno-unused-result -Wno-unused-parameter -Wno-missing-field-initializ
ers -Werror-implicit-function-declaration -fvisibility=hidden -I./Include/internal -DSQLite_OMIT_LOAD_EXTENSION=1 -IModules/_sqlite -I./Include -I. -I/home/sway
am/local/include -I/usr/include/x86_64-linux-gnu -I/usr/local/include -I/home/swayam/libraries/Python-3.10.16/Include -I/home/swayam/libraries/Python-3.10.16 -c
/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/util.c -o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/util.o
gcc -shared -L/home/swayam/local/lib build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/cache.o build/temp.linux-x86_64-3.10/home
/swayam/libraries/Python-3.10.16/Modules/_sqlite/connection.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/cursor.o build/t
emp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/module.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_s
qlite/microprotocols.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/prepare_protocol.o build/temp.linux-x86_64-3.10/home/sw
ayam/libraries/Python-3.10.16/Modules/_sqlite/row.o build/temp.linux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/statement.o build/temp.lin
ux-x86_64-3.10/home/swayam/libraries/Python-3.10.16/Modules/_sqlite/util.o -L/home/swayam/local/lib -L/usr/lib/x86_64-linux-gnu -L/usr/local/lib -lsqlite3 -o bu
ild/lib.linux-x86_64-3.10/_sqlite3.cpython-310-x86_64-linux-gnu.so

The necessary bits to build these optional modules were not found:
_bz2 _curses _curses_panel
_dbm _gdbm _lzma
_tkinter _uuid nis

To find the necessary bits, look in setup.py in detect_modules() for the module's name.

The following modules found by detect_modules() in setup.py, have been
built by the Makefile instead, as configured by the Setup files:
_abc pwd time

running build_scripts
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/pydoc3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/idle3 -> build/scripts-3.10
copying and adjusting /home/swayam/libraries/Python-3.10.16/Tools/scripts/2to3 -> build/scripts-3.10
changing mode of build/scripts-3.10/pydoc3 from 664 to 775
changing mode of build/scripts-3.10/idle3 from 664 to 775
changing mode of build/scripts-3.10/2to3 from 664 to 775
renaming build/scripts-3.10/pydoc3 to build/scripts-3.10/pydoc3.10
renaming build/scripts-3.10/idle3 to build/scripts-3.10/idle3.10
renaming build/scripts-3.10/2to3 to build/scripts-3.10/2to3.10
gcc -L/home/swayam/local/lib -Xlinker -export-dynamic -o Programs/_testembed libpython3.10.a -lcrypt -ldl -lm -lm
sed -e 's,0EXENAME0,/home/swayam/local/bin/python3.10,' < ./Misc/python-config.in >python-config.py
LC_ALL=C sed -e 's,\${([A-Za-z0-9_]*)},\${\1},g' < Misc/python-config.sh >python-config
swayam@swayam:~/libraries/Python-3.10.16$ make install

echo "Creating directory /home/swayam/local/lib/pkgconfig"; \
/usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \

fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
rm -f /home/swayam/local/bin/python3.10-config; \
(cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3.10 2to3)
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-32; \
(cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
rm -f /home/swayam/local/bin/python3-intel64; \
(cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
case install in \
upgrade) ensurepip="--upgrade" ;; \
install|*) ensurepip="" ;; \
esac; \
./python -E -m ensurepip \
$ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpx12up6w0
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$

```

87. `python3.10 -m venv my`: Creates a new virtual environment named "my" using the Python 3.10 installation.

```

/usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \
fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/; \
fi
Looking in links: /tmp/tmpx12up6wup
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my

```

```

/usr/bin/install -c -d -m 755 /home/swayam/local/lib/pkgconfig; \
fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/; \
fi
Looking in links: /tmp/tmpx12up6wup
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my

```

88. source my/bin/activate: Activates the virtual environment.

```

fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpx12up6wup
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my
swayam@swayam:~/libraries/Python-3.10.16$ source my/bin/activate_

```

```

fi
if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3-3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install|*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 $ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpx12up6wup
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my
swayam@swayam:~/libraries/Python-3.10.16$ source my/bin/activate
(my) swayam@swayam:~/libraries/Python-3.10.16$ _

```

89. pip install label-studio: Installs the label-studio package using pip.

```

if test -f /home/swayam/local/bin/python3 -o -h /home/swayam/local/bin/python3; \
then rm -f /home/swayam/local/bin/python3; \
else true; \
fi
(cd /home/swayam/local/bin; ln -s python3.10 python3)
if test "3.10" != "3.10"; then \
 rm -f /home/swayam/local/bin/python3.10-config; \
 (cd /home/swayam/local/bin; ln -s python3.10-config python3.10-config); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python-3.10.pc); \
 rm -f /home/swayam/local/lib/pkgconfig/python-3.10-embed.pc; \
 (cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python-3.10-embed.pc); \
fi
rm -f /home/swayam/local/bin/python3-config
(cd /home/swayam/local/bin; ln -s python3.10-config python3-config)
rm -f /home/swayam/local/lib/pkgconfig/python3.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10.pc python3.pc)
rm -f /home/swayam/local/lib/pkgconfig/python3-embed.pc
(cd /home/swayam/local/lib/pkgconfig; ln -s python-3.10-embed.pc python3-embed.pc)
rm -f /home/swayam/local/bin/idle3
(cd /home/swayam/local/bin; ln -s idle3.10 idle3)
rm -f /home/swayam/local/bin/pydoc3
(cd /home/swayam/local/bin; ln -s pydoc3.10 pydoc3)
rm -f /home/swayam/local/bin/2to3
(cd /home/swayam/local/bin; ln -s 2to3.3.10 2to3)
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-32; \
 (cd /home/swayam/local/bin; ln -s python3.10-32 python3-32) \
fi
if test "x" != "x"; then \
 rm -f /home/swayam/local/bin/python3-intel64; \
 (cd /home/swayam/local/bin; ln -s python3.10-intel64 python3-intel64) \
fi
rm -f /home/swayam/local/share/man/man1/python3.1
(cd /home/swayam/local/share/man/man1; ln -s python3.10.1 python3.1)
if test "xinstall" != "xno" ; then \
 case install in \
 upgrade) ensurepip="--upgrade" ;; \
 install*) ensurepip="" ;; \
 esac; \
 ./python -E -m ensurepip \
 --ensurepip --root=/ ; \
fi
Looking in links: /tmp/tmpxi2up6w
Requirement already satisfied: setuptools in /home/swayam/local/lib/python3.10/site-packages (65.5.0)
Requirement already satisfied: pip in /home/swayam/local/lib/python3.10/site-packages (23.0.1)
swayam@swayam:~/libraries/Python-3.10.16$ python3.10 -m venv my
swayam@swayam:~/libraries/Python-3.10.16$ source my/bin/activate
(my) swayam@swayam:~/libraries/Python-3.10.16$ pip install label-studio_

Requirement already satisfied: cachetools<6.0,>=2.0.0 in ./my/lib/python3.10/site-packages (from google-auth!=2.24.0,!=2.25.0,<3.0.0dev,>=2.14.1->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (5.5.1)
Requirement already satisfied: httpcore==1.* in ./my/lib/python3.10/site-packages (from httpx==0.21.2->label-studio-sdk==1.0.8->label-studio) (1.0.7)
Requirement already satisfied: h11<0.15,>=0.13 in ./my/lib/python3.10/site-packages (from httpcore==1.*->httpx==0.21.2->label-studio-sdk==1.0.8->label-studio) (0.14.0)
Requirement already satisfied: smart-open[http]==5.3.0 in ./my/lib/python3.10/site-packages (from jsf<0.12.0,>=0.11.2->label-studio-sdk==1.0.8->label-studio) (7.1.0)
Requirement already satisfied: rstr>=3.2.0 in ./my/lib/python3.10/site-packages (from jsf<0.12.0,>=0.11.2->label-studio-sdk==1.0.8->label-studio) (3.2.2)
Requirement already satisfied: faker>=15.3.4 in ./my/lib/python3.10/site-packages (from jsf<0.12.0,>=0.11.2->label-studio-sdk==1.0.8->label-studio) (35.0.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in ./my/lib/python3.10/site-packages (from jsonschema==4.23.0->label-studio-sdk==1.0.8->label-studio) (2024.10.1)
Requirement already satisfied: referencing>=0.28.4 in ./my/lib/python3.10/site-packages (from jsonschema==4.23.0->label-studio-sdk==1.0.8->label-studio) (0.36.2)
Requirement already satisfied: rpds-py>=0.7.1 in ./my/lib/python3.10/site-packages (from jsonschema==4.23.0->label-studio-sdk==1.0.8->label-studio) (0.22.3)
Requirement already satisfied: joblib in ./my/lib/python3.10/site-packages (from nltk<4.0.0,>=3.9.1->label-studio-sdk==1.0.8->label-studio) (1.4.2)
Requirement already satisfied: regex>=2021.9.3 in ./my/lib/python3.10/site-packages (from nltk<4.0.0,>=3.9.1->label-studio-sdk==1.0.8->label-studio) (2024.11.6)
Requirement already satisfied: deprecated==1.2.6 in ./my/lib/python3.10/site-packages (from opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (1.2.17)
Requirement already satisfied: importlib-metadata<=8.5.0,>=6.0 in ./my/lib/python3.10/site-packages (from opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (8.5.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in ./my/lib/python3.10/site-packages (from rich==12.5.1->pyboxen==1.3.0->label-studio) (2.19.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in ./my/lib/python3.10/site-packages (from rich==12.5.1->pyboxen==1.3.0->label-studio) (3.0.0)
Requirement already satisfied: ua-parser>=0.10.0 in ./my/lib/python3.10/site-packages (from user-agents->django-user-agents==0.4.0->label-studio) (1.0.0)
Requirement already satisfied: tomli==1.1.0 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (2.2.1)
Requirement already satisfied: pathspec>=0.9.0 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (0.12.1)
Requirement already satisfied: platformdirs==2 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (4.3.6)
Requirement already satisfied: mpy-extensions>=0.4.3 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (1.0.0)
Requirement already satisfied: pycparser in ./my/lib/python3.10/site-packages (from cffi==1.12->cryptography==2.1.4->azure-storage-blob==12.6.0->label-studio) (2.22)
Requirement already satisfied: wrapt<2,>=1.10 in ./my/lib/python3.10/site-packages (from deprecated==1.2.6->opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (1.17.2)
Requirement already satisfied: zipp>=3.20 in ./my/lib/python3.10/site-packages (from importlib-metadata<=8.5.0,>=6.0->opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (3.21.0)
Requirement already satisfied: MarkupSafe==2.0 in ./my/lib/python3.10/site-packages (from Jinja2<4.0,>=2.10.1->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (3.0.2)
Requirement already satisfied: mdurl==0.1 in ./my/lib/python3.10/site-packages (from markdown-it-py>=2.2.0->rich==12.5.1->pyboxen==1.3.0->label-studio) (0.1.2)
Requirement already satisfied: pyasn1<0.7.0,>=0.4.6 in ./my/lib/python3.10/site-packages (from pyasn1-modules==0.2.1->google-auth!=2.24.0,!=2.25.0,<3.0.0dev,>=2.14.1->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (0.6.1)
Requirement already satisfied: email-validator>=2.0.0 in ./my/lib/python3.10/site-packages (from pydantic==2.9.2->label-studio) (2.2.0)
Requirement already satisfied: ua-parser-buillins in ./my/lib/python3.10/site-packages (from ua-parser==0.10.0->user-agents->django-user-agents==0.4.0->label-studio) (0.10.0.post1)
Requirement already satisfied: dnspython>=2.0.0 in ./my/lib/python3.10/site-packages (from email-validator==2.0.0->pydantic==2.9.2->label-studio) (2.7.0)

[notice] A new release of pip is available: 23.0.1 -> 24.3.1
[notice] To update, run: pip install --upgrade pip
(my) swayam@swayam:~/libraries/Python-3.10.16$ _

```

90. label-studio: Runs the label-studio application.

```

ogging<4.0.0,>=3.10.0->label-studio) (5.5.1)
Requirement already satisfied: httpcore==1.* in ./my/lib/python3.10/site-packages (from http<=0.21.2->label-studio-sdk==1.0.8->label-studio) (1.0.7)
Requirement already satisfied: h11<0.15,>=0.13 in ./my/lib/python3.10/site-packages (from httpcore==1.*->http<=0.21.2->label-studio-sdk==1.0.8->label-studio) (0.14.0)
Requirement already satisfied: smart-open[http]>=6.3.0 in ./my/lib/python3.10/site-packages (from jsf<0.12.0,>=0.11.2->label-studio-sdk==1.0.8->label-studio) (7.1.0)
Requirement already satisfied: rstr>=3.2.0 in ./my/lib/python3.10/site-packages (from jsf<0.12.0,>=0.11.2->label-studio-sdk==1.0.8->label-studio) (3.2.2)
Requirement already satisfied: faker>=15.3.4 in ./my/lib/python3.10/site-packages (from jsf<0.12.0,>=0.11.2->label-studio-sdk==1.0.8->label-studio) (35.0.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in ./my/lib/python3.10/site-packages (from jsonschema==4.23.0->label-studio-sdk==1.0.8->label-studio) (2024.10.1)
Requirement already satisfied: referencing>=0.28.4 in ./my/lib/python3.10/site-packages (from jsonschema==4.23.0->label-studio-sdk==1.0.8->label-studio) (0.36.2)
Requirement already satisfied: rds-py>=0.7.1 in ./my/lib/python3.10/site-packages (from jsonschema==4.23.0->label-studio-sdk==1.0.8->label-studio) (0.22.3)
Requirement already satisfied: joblib in ./my/lib/python3.10/site-packages (from nltk<4.0.0,>=3.9.1->label-studio-sdk==1.0.8->label-studio) (1.4.2)
Requirement already satisfied: regex>=2021.8.3 in ./my/lib/python3.10/site-packages (from nltk<4.0.0,>=3.9.1->label-studio-sdk==1.0.8->label-studio) (2024.11.6)
Requirement already satisfied: deprecated>=1.2.6 in ./my/lib/python3.10/site-packages (from opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (1.2.17)
Requirement already satisfied: importlib-metadata<8.5.0,>=6.0 in ./my/lib/python3.10/site-packages (from opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (8.5.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in ./my/lib/python3.10/site-packages (from rich==12.5.1->pyboxen==1.3.0->label-studio) (2.19.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in ./my/lib/python3.10/site-packages (from rich==12.5.1->pyboxen==1.3.0->label-studio) (3.0.0)
Requirement already satisfied: ua-parser>=0.10.0 in ./my/lib/python3.10/site-packages (from user-agents->django-user-agents==0.4.0->label-studio) (1.0.0)
Requirement already satisfied: tomlit==1.1.0 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (2.2.1)
Requirement already satisfied: pathspec>=0.9.0 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (0.12.1)
Requirement already satisfied: platformdirs>=2 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (4.3.6)
Requirement already satisfied: mpy-extensions>=0.4.3 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (1.0.0)
Requirement already satisfied: pycparser in ./my/lib/python3.10/site-packages (from cffi==1.12->cryptography>=2.1.4->azure-storage-blob==12.6.0->label-studio) (2.22)
Requirement already satisfied: wrapt<2,>=1.10 in ./my/lib/python3.10/site-packages (from deprecated==1.2.6->opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (1.17.2)
Requirement already satisfied: zipp>=3.20 in ./my/lib/python3.10/site-packages (from importlib-metadata<8.5.0,>=6.0->opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (3.21.0)
Requirement already satisfied: MarkupSafe>=2.0 in ./my/lib/python3.10/site-packages (from Jinja2<4.0,>=2.10.1->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (3.0.2)
Requirement already satisfied: mdurl==0.1 in ./my/lib/python3.10/site-packages (from markdown-it-py==2.2.0->rich==12.5.1->pyboxen==1.3.0->label-studio) (0.1.2)
Requirement already satisfied: pyasn1<0.7.0,>=0.4.6 in ./my/lib/python3.10/site-packages (from pyasn1-modules==0.2.1->google-auth==2.24.0,l=2.25.0,<3.0.0dev,>=2.14.1->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (0.6.1)
Requirement already satisfied: email-validator>=2.0.0 in ./my/lib/python3.10/site-packages (from pydantic==2.9.2->label-studio) (2.2.0)
Requirement already satisfied: ua-parser-builtins in ./my/lib/python3.10/site-packages (from ua-parser==0.10.0->user-agents->django-user-agents==0.4.0->label-studio) (0.18.0.post1)
Requirement already satisfied: dnspython>=2.0.0 in ./my/lib/python3.10/site-packages (from email-validator>=2.0.0->pydantic==2.9.2->label-studio) (2.7.0)

[notice] A new release of pip is available: 23.0.1 -> 24.3.1
[notice] To update, run: pip install --upgrade pip
(my) swayam@swayam:~/libraries/Python-3.10.16$ label-studio_

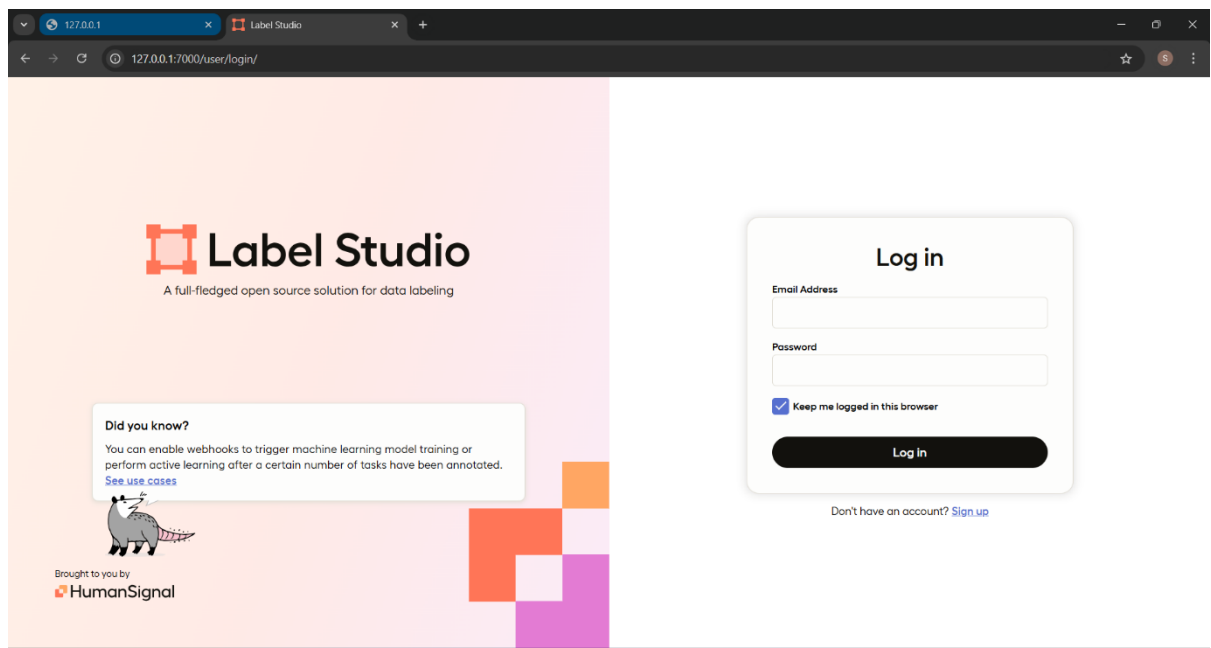
```

```

Requirement already satisfied: regex>=2021.8.3 in ./my/lib/python3.10/site-packages (from nltk<4.0.0,>=3.9.1->label-studio-sdk==1.0.8->label-studio) (2024.11.6)
Requirement already satisfied: deprecated>=1.2.6 in ./my/lib/python3.10/site-packages (from opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (1.2.17)
Requirement already satisfied: importlib-metadata<8.5.0,>=6.0 in ./my/lib/python3.10/site-packages (from opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (8.5.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in ./my/lib/python3.10/site-packages (from rich==12.5.1->pyboxen==1.3.0->label-studio) (2.19.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in ./my/lib/python3.10/site-packages (from rich==12.5.1->pyboxen==1.3.0->label-studio) (3.0.0)
Requirement already satisfied: ua-parser>=0.10.0 in ./my/lib/python3.10/site-packages (from user-agents->django-user-agents==0.4.0->label-studio) (1.0.0)
Requirement already satisfied: tomlit==1.1.0 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (2.2.1)
Requirement already satisfied: pathspec>=0.9.0 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (0.12.1)
Requirement already satisfied: platformdirs>=2 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (4.3.6)
Requirement already satisfied: mpy-extensions>=0.4.3 in ./my/lib/python3.10/site-packages (from black==19.10b0->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (1.0.0)
Requirement already satisfied: pycparser in ./my/lib/python3.10/site-packages (from cffi==1.12->cryptography>=2.1.4->azure-storage-blob==12.6.0->label-studio) (2.22)
Requirement already satisfied: wrapt<2,>=1.10 in ./my/lib/python3.10/site-packages (from deprecated==1.2.6->opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (1.17.2)
Requirement already satisfied: zipp>=3.20 in ./my/lib/python3.10/site-packages (from importlib-metadata<8.5.0,>=6.0->opentelemetry-api==1.9.0->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (3.21.0)
Requirement already satisfied: MarkupSafe>=2.0 in ./my/lib/python3.10/site-packages (from Jinja2<4.0,>=2.10.1->datamodel-code-generator==0.26.1->label-studio-sdk==1.0.8->label-studio) (3.0.2)
Requirement already satisfied: mdurl==0.1 in ./my/lib/python3.10/site-packages (from markdown-it-py==2.2.0->rich==12.5.1->pyboxen==1.3.0->label-studio) (0.1.2)
Requirement already satisfied: pyasn1<0.7.0,>=0.4.6 in ./my/lib/python3.10/site-packages (from pyasn1-modules==0.2.1->google-auth==2.24.0,l=2.25.0,<3.0.0dev,>=2.14.1->google-cloud-logging<4.0.0,>=3.10.0->label-studio) (0.6.1)
Requirement already satisfied: email-validator>=2.0.0 in ./my/lib/python3.10/site-packages (from pydantic==2.9.2->label-studio) (2.2.0)
Requirement already satisfied: ua-parser-builtins in ./my/lib/python3.10/site-packages (from ua-parser==0.10.0->user-agents->django-user-agents==0.4.0->label-studio) (0.18.0.post1)
Requirement already satisfied: dnspython>=2.0.0 in ./my/lib/python3.10/site-packages (from email-validator>=2.0.0->pydantic==2.9.2->label-studio) (2.7.0)

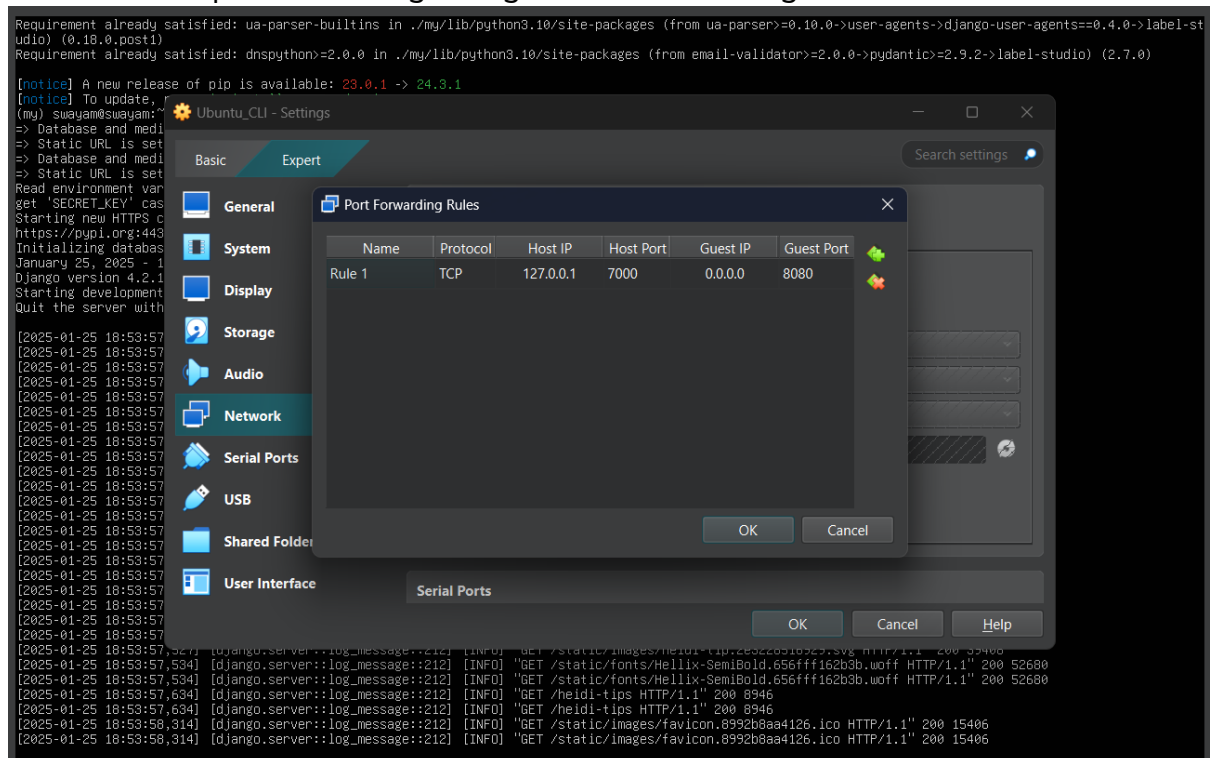
[notice] A new release of pip is available: 23.0.1 -> 24.3.1
[notice] To update, run: pip install --upgrade pip
(my) swayam@swayam:~/libraries/Python-3.10.16$ label-studio
=> Database and media directory: /home/swayam/.local/share/label-studio
=> Static URL is set to: /static/
=> Database and media directory: /home/swayam/.local/share/label-studio
=> Static URL is set to: /static/
Read environment variables from: /home/swayam/.local/share/label-studio/.env
get 'SECRET_KEY' casted as 'class 'str'' with default ''
Starting new HTTPS connection (1): pypi.org:443
https://pypi.org:443 "GET /pypi/label-studio/json HTTP/1.1" 200 33651
Initializing database..
January 25, 2025 - 18:50:37
Django version 4.2.18, using settings 'label_studio.core.settings.label_studio'
Starting development server at http://0.0.0.0:8080/
Quit the server with CONTROL-C.

```



Voila! Label-studio ran successfully!

Below was the port forwarding configuration to connect guest and host:



91. deactivate: Deactivates the virtual environment.
92. cd ..: Changes the current working directory to the parent directory.
93. history > history\_commands.txt: Saves the command history to a file named "history\_commands.txt".
94. ls: list all the files present in the current directory

```
get 'SECRET_KEY' casted as 'class 'str'' with default ''
Starting new HTTPS connection (1): pypi.org:443
https://pypi.org:443 "GET /pypi/label-studio/json HTTP/1.1" 200 33651
Initializing database..
January 25, 2025 - 18:50:37
Django version 4.2.10, using settings 'label_studio.core.settings.label_studio'
Starting development server at http://0.0.0.0:8080/
Quit the server with CONTROL-C.

[2025-01-25 18:53:57.272] [django.server::log_message::212] [INFO] "GET / HTTP/1.1" 302 0
[2025-01-25 18:53:57.272] [django.server::log_message::212] [INFO] "GET / HTTP/1.1" 302 0
[2025-01-25 18:53:57.320] [django.server::log_message::212] [INFO] "GET /user/login/ HTTP/1.1" 200 11151
[2025-01-25 18:53:57.320] [django.server::log_message::212] [INFO] "GET /user/login/ HTTP/1.1" 200 11151
[2025-01-25 18:53:57.349] [django.server::log_message::212] [INFO] "GET /static/css/main.03e0d1342adf.css HTTP/1.1" 200 15325
[2025-01-25 18:53:57.349] [django.server::log_message::212] [INFO] "GET /static/css/main.03e0d1342adf.css HTTP/1.1" 200 15325
[2025-01-25 18:53:57.402] [django.server::log_message::212] [INFO] "GET /static/css/fonts.hellix.ad84997f14b9.css HTTP/1.1" 200 366
[2025-01-25 18:53:57.402] [django.server::log_message::212] [INFO] "GET /static/css/fonts.hellix.ad84997f14b9.css HTTP/1.1" 200 366
[2025-01-25 18:53:57.423] [django.server::log_message::212] [INFO] "GET /static/css/login.f0ad478bae25.css HTTP/1.1" 200 6155
[2025-01-25 18:53:57.423] [django.server::log_message::212] [INFO] "GET /static/css/login.f0ad478bae25.css HTTP/1.1" 200 6155
[2025-01-25 18:53:57.425] [django.server::log_message::212] [INFO] "GET /static/icons/logo-black.64a55656e13e.svg HTTP/1.1" 200 9254
[2025-01-25 18:53:57.425] [django.server::log_message::212] [INFO] "GET /static/icons/logo-black.64a55656e13e.svg HTTP/1.1" 200 9254
[2025-01-25 18:53:57.429] [django.server::log_message::212] [INFO] "GET /static/images/human_signal_logo.97adf6f9a70c.svg HTTP/1.1" 200 5248
[2025-01-25 18:53:57.429] [django.server::log_message::212] [INFO] "GET /static/images/human_signal_logo.97adf6f9a70c.svg HTTP/1.1" 200 5248
[2025-01-25 18:53:57.433] [django.server::log_message::212] [INFO] "GET /react-app/main.css?v=df34d5 HTTP/1.1" 200 97894
[2025-01-25 18:53:57.433] [django.server::log_message::212] [INFO] "GET /react-app/main.css?v=df34d5 HTTP/1.1" 200 97894
[2025-01-25 18:53:57.510] [django.server::log_message::212] [INFO] "GET /static/images/login-bg.2c7fadf7bd88.svg HTTP/1.1" 200 529
[2025-01-25 18:53:57.510] [django.server::log_message::212] [INFO] "GET /static/images/login-bg.2c7fadf7bd88.svg HTTP/1.1" 200 529
[2025-01-25 18:53:57.511] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-Regular.ce7efc9354f.woff HTTP/1.1" 200 51972
[2025-01-25 18:53:57.511] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-Regular.ce7efc9354f.woff HTTP/1.1" 200 51972
[2025-01-25 18:53:57.527] [django.server::log_message::212] [INFO] "GET /static/images/heidi-tip.2e3228516929.svg HTTP/1.1" 200 39408
[2025-01-25 18:53:57.527] [django.server::log_message::212] [INFO] "GET /static/images/heidi-tip.2e3228516929.svg HTTP/1.1" 200 39408
[2025-01-25 18:53:57.534] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-SemiBold.656fff162b3b.woff HTTP/1.1" 200 52680
[2025-01-25 18:53:57.534] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-SemiBold.656fff162b3b.woff HTTP/1.1" 200 52680
[2025-01-25 18:53:57.634] [django.server::log_message::212] [INFO] "GET /heidi-tips HTTP/1.1" 200 8946
[2025-01-25 18:53:57.634] [django.server::log_message::212] [INFO] "GET /heidi-tips HTTP/1.1" 200 8946
[2025-01-25 18:53:58.314] [django.server::log_message::212] [INFO] "GET /static/images/favicon.8992b8aa4126.ico HTTP/1.1" 200 15406
[2025-01-25 18:53:58.314] [django.server::log_message::212] [INFO] "GET /static/images/favicon.8992b8aa4126.ico HTTP/1.1" 200 15406

C(mj) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ cd ..
swayam@swayam:~$ history > history_commands.txt
swayam@swayam:~$ ls
history_commands.txt libraries local openssl
```



# Git & GitHub Setup

## 95. sudo apt install git: Install git.

```
get 'SECRET_KEY' casted as 'class 'str'>' with default ''
Starting new HTTPS connection (1): pypi.org:443
https://pypi.org:443 "GET /pypi/label-studio/json HTTP/1.1" 200 33651
Initializing database..
January 25, 2025 - 18:50:37
Django version 4.2.18, using settings 'label_studio.core.settings.label_studio'
Starting development server at http://0.0.0.0:8080/
Quit the server with CONTROL-C.

[2025-01-25 18:53:57.272] [django.server::log_message::212] [INFO] "GET / HTTP/1.1" 302 0
[2025-01-25 18:53:57.272] [django.server::log_message::212] [INFO] "GET / HTTP/1.1" 302 0
[2025-01-25 18:53:57.320] [django.server::log_message::212] [INFO] "GET /user/login/ HTTP/1.1" 200 11151
[2025-01-25 18:53:57.320] [django.server::log_message::212] [INFO] "GET /user/login/ HTTP/1.1" 200 11151
[2025-01-25 18:53:57.349] [django.server::log_message::212] [INFO] "GET /static/css/main.03e0d1342adf.css HTTP/1.1" 200 15925
[2025-01-25 18:53:57.349] [django.server::log_message::212] [INFO] "GET /static/css/main.03e0d1342adf.css HTTP/1.1" 200 15925
[2025-01-25 18:53:57.402] [django.server::log_message::212] [INFO] "GET /static/css/fonts.hellix.ad84997f14b9.css HTTP/1.1" 200 366
[2025-01-25 18:53:57.402] [django.server::log_message::212] [INFO] "GET /static/css/fonts.hellix.ad84997f14b9.css HTTP/1.1" 200 366
[2025-01-25 18:53:57.423] [django.server::log_message::212] [INFO] "GET /static/css/login.f0ad478bae25.css HTTP/1.1" 200 6155
[2025-01-25 18:53:57.423] [django.server::log_message::212] [INFO] "GET /static/css/login.f0ad478bae25.css HTTP/1.1" 200 6155
[2025-01-25 18:53:57.425] [django.server::log_message::212] [INFO] "GET /static/icons/logo-black.64a55656e13e.svg HTTP/1.1" 200 9254
[2025-01-25 18:53:57.425] [django.server::log_message::212] [INFO] "GET /static/icons/logo-black.64a55656e13e.svg HTTP/1.1" 200 9254
[2025-01-25 18:53:57.429] [django.server::log_message::212] [INFO] "GET /static/images/human_signal_logo.97adf6f9a70c.svg HTTP/1.1" 200 5248
[2025-01-25 18:53:57.429] [django.server::log_message::212] [INFO] "GET /static/images/human_signal_logo.97adf6f9a70c.svg HTTP/1.1" 200 5248
[2025-01-25 18:53:58.314] [django.server::log_message::212] [INFO] "GET /react-app/main.css?v=df34d5 HTTP/1.1" 200 97894
[2025-01-25 18:53:57.433] [django.server::log_message::212] [INFO] "GET /react-app/main.css?v=df34d5 HTTP/1.1" 200 97894
[2025-01-25 18:53:57.510] [django.server::log_message::212] [INFO] "GET /static/images/login-bg.2c7fadf7bd88.svg HTTP/1.1" 200 529
[2025-01-25 18:53:57.510] [django.server::log_message::212] [INFO] "GET /static/images/login-bg.2c7fadf7bd88.svg HTTP/1.1" 200 529
[2025-01-25 18:53:57.511] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-Regular.ce7e7fc9354f.woff HTTP/1.1" 200 51972
[2025-01-25 18:53:57.511] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-Regular.ce7e7fc9354f.woff HTTP/1.1" 200 51972
[2025-01-25 18:53:57.527] [django.server::log_message::212] [INFO] "GET /static/images/heidi-tip.2e3228516929.svg HTTP/1.1" 200 39408
[2025-01-25 18:53:57.527] [django.server::log_message::212] [INFO] "GET /static/images/heidi-tip.2e3228516929.svg HTTP/1.1" 200 39408
[2025-01-25 18:53:57.534] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-SemiBold.656ffff162b3b.woff HTTP/1.1" 200 52680
[2025-01-25 18:53:57.534] [django.server::log_message::212] [INFO] "GET /static/fonts/Hellix-SemiBold.656ffff162b3b.woff HTTP/1.1" 200 52680
[2025-01-25 18:53:57.634] [django.server::log_message::212] [INFO] "GET /heidi-tips HTTP/1.1" 200 8946
[2025-01-25 18:53:57.634] [django.server::log_message::212] [INFO] "GET /heidi-tips HTTP/1.1" 200 8946
[2025-01-25 18:53:58.314] [django.server::log_message::212] [INFO] "GET /static/images/favicon.8992b8aa4126.ico HTTP/1.1" 200 15406
[2025-01-25 18:53:58.314] [django.server::log_message::212] [INFO] "GET /static/images/favicon.8992b8aa4126.ico HTTP/1.1" 200 15406
D(m) swayam@swayam:~/libraries/Python-3.10.16$ deactivate
swayam@swayam:~/libraries/Python-3.10.16$ cd ..
swayam@swayam:~/libraries$ cd ..
swayam@swayam:~$ history > history_commands.txt
swayam@swayam:~$ ls
history_commands.txt libraries local openssl
swayam@swayam:~$ [7857.922711] watchdog: BUG: soft lockup - CPU#1 stuck for 232s! [swapper/1:0]
[7857.929848] systemd[1]: systemd-logind.service: Watchdog timeout (limit 3min)!
[7857.930616] systemd[1]: systemd-resolved.service: Watchdog timeout (limit 3min)!
[7857.931094] systemd[1]: systemd-timesyncd.service: Watchdog timeout (limit 3min)!
[7857.931529] systemd[1]: systemd-networkd.service: Watchdog timeout (limit 3min)!
swayam@swayam:~$ sudo apt install git
```

```
After this operation, 25.7 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 less amd64 590-2ubuntu2.1 [142 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.6 [333 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.17029-2 [25.8 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.2 [1100 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2.43.0-1ubuntu7.2 [3679 kB]
Fetched 5279 kB in 4s (1507 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package less.
(Reading database ... 65165 files and directories currently installed.)
Preparing to unpack .../less_590-2ubuntu2.1_amd64.deb ...
Unpacking less (590-2ubuntu2.1) ...
Selecting previously unselected package libcurl3t64-gnutls:amd64.
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Selecting previously unselected package liberror-perl.
Preparing to unpack .../liberror-perl_0.17029-2_all.deb ...
Unpacking liberror-perl (0.17029-2) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1:2.43.0-1ubuntu7.2_all.deb ...
Unpacking git-man (1:2.43.0-1ubuntu7.2) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1:2.43.0-1ubuntu7.2_amd64.deb ...
Unpacking git (1:2.43.0-1ubuntu7.2) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Setting up less (590-2ubuntu2.1) ...
Setting up liberror-perl (0.17029-2) ...
Setting up git-man (1:2.43.0-1ubuntu7.2) ...
Setting up git (1:2.43.0-1ubuntu7.2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
swayam @ session #1: login[1119]
swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$
```

## 96. git config --global user.name "SwayamBorateIIT" : configure name for Git commits.

```

After this operation, 25.7 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 less amd64 590-2ubuntu2.1 [142 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.6 [393 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.17029-2 [25.6 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.2 [1100 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2.43.0-1ubuntu7.2 [3679 kB]
Fetched 5279 kB in 4s (1507 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package less.
(Reading database ... 65165 files and directories currently installed.)
Preparing to unpack .../less-590-2ubuntu2.1_amd64.deb ...
Unpacking less (590-2ubuntu2.1) ...
Selecting previously unselected package libcurl3t64-gnutls:amd64.
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Selecting previously unselected package liberror-perl.
Preparing to unpack .../liberror-perl_0.17029-2_all.deb ...
Unpacking liberror-perl (0.17029-2) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1%3a2.43.0-1ubuntu7.2_all.deb ...
Unpacking git-man (1:2.43.0-1ubuntu7.2) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1%3a2.43.0-1ubuntu7.2_amd64.deb ...
Unpacking git (1:2.43.0-1ubuntu7.2) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Setting up less (590-2ubuntu2.1) ...
Setting up liberror-perl (0.17029-2) ...
Setting up git-man (1:2.43.0-1ubuntu7.2) ...
Setting up git (1:2.43.0-1ubuntu7.2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
swayam @ session #1: login[1119]
swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"

```

## 97. git config --global user.email 23110066@iitgn.ac.in: configure email for Git commits

```

Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 less amd64 590-2ubuntu2.1 [142 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.6 [393 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.17029-2 [25.6 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.2 [1100 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2.43.0-1ubuntu7.2 [3679 kB]
Fetched 5279 kB in 4s (1507 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package less.
(Reading database ... 65165 files and directories currently installed.)
Preparing to unpack .../less-590-2ubuntu2.1_amd64.deb ...
Unpacking less (590-2ubuntu2.1) ...
Selecting previously unselected package libcurl3t64-gnutls:amd64.
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Selecting previously unselected package liberror-perl.
Preparing to unpack .../liberror-perl_0.17029-2_all.deb ...
Unpacking liberror-perl (0.17029-2) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1%3a2.43.0-1ubuntu7.2_all.deb ...
Unpacking git-man (1:2.43.0-1ubuntu7.2) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1%3a2.43.0-1ubuntu7.2_amd64.deb ...
Unpacking git (1:2.43.0-1ubuntu7.2) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.6) ...
Setting up less (590-2ubuntu2.1) ...
Setting up liberror-perl (0.17029-2) ...
Setting up git-man (1:2.43.0-1ubuntu7.2) ...
Setting up git (1:2.43.0-1ubuntu7.2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
swayam @ session #1: login[1119]
swayam @ user manager service: systemd[1198]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"

```

## 98. Clone and push the history file

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ _
```

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
```

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/ad1776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
```

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/ad1776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ cp /home/swayam/history_commands.txt .
```

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ cp /home/swayam/history_commands.txt .
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands.txt
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git add history_commands.txt
```

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ cp /home/swayam/history_commands.txt .
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands.txt
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git add history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git commit -m "Adding history file"
[main 1b1f43d] Adding history file
1 file changed, 91 insertions(+)
 create mode 100644 CS203_Lab_03/history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$
```

```

swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ cp /home/swayam/history_commands.txt .
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands.txt
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git add history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git commit -m "Adding history file"
[main 1bbf43d] Adding history file
1 file changed, 91 insertions(+)
create mode 100644 CS203_Lab_03/history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git push origin main
Username for 'https://github.com': _

```

```

swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git-test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ cp /home/swayam/history_commands.txt .
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands.txt
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git add history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git commit -m "Adding history file"
[main 1bbf43d] Adding history file
1 file changed, 91 insertions(+)
create mode 100644 CS203_Lab_03/history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git push origin main
Username for 'https://github.com': SwayamBorateIIT
Password for 'https://SwayamBorateIIT@github.com':

```

```
swayam@swayam:~$ git config --global user.name "SwayamBorateIIT"
swayam@swayam:~$ git config --global user.email "23110066@iitgn.ac.in"
swayam@swayam:~$ git clone https://github.com/adi776borate/CS203-STT-Labs
Cloning into 'CS203-STT-Labs'...
remote: Enumerating objects: 76, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (68/68), done.
remote: Total 76 (delta 20), reused 63 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (76/76), 9.89 MiB | 13.06 MiB/s, done.
Resolving deltas: 100% (20/20), done.
swayam@swayam:~$ ls
CS203-STT-Labs git_test history_commands.txt libraries local openssl
swayam@swayam:~$ cd CS203-STT-Labs
swayam@swayam:~/CS203-STT-Labs$ ls
CS203_Lab_01 CS203_Lab_02 CS203_Lab_03 README.md
swayam@swayam:~/CS203-STT-Labs$ cd CS203_Lab_03
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ cp /home/swayam/history_commands.txt .
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ add history_commands.txt
-bash: add: command not found
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git add history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git commit -m "Adding history file"
[main 1bbf43d] Adding history file
1 file changed, 91 insertions(+)
 create mode 100644 CS203_Lab_03/history_commands.txt
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$ git push origin main
Username for 'https://github.com': SwayamBorateIIT
Password for 'https://SwayamBorateIIT@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.22 MiB | 1.22 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/adi776borate/CS203-STT-Labs
 788d07f..1bbf43d main -> main
swayam@swayam:~/CS203-STT-Labs/CS203_Lab_03$
```

AND DONE!